Stochastic Calculus and Quantitative Finance

A rigorous companion, with intuition

Companion to the *Stochastic Calculus Notes* of Lokesh Parihar

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How to use this booklet

The class notes build one long argument: random walk, Wiener process, Itô calculus, change of measure, and the fundamental theorems. They are a fast road, and they leave several things to faith. Why does the Radon–Nikodym derivative exist? Why does the proof of the first fundamental theorem need more care than it seems to? Why does the market quote a whole smile when Black–Scholes has one volatility? This booklet goes back over that road carefully and then continues past where the class stopped.

It is meant to be *read with a pencil*. The structure is:

- **Part I** gives the measure-theoretic probability the notes use implicitly, and the theory of martingales.
- **Part II** redoes stochastic calculus with proofs: Brownian path properties, the Itô integral, Girsanov, martingale representation, SDEs, Feynman–Kac, optimal stopping, then jumps.
- **Part III** takes up what practice actually needs: the volatility smile, local and stochastic volatility, and interest rates.
- **Part IV** covers how to compute: Monte Carlo, finite differences, trees.
- **Part V** covers risk measures, portfolio theory, financial statistics, and market conventions.

Three kinds of box. Blue boxes give the idea behind a result. Orange boxes flag a mistake that is easy to make. Green boxes report a numerical experiment.

About the numbers. Every number in a green box or a table of results was produced by `code/make_data.py`, which sits beside this document and needs only NumPy. Re-running it reproduces `data/results.txt` exactly. Monte Carlo figures are quoted with their standard errors, and you should expect a closed form to fall within two or three standard errors of its estimate, not exactly on it. The parameter sets are chosen for illustration; none are calibrated to market data.

How rigorous is it? Theorems are stated with their hypotheses. Proofs are complete where a complete proof is short enough to teach something; where the full argument is long (the Radon–Nikodym theorem, the martingale representation theorem, Lévy–Itô) the text says so, gives the main idea, and points to a reference. A statement marked "sketch" is exactly that.

Notation compared with the class notes. The notes use blackboard letters for random objects, which works by hand but gets heavy in print. Here:

Class notes	This booklet	Meaning
$\mathbb{W}(t)$, $\tilde{\mathbb{W}}(t)$	W_t, $W_t^{\mathbb{Q}}$	Wiener process under $\mathbb{P}$, $\mathbb{Q}$
$\mathbb{S}(t)$, $\mathbb{X}(t)$	S_t, X_t	stock, wealth
$\tilde{\mathbb{P}}$, $\tilde{\mathbb{E}}$	$\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}$	risk-neutral measure, expectation
$\mathbb{Z}(t)$	Z_t	Radon–Nikodym derivative process
$\mathbb{D}(t)$	D_t	discount factor
$[\mathbb{W}, \mathbb{W}](t)$	$[W]_t$	quadratic variation
∂_x, ∂_x^2	∂_x, ∂_{xx}	partial derivatives

If you are short of time. A first pass that keeps the logical thread is: Chapters 1 (conditional expectation only), 3, 4, 5, 6, then 8 and 10, then 11, then 12. Chapters 14–17 can be read independently of everything else.

Part I

Foundations of Probability

Chapter 1

Measure-Theoretic Probability

1.1 Why bother?

In the class notes, the coin-toss expectation was $\mathbb{E}[X] = \sum_\omega X(\omega)\mathbb{P}\{\omega\}$. For Brownian motion this formula is useless: Ω is a space of continuous paths, and every individual path has probability zero, so the sum is $0 + 0 + \dots$ over an uncountable set. Something must replace it. Three separate needs drive the machinery.

1. **Integrals over path space.** We need $\mathbb{E}[X]$ when no single outcome carries mass.
2. **Information.** "What we know at time t" must be a mathematical object, and $\mathbb{E}[X \mid \text{what we know}]$ must make sense even when we condition on events of probability zero, such as $\{W_t = x\}$.
3. **Limits.** The Itô integral is a limit of random variables. We must know in which sense it converges and when limits and expectations can be swapped.

There is also a reason we cannot just assign a probability to every subset of Ω. Under the uniform distribution on $[0, 1)$ with translation invariance, a Vitali set has no consistent probability: translated copies of it partition $[0, 1)$ into countably many pieces of equal size, and no number works as that size. So we restrict attention to a well-behaved family of events.

1.2 Probability spaces

Definition 1.1. A *σ-algebra* on Ω is a collection $\mathcal{F}$ of subsets with $\Omega \in \mathcal{F}$, closed under complements and countable unions. A *probability measure* is a map $\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(A_n)$ for pairwise disjoint $A_n \in \mathcal{F}$. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*. The *Borel σ-algebra* $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing all open intervals.

> **Intuition.** Read a σ-algebra as a *list of yes/no questions you can answer*. If you can answer "did A happen?", you can answer "did A not happen?". If you can answer each of a sequence of questions, you can answer "did at least one happen?". For two coin tosses, $\mathcal{F}_1$ answers questions about the first toss only: it cannot separate HH from HT. This is exactly why the class notes' $\mathcal{F}_n$ form a growing family.

Countable additivity gives *continuity*: if $A_n \uparrow A$ then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$, and if $A_n \downarrow A$ then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$. To check that two measures agree, it is enough to check them on a generating family closed under intersections:

Theorem 1.2 (π–λ uniqueness)**.** *If two probability measures agree on a π-system $\mathcal{C}$ (a family closed under finite intersection), they agree on $\sigma(\mathcal{C})$.*

So the law of a real random variable is determined by its CDF, since half-lines $(-\infty, x]$ form a π-system generating $\mathcal{B}(\mathbb{R})$. The law of a process on path space is determined by its finite-dimensional distributions. This is the reason it was enough, in the class notes, to specify W through its increments.

1.3 Random variables and information

Definition 1.3. A *random variable* is a map $X : \Omega \to \mathbb{R}$ with $\{X \in B\} \in \mathcal{F}$ for every Borel B. The σ-algebra $\sigma(X) = \{\{X \in B\} : B \in \mathcal{B}(\mathbb{R})\}$ is the information revealed by X. The *law* of X is $\mu_X(B) = \mathbb{P}(X \in B)$.

Theorem 1.4 (Doob–Dynkin)**.** *Y is $\sigma(X)$-measurable if and only if $Y = g(X)$ for some Borel function g.*

This turns a slogan into a theorem: "Y is known once X is known" means Y is a function of X. It is what lets us write a conditional expectation $\mathbb{E}[h(S_T) \mid S_t]$ as an ordinary function $g(t, S_t)$, the step at the heart of Feynman–Kac.

1.4 The Lebesgue integral

Define $\mathbb{E}[X] = \int_\Omega X \, \mathrm{d}\mathbb{P}$ in three steps.

1. For a simple $X = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ with $a_i \geq 0$: $\mathbb{E}[X] = \sum_i a_i \mathbb{P}(A_i)$.
2. For $X \geq 0$: $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : Y \text{ simple}, \ 0 \leq Y \leq X\} \in [0, \infty]$.
3. For general X: $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$ when at least one is finite.

> **Intuition.** The Riemann integral slices the *domain* and needs X to be nearly constant on each small slice; on path space there is no useful notion of a small slice. Lebesgue slices the *range*: it asks how much probability sits on $\{v \leq X < v + \mathrm{d}v\}$, whatever shape that set has. This is the "sum over range instead of domain" shift mentioned in the class notes.

Three theorems let you exchange limit and expectation. Each is derived from the previous one.

Theorem 1.5 (Monotone convergence)**.** *If $0 \leq X_n \uparrow X$ a.s., then $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

Theorem 1.6 (Fatou's lemma)**.** *If $X_n \geq 0$, then $\mathbb{E}[\liminf_n X_n] \leq \liminf_n \mathbb{E}[X_n]$.*

Proof. Let $Y_n = \inf_{k \geq n} X_k$. Then $Y_n \uparrow \liminf X_n$ and $Y_n \leq X_n$. Monotone convergence gives $\mathbb{E}[\liminf X_n] = \lim \mathbb{E}[Y_n] \leq \liminf \mathbb{E}[X_n]$. □

Theorem 1.7 (Dominated convergence)**.** *If $X_n \to X$ a.s. and $|X_n| \le Y$ with $\mathbb{E}[Y] < \infty$, then $\mathbb{E}[X_n] \to \mathbb{E}[X]$.*

Proof. Apply Fatou to $Y + X_n \ge 0$ and to $Y - X_n \ge 0$. This gives $\mathbb{E}[X] \le \liminf \mathbb{E}[X_n]$ and $\mathbb{E}[X] \ge \limsup \mathbb{E}[X_n]$. □

> **Pitfall.** Mass can escape to infinity. On $[0,1]$ with Lebesgue measure, $X_n = n\mathbf{1}_{(0,1/n)} \to 0$ everywhere while $\mathbb{E}[X_n] = 1$. In finance the same phenomenon appears as the *doubling strategy*: a process can be almost surely eventually at +1 while its expected value stays at zero along the way, because of an unbounded potential loss. The inequality in Fatou is why positive local martingales are only supermartingales (Proposition 2.12).

Law of the unconscious statistician. $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x)\,\mu_X(\mathrm{d}x)$. Prove it for indicators, extend to simple functions, then pass to limits with monotone convergence. This "standard machine" proves most identities about integrals.

1.5 Inequalities and L^p spaces

Write $\|X\|_p = \mathbb{E}[|X|^p]^{1/p}$ and let L^p be the space of X with $\|X\|_p < \infty$ (identifying a.s. equal variables).

- **Markov:** for $X \ge 0$, $a > 0$: $\mathbb{P}(X \ge a) \le \mathbb{E}[X]/a$. Proof: $a\mathbf{1}_{\{X \ge a\}} \le X$.
- **Chebyshev:** $\mathbb{P}(|X - \mathbb{E}X| \ge a) \le \mathrm{Var}(X)/a^2$.
- **Jensen:** for convex φ, $\varphi(\mathbb{E}X) \le \mathbb{E}\varphi(X)$. Proof: take a supporting line $\ell(x) = \varphi(\mathbb{E}X) + c(x - \mathbb{E}X) \le \varphi(x)$, then take expectations.
- **Hölder:** $\mathbb{E}|XY| \le \|X\|_p \|Y\|_q$ for $1/p + 1/q = 1$. With $p = q = 2$ this is Cauchy–Schwarz.
- **Minkowski:** $\|X + Y\|_p \le \|X\|_p + \|Y\|_p$.

The important fact is that L^2 is a complete inner product space (a Hilbert space) with $\langle X, Y\rangle = \mathbb{E}[XY]$. Completeness is exactly why the Itô integral exists: Cauchy sequences of approximations must have a limit.

1.6 Modes of convergence

Definition 1.8. Let X_n, X be random variables.

- $X_n \to X$ *almost surely* if $\mathbb{P}(\lim X_n = X) = 1$.
- $X_n \to X$ *in probability* if $\mathbb{P}(|X_n - X| > \epsilon) \to 0$ for all $\epsilon > 0$.
- $X_n \to X$ *in L^p* if $\mathbb{E}|X_n - X|^p \to 0$.
- $X_n \to X$ *in distribution* if $\mathbb{E}f(X_n) \to \mathbb{E}f(X)$ for all bounded continuous f.

The implications are: a.s. $\Rightarrow$ in probability $\Rightarrow$ in distribution, and $L^p \Rightarrow$ in probability. Neither a.s. nor L^p implies the other. Two counterexamples on $[0,1]$ are worth remembering. $n\mathbf{1}_{(0,1/n)} \to 0$ a.s. but not in L^1. The "typewriter" sequence $\mathbf{1}_{[0,1]}, \mathbf{1}_{[0,1/2]}, \mathbf{1}_{[1/2,1]}, \mathbf{1}_{[0,1/4]}, \ldots$

tends to 0 in every L^p but converges at no point. It is true, though, that convergence in probability implies a.s. convergence along a subsequence.

The Itô integral is built as an L^2 limit, and quadratic variation is an L^2 limit. Almost sure statements are recovered along fast subsequences, using the next lemma.

Lemma 1.9 (Borel–Cantelli)**.** *Let* $\{A_n \text{ i.o.}\} = \bigcap_m \bigcup_{n\ge m} A_n$.

1. *If* $\sum_n \mathbb{P}(A_n) < \infty$, *then* $\mathbb{P}(A_n \text{ i.o.}) = 0$.
2. *If the* A_n *are independent and* $\sum_n \mathbb{P}(A_n) = \infty$, *then* $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof. (1) $\mathbb{P}(\bigcup_{n\ge m} A_n) \le \sum_{n\ge m} \mathbb{P}(A_n) \to 0$. (2) By independence and $1 - x \le e^{-x}$, $\mathbb{P}(\bigcap_{n=m}^{M} A_n^c) = \prod_{n=m}^{M}(1 - \mathbb{P}(A_n)) \le \exp(-\sum_{n=m}^{M} \mathbb{P}(A_n)) \to 0$ as $M \to \infty$. □

Definition 1.10. A family $\{X_i\}$ is *uniformly integrable* (UI) if $\lim_{c\to\infty} \sup_i \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i|>c\}}] = 0$.

Theorem 1.11 (Vitali)**.** *$X_n \to X$ in L^1 if and only if $X_n \to X$ in probability and $\{X_n\}$ is UI.*

UI is the sharp condition that rules out mass escaping to infinity. Bounded families in L^p for $p > 1$ are UI, and so is every family $\{\mathbb{E}[X \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{F}\}$ with $X \in L^1$. This is why a true martingale on $[0, T]$ closed by X_T is well behaved, while a strict local martingale is not.

1.7 Independence, product measures and limit theorems

σ-algebras $\mathcal{G}_1, \mathcal{G}_2$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for $A \in \mathcal{G}_1, B \in \mathcal{G}_2$. By Theorem 1.2, it suffices to check this on generating π-systems. If X, Y are independent and integrable, then $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$.

Theorem 1.12 (Fubini–Tonelli)**.** *If $f \ge 0$ or $\mathbb{E}|f| < \infty$ on a product space, the integral can be computed iteratively in either order.*

Practically: $\mathbb{E}\int_0^T X_t \,\mathrm{d}t = \int_0^T \mathbb{E}[X_t]\,\mathrm{d}t$ whenever $\mathbb{E}\int_0^T |X_t|\,\mathrm{d}t < \infty$. We use this constantly.

Theorem 1.13 (Strong law)**.** *If the X_i are i.i.d. with $\mathbb{E}|X_1| < \infty$, then $\frac{1}{n}\sum_{i\le n} X_i \to \mathbb{E}X_1$ a.s.*

Theorem 1.14 (Central limit theorem)**.** *If the X_i are i.i.d. with mean μ and variance $\sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \Rightarrow \mathcal{N}(0,1)$.*

Sketch. Let $\varphi(u) = \mathbb{E}e^{iu(X_1-\mu)/\sigma} = 1 - u^2/2 + o(u^2)$. The characteristic function of the normalised sum is $\varphi(u/\sqrt{n})^n \to e^{-u^2/2}$. Lévy's continuity theorem converts convergence of characteristic functions into convergence in distribution. □

These two results are the theory behind Monte Carlo (Chapter 12), and the CLT is why the scaled random walk has a Gaussian limit. A functional version (Donsker's theorem) says the whole path $W^{(p)}$ converges in distribution to Brownian motion on $C[0, T]$. That is the rigorous meaning of the limit written in the class notes.

1.8 Absolute continuity and the Radon–Nikodym theorem

Definition 1.15. $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$, written $\mathbb{Q} \ll \mathbb{P}$, if $\mathbb{P}(A) = 0 \Rightarrow \mathbb{Q}(A) = 0$. They are *equivalent*, $\mathbb{Q} \sim \mathbb{P}$, if each is absolutely continuous with respect to the other. They are *mutually singular* if there is an A with $\mathbb{P}(A) = 1$ and $\mathbb{Q}(A) = 0$.

Theorem 1.16 (Radon–Nikodym)**.** *If $\mathbb{Q} \ll \mathbb{P}$ on $(\Omega, \mathcal{F})$, there is an a.s. unique $Z \geq 0$ with $\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A]$ for all $A \in \mathcal{F}$. We write $Z = \mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$. Moreover $\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[XZ]$ whenever either side is defined, and $\mathbb{Q} \sim \mathbb{P}$ if and only if $Z > 0$ $\mathbb{P}$-a.s.*

Sketch (von Neumann). Let $\mu = \mathbb{P} + \mathbb{Q}$. The map $X \mapsto \mathbb{E}^{\mathbb{Q}}[X]$ is a bounded linear functional on $L^2(\mu)$, because $|\mathbb{E}^{\mathbb{Q}}X| \leq \|X\|_{L^2(\mathbb{Q})} \leq \|X\|_{L^2(\mu)}$. By the Riesz representation theorem there is $g \in L^2(\mu)$ with $\mathbb{E}^{\mathbb{Q}}[X] = \int Xg \,\mathrm{d}\mu$. Taking indicators gives $0 \leq g \leq 1$ μ-a.e. Rearranging, $\int X(1-g)\,\mathrm{d}\mathbb{Q} = \int Xg\,\mathrm{d}\mathbb{P}$. Absolute continuity forces $g < 1$ $\mathbb{Q}$-a.e.; putting $X = Y/(1-g)$ gives $Z = g/(1-g)$. □

> **Intuition.** $Z(\omega)$ is the factor by which $\mathbb{Q}$ inflates or deflates the likelihood of ω relative to $\mathbb{P}$. In a finite world it is just the ratio $\mathbb{Q}\{\omega\}/\mathbb{P}\{\omega\}$ from the class notes. It is a *likelihood ratio*, the same object a statistician uses to compare hypotheses.

Example 1.17 (The homework in the class notes: changing the variance)**.** Let $X \sim \mathcal{N}(0, \sigma^2)$ under $\mathbb{P}$ and let $\mathbb{Q}$ make $X \sim \mathcal{N}(0, s^2)$. The ratio of the two densities gives

$$Z = \frac{\sigma}{s} \exp\Big(-\frac{X^2}{2}\Big(\frac{1}{s^2} - \frac{1}{\sigma^2}\Big)\Big).$$

This is a valid derivative: $Z > 0$ and $\mathbb{E}^{\mathbb{P}}Z = 1$. So for a *single* Gaussian, the variance can be changed.

For a *path* it cannot. Take n i.i.d. increments; the likelihood ratio is a product of n such terms. By the strong law, under $\mathbb{P}$ the realised variance $\frac{1}{n}\sum X_i^2 \to \sigma^2$, and under $\mathbb{Q}$ it tends to s^2. Those two events are disjoint, so the limiting measures are mutually singular. Continuous Brownian paths with volatility σ and with volatility $s \neq \sigma$ live on disjoint sets, distinguished by quadratic variation. That is the deep reason Girsanov's theorem changes the drift only, and why volatility is the same under $\mathbb{P}$ and $\mathbb{Q}$.

1.9 Conditional expectation

This is the most important definition in the booklet.

Definition 1.18. Let $X \in L^1(\mathcal{F})$ and $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. $\mathbb{E}[X \mid \mathcal{G}]$ is a random variable Y with

1. Y is $\mathcal{G}$-measurable,
2. $\mathbb{E}[Y\mathbf{1}_G] = \mathbb{E}[X\mathbf{1}_G]$ for every $G \in \mathcal{G}$.

Theorem 1.19. *$\mathbb{E}[X \mid \mathcal{G}]$ exists and is a.s. unique.*

Proof. Uniqueness. If Y, Y' both work, then $G = \{Y > Y'\} \in \mathcal{G}$ and $\mathbb{E}[(Y - Y')\mathbf{1}_G] = 0$ with a positive integrand on G, so $\mathbb{P}(G) = 0$. Symmetrically, $\mathbb{P}(Y < Y') = 0$.

Existence. Assume $X \geq 0$ (split into $X^{\pm}$ otherwise). On $(\Omega, \mathcal{G})$ define $\nu(G) = \mathbb{E}[X\mathbf{1}_G]$. Then ν is a finite measure with $\nu \ll \mathbb{P}|_{\mathcal{G}}$, and Radon–Nikodym on $\mathcal{G}$ supplies a $\mathcal{G}$-measurable density Y. □

Intuition. For $X \in L^2$, $\mathbb{E}[X \mid \mathcal{G}]$ is the *orthogonal projection* of X onto the closed subspace $L^2(\mathcal{G})$. Condition (2) says $X - Y$ is orthogonal to every indicator in $\mathcal{G}$, hence to all of $L^2(\mathcal{G})$. So $\mathbb{E}[X \mid \mathcal{G}]$ minimises $\mathbb{E}[(X-Y)^2]$ over all $\mathcal{G}$-measurable Y: it is the *best forecast of X given the information $\mathcal{G}$*. This picture also explains the following properties, since a projection onto a smaller subspace after projecting onto a bigger one is the same as projecting straight onto the smaller one.

Proposition 1.20. *For integrable X, Y, sub-σ-algebras $\mathcal{H} \subseteq \mathcal{G}$:*

1. ***Linearity and positivity:*** $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$, *and* $X \geq 0 \Rightarrow \mathbb{E}[X \mid \mathcal{G}] \geq 0$.
2. ***Tower:*** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$. *In particular* $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.
3. ***Taking out what is known:*** *if Y is $\mathcal{G}$-measurable and XY is integrable,* $\mathbb{E}[XY \mid \mathcal{G}] = Y\mathbb{E}[X \mid \mathcal{G}]$.
4. ***Independence:*** *if $\sigma(X)$ is independent of $\mathcal{G}$,* $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.
5. ***Jensen:*** *for convex φ,* $\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}]$.
6. ***Freezing:*** *if X is independent of $\mathcal{G}$, Y is $\mathcal{G}$-measurable and g is bounded, then* $\mathbb{E}[g(X,Y) \mid \mathcal{G}] = h(Y)$, *where* $h(y) = \mathbb{E}[g(X,y)]$.

Proof. All follow by checking Definition 1.18. For (2): for $H \in \mathcal{H} \subseteq \mathcal{G}$, $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbf{1}_H] = \mathbb{E}[X\mathbf{1}_H]$ by the definition for $\mathcal{G}$, and that is the defining identity for $\mathbb{E}[X \mid \mathcal{H}]$. For (3): take $Y = \mathbf{1}_{G'}$ first, where it reduces to the definition with $G \cap G'$, then extend by the standard machine. For (4): $\mathbb{E}[X\mathbf{1}_G] = \mathbb{E}[X]\mathbb{P}(G) = \mathbb{E}[\mathbb{E}[X]\mathbf{1}_G]$. For (6): check it for products $g(x,y) = a(x)b(y)$ using (3) and (4), then extend by a monotone class argument. □

Property (6) is how we actually compute. When the class notes write $\mathbb{E}[(S_T - K)^+ \mid \mathcal{F}_t]$ and treat S_t as a constant x while integrating over the independent increment, that is the freezing lemma.

Theorem 1.21 (Bayes rule for measure changes)**.** *Let $\mathbb{Q} \sim \mathbb{P}$ on $\mathcal{F}$ with density Z, let $\mathcal{G} \subseteq \mathcal{F}$, and let $Z_{\mathcal{G}} = \mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]$. For X with $\mathbb{E}^{\mathbb{Q}}|X| < \infty$,*

$$\mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{G}] = \frac{\mathbb{E}^{\mathbb{P}}[XZ \mid \mathcal{G}]}{Z_{\mathcal{G}}}.$$

Proof. The right side, call it Y, is $\mathcal{G}$-measurable. For $G \in \mathcal{G}$, using the tower property and taking out what is known,

$$\mathbb{E}^{\mathbb{Q}}[Y\mathbf{1}_G] = \mathbb{E}^{\mathbb{P}}[YZ\mathbf{1}_G] = \mathbb{E}^{\mathbb{P}}[Y\mathbf{1}_G\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]] = \mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[XZ \mid \mathcal{G}]\mathbf{1}_G] = \mathbb{E}^{\mathbb{P}}[XZ\mathbf{1}_G] = \mathbb{E}^{\mathbb{Q}}[X\mathbf{1}_G].$$

□

Example 1.22 (Gaussian conditioning)**.** If (X, Y) is jointly Gaussian with $\operatorname{Var} Y > 0$, set $\beta = \operatorname{Cov}(X,Y)/\operatorname{Var}(Y)$. Then $X - \beta Y$ is uncorrelated with Y, hence independent of Y (jointly Gaussian). So

$$\mathbb{E}[X \mid Y] = \mathbb{E}X + \beta(Y - \mathbb{E}Y), \qquad \operatorname{Var}(X \mid Y) = \operatorname{Var} X - \beta^2 \operatorname{Var} Y.$$

Conditional expectation is linear regression, and the conditional variance does not depend on Y. Applied to the Brownian bridge this gives $\mathbb{E}[W_s \mid W_t] = (s/t)W_t$ for $s < t$. The Kalman filter (Section 16.6) is a repeated application of this one formula.

Exercises

1.1 Let $\Omega = \{HH, HT, TH, TT\}$. List $\sigma(M_2)$, where M_2 is the symmetric walk at time 2. Show that $\sigma(M_1, M_2)$ has 16 elements and is strictly larger.

1.2 Prove $\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid \mathcal{G})] + \operatorname{Var}(\mathbb{E}[X \mid \mathcal{G}])$. Use the result to show that conditional expectation reduces variance. This fact is the basis of conditional Monte Carlo.

1.3 Show that if $X_n \to X$ in probability then a subsequence converges a.s. *Hint:* pick n_k with $\mathbb{P}(|X_{n_k} - X| > 2^{-k}) < 2^{-k}$ and use Borel–Cantelli.

1.4 Let $X \sim \mathcal{N}(0,1)$ under $\mathbb{P}$. Find the density Z turning X into $\mathcal{N}(\mu, 1)$. Compute $\mathbb{E}^{\mathbb{P}}[Z^2]$, and interpret it for importance sampling.

1.5 Prove that if $X_n \geq 0$, $X_n \to X$ a.s. and $\mathbb{E}X_n \to \mathbb{E}X < \infty$, then $X_n \to X$ in L^1 (Scheffé). *Hint:* apply dominated convergence to $(X - X_n)^+$.

1.6 Compute $\mathbb{E}[W_s \mid W_t, W_u]$ for $s < t < u$ and for $t < s < u$.

Chapter 2

Martingales

The class notes define martingales and then use one fact: Itô integrals are martingales. But martingales carry far more than that. Their theory gives sharp inequalities, tells us when strategies cannot beat a fair game, and provides the convergence results behind stochastic integration.

2.1 Discrete-time definitions

A *filtration* is an increasing sequence $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots$. A process X is *adapted* if each X_n is $\mathcal{F}_n$-measurable, and *predictable* if each X_n is $\mathcal{F}_{n-1}$-measurable.

Definition 2.1. An adapted integrable M is a *martingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$. With $\leq$ it is a *supermartingale* and with $\geq$ a *submartingale*.

By Jensen, a convex function of a martingale is a submartingale (when integrable). Thus $|M|$, M^2 and $(M-K)^+$ are submartingales. This is the discrete shadow of why a call option's value, discounted, rises with maturity under zero rates.

Theorem 2.2 (You cannot beat a fair game)**.** *Let M be a martingale and H a bounded predictable process. Then the* martingale transform

$$(H \cdot M)_n = \sum_{k=1}^{n} H_k (M_k - M_{k-1})$$

is a martingale.

Proof. $\mathbb{E}[H_k(M_k - M_{k-1}) \mid \mathcal{F}_{k-1}] = H_k \mathbb{E}[M_k - M_{k-1} \mid \mathcal{F}_{k-1}] = 0$, because H_k is known at time $k-1$. □

> **Intuition.** H_k is your stake on round k, decided before the round is played. The transform is your winnings. Whatever you do with the information you have, your expected winnings stay zero. The Itô integral is the continuous-time limit of this sum; predictability of H is why Itô chose the left endpoint. Allow H_k to depend on M_k (peeking), and the theorem fails.

Theorem 2.3 (Doob decomposition)**.** *An integrable adapted X has a unique decomposition $X_n = X_0 + M_n + A_n$, with M a martingale, A predictable and $M_0 = A_0 = 0$. Here $A_n = \sum_{k\le n} \mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}]$. X is a submartingale if and only if A is increasing.*

Proof. Set A as stated and $M = X - X_0 - A$; M has zero conditional increments by construction. For uniqueness, if $M + A = M' + A'$, then $M - M' = A' - A$ is a predictable martingale, so $N_n = \mathbb{E}[N_n \mid \mathcal{F}_{n-1}] = N_{n-1}$, and it starts at zero. □

Apply this to M^2 for a square-integrable martingale M. The increasing part is the *predictable quadratic variation*

$$\langle M\rangle_n = \sum_{k\le n} \mathbb{E}[(M_k - M_{k-1})^2 \mid \mathcal{F}_{k-1}],$$

and $M^2 - \langle M\rangle$ is a martingale. For the symmetric random walk, $\langle M\rangle_n = n$, which recovers $\mathbb{E}M_n^2 = n$. Compare the *optional* quadratic variation $[M]_n = \sum_k (M_k - M_{k-1})^2$: it is not predictable, but $M^2 - [M]$ is also a martingale. For Brownian motion the two coincide, $\langle W\rangle_t = [W]_t = t$, but not for a Poisson process, where $[N]_t = N_t$ and $\langle N\rangle_t = \lambda t$ (Chapter 7).

Proposition 2.4 (Orthogonal increments)**.** *If M is a square-integrable martingale and $m \le n$, then $\mathbb{E}[(M_n - M_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[M_n^2 - M_m^2 \mid \mathcal{F}_m]$, and*

$$\mathbb{E}[M_n^2] = \mathbb{E}[M_0^2] + \sum_{k\le n} \mathbb{E}[(M_k - M_{k-1})^2].$$

Proof. Expand the square and use $\mathbb{E}[M_n M_m \mid \mathcal{F}_m] = M_m^2$. □

This Pythagoras identity is the seed of the Itô isometry.

2.2 Stopping times and optional stopping

Definition 2.5. $\tau : \Omega \to \{0, 1, \dots\} \cup \{\infty\}$ is a *stopping time* if $\{\tau \le n\} \in \mathcal{F}_n$ for all n. You can decide whether to stop using only information available now. The first time a process hits a level is a stopping time; the last time it does so is not.

Theorem 2.6 (Optional stopping)**.** *Let M be a martingale and τ a stopping time. Then the stopped process $M_n^\tau = M_{\tau\wedge n}$ is a martingale. If τ is bounded, or if M^τ is UI and $\tau < \infty$ a.s., then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*

Proof. Write $M_{\tau\wedge n} = M_0 + \sum_{k\le n} \mathbf{1}_{\{\tau\ge k\}}(M_k - M_{k-1})$. Since $\{\tau \ge k\} = \{\tau \le k-1\}^c \in \mathcal{F}_{k-1}$, this is a martingale transform, so Theorem 2.2 applies. If $\tau \le N$ then $M_\tau = M_N^\tau$, and taking expectations gives the result. In the UI case, $M_{\tau\wedge n} \to M_\tau$ a.s., hence in L^1 by Vitali's theorem. □

> **Pitfall. The doubling strategy.** Let M be a symmetric walk started at zero and $\tau = \inf\{n : M_n = 1\}$. Then $\tau < \infty$ a.s. and $M_\tau = 1$, so $\mathbb{E}M_\tau = 1 \ne 0 = \mathbb{E}M_0$. Optional stopping fails because M^τ is not UI: before hitting 1 the walk can fall arbitrarily far. The moral for finance: a credit limit (a lower bound on wealth) is what restores the theorem. Hence the admissibility condition in continuous time (Section 5.5).

Example 2.7 (Gambler's ruin)**.** A symmetric walk starts at 0; let τ be the first exit time from $(-a, b)$ with $a, b > 0$. Since M^τ is bounded, $\mathbb{E}M_\tau = 0$, so $-a\,\mathbb{P}(M_\tau = -a) + b\,\mathbb{P}(M_\tau = b) = 0$, giving

$$\mathbb{P}(M_\tau = b) = \frac{a}{a+b}.$$

Next, $M_n^2 - n$ is a martingale. Stopping and using monotone and bounded convergence yields

$$\mathbb{E}\tau = \mathbb{E}M_\tau^2 = a^2\frac{b}{a+b} + b^2\frac{a}{a+b} = ab.$$

Letting $b \to \infty$ gives $\mathbb{P}(\text{ever hit } b) = 1$ but $\mathbb{E}[\text{hitting time of } b] = \infty$. For an asymmetric walk with up-probability $p \neq \frac{1}{2}$, the *exponential martingale* $(q/p)^{M_n}$ gives $\mathbb{P}(M_\tau = b) = \frac{1-(q/p)^{-a}}{(q/p)^b-(q/p)^{-a}}$, and $M_n - n(p-q)$ gives $\mathbb{E}\tau = \mathbb{E}M_\tau/(p-q)$. Questions of this type are staples of interviews; the pattern is always to *find a martingale, stop it, and solve*.

2.3 Maximal inequalities

Theorem 2.8 (Doob's maximal inequalities)**.** *Let M be a martingale or nonnegative submartingale and $M_n^* = \max_{k\le n}|M_k|$. For $\lambda > 0$,*

$$\lambda\mathbb{P}(M_n^* \ge \lambda) \le \mathbb{E}[|M_n|\mathbf{1}_{\{M_n^*\ge\lambda\}}] \le \mathbb{E}|M_n|.$$

For $p > 1$, $\|M_n^*\|_p \le \frac{p}{p-1}\|M_n\|_p$. In particular $\mathbb{E}[(M_n^*)^2] \le 4\mathbb{E}[M_n^2]$.

Proof. $|M|$ is a submartingale. Let $\tau = \inf\{k : |M_k| \ge \lambda\}$ and $A_k = \{\tau = k\} \in \mathcal{F}_k$. Then

$$\lambda\mathbb{P}(M_n^* \ge \lambda) = \sum_{k\le n}\lambda\mathbb{P}(A_k) \le \sum_{k\le n}\mathbb{E}[|M_k|\mathbf{1}_{A_k}] \le \sum_{k\le n}\mathbb{E}[|M_n|\mathbf{1}_{A_k}] = \mathbb{E}[|M_n|\mathbf{1}_{\{M_n^*\ge\lambda\}}].$$

For the L^p bound, write $\mathbb{E}[(M_n^*)^p] = \int_0^\infty p\lambda^{p-1}\mathbb{P}(M_n^* \ge \lambda)\,\mathrm{d}\lambda$, insert the first inequality, use Fubini, and then Hölder:

$$\mathbb{E}[(M_n^*)^p] \le \frac{p}{p-1}\mathbb{E}[|M_n|(M_n^*)^{p-1}] \le \frac{p}{p-1}\|M_n\|_p\,\|M_n^*\|_p^{p-1}.$$ □

> **Intuition.** The maximum of a martingale over the whole interval is controlled by its value at the *end*. A martingale cannot make a large excursion and then reliably come back, since it would have to have "known" it was going to. This is why, in constructing the Itô integral, an L^2 bound at the final time upgrades to *uniform in time* convergence of paths, which gives path continuity.

2.4 Convergence

Theorem 2.9 (Martingale convergence)**.** *If M is a submartingale with $\sup_n \mathbb{E}M_n^+ < \infty$, then M_n converges a.s. to an integrable limit. If moreover M is a UI martingale, convergence holds in L^1 and $M_n = \mathbb{E}[M_\infty \mid \mathcal{F}_n]$. If M is bounded in L^p with $p > 1$, convergence holds in L^p.*

Idea. Doob's upcrossing inequality: if $U_n[a,b]$ counts completed upcrossings of $[a,b]$ by time n, then $(b-a)\mathbb{E}U_n[a,b] \le \mathbb{E}(M_n - a)^+$. Proof: the strategy "buy at or below a, sell at or above b" is a predictable transform of a submartingale, and its profit is at least $(b-a)U_n$ minus the final open position. So $U_\infty[a,b] < \infty$ a.s. for each rational pair $a < b$, and a sequence with finitely many upcrossings of all rational intervals must converge. □

A striking consequence: a positive martingale (or supermartingale) converges a.s. For $Z_t = \exp(\theta W_t - \frac{1}{2}\theta^2 t)$ with $\theta \neq 0$, the limit is 0 since $W_t/t \to 0$. So $\mathbb{E}Z_t = 1$ for every t yet $Z_t \to 0$ a.s.: the family is not UI. This gives a warning about changes of measure: the measures $\mathbb{Q}_T$ with density Z_T are each equivalent to $\mathbb{P}$ on $\mathcal{F}_T$, but they do not extend to an equivalent measure on $\mathcal{F}_\infty$. The drift θ is detectable from an infinite path.

Theorem 2.10 (Lévy upward)**.** *If $X \in L^1$ and $\mathcal{F}_n \uparrow \mathcal{F}_\infty$, then $\mathbb{E}[X \mid \mathcal{F}_n] \to \mathbb{E}[X \mid \mathcal{F}_\infty]$ a.s. and in L^1.*

This is the discrete Radon–Nikodym theorem: with $\mathcal{F}_n$ generated by finite partitions and $\mathbb{Q} \ll \mathbb{P}$, the ratios $Z_n = \mathbb{Q}(A)/\mathbb{P}(A)$ on atoms A of $\mathcal{F}_n$ form a UI martingale whose limit is $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$. Densities on path space are thus limits of the elementary coin-toss formula $(\tilde{p}/p)^{n_H}(\tilde{q}/q)^{n_T}$ from the class notes.

2.5 Continuous time

Let $(\mathcal{F}_t)_{t\ge 0}$ be a filtration satisfying the *usual conditions*: right-continuous, and $\mathcal{F}_0$ contains all null sets. These technical conditions ensure that hitting times of open and closed sets are stopping times and that martingales have càdlàg (right-continuous with left limits) versions. Doob's inequalities and optional stopping transfer directly by discretising along dyadic times and using path regularity.

Definition 2.11. An adapted continuous process M is a *local martingale* if there are stopping times $\tau_n \uparrow \infty$ a.s. such that M^{τ_n} is a martingale for each n. A local martingale that is not a martingale is a *strict* local martingale.

Proposition 2.12. *A nonnegative local martingale with $\mathbb{E}M_0 < \infty$ is a supermartingale. It is a martingale on $[0,T]$ if and only if $\mathbb{E}M_T = \mathbb{E}M_0$.*

Proof. For $s \le t$, by conditional Fatou, $\mathbb{E}[M_t \mid \mathcal{F}_s] = \mathbb{E}[\lim_n M_{t\wedge\tau_n} \mid \mathcal{F}_s] \le \liminf_n \mathbb{E}[M_{t\wedge\tau_n} \mid \mathcal{F}_s] = \liminf_n M_{s\wedge\tau_n} = M_s$. If $\mathbb{E}M_T = \mathbb{E}M_0$, then $M_s - \mathbb{E}[M_T \mid \mathcal{F}_s] \ge 0$ has zero mean, so it vanishes. □

Example 2.13 (Strict local martingale)**.** Let B be a three-dimensional Brownian motion started at $x \neq 0$ and $M_t = 1/|B_t|$. Off the origin, $1/|x|$ is harmonic in $\mathbb{R}^3$, and B a.s. never hits the origin, so Itô's formula gives $\mathrm{d}M = -M^2 \frac{B}{|B|}\cdot \mathrm{d}B$, a local martingale. But $\mathbb{E}M_t \to 0$ as $t \to \infty$ while $M_0 > 0$, so it is not a martingale. Such processes occur as asset prices in models of *bubbles*, and a stock like this breaks put–call parity. They are a reason to take care in checking whether an exponential local martingale is a true martingale (Novikov's condition, Theorem 5.3).

2.6 The Snell envelope: optimal stopping in discrete time

Let G_n be adapted and integrable (for example, the discounted exercise value of an American option), on horizon N. What is $\sup_\tau \mathbb{E}[G_\tau]$ over stopping times $\tau \le N$? Define by backward induction

$$U_N = G_N, \qquad U_n = \max\bigl(G_n,\ \mathbb{E}[U_{n+1} \mid \mathcal{F}_n]\bigr).$$

Theorem 2.14. *U is the smallest supermartingale dominating G. The time $\tau^* = \min\{n : U_n = G_n\}$ is optimal, and $\mathbb{E}[G_{\tau^*}] = U_0 = \sup_\tau \mathbb{E}[G_\tau]$.*

Proof. By construction $U \ge G$ and $U_n \ge \mathbb{E}[U_{n+1} \mid \mathcal{F}_n]$. If $V \ge G$ is another supermartingale, backward induction gives $V_n \ge \max(G_n, \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]) \ge U_n$. For any τ, optional stopping for supermartingales gives $\mathbb{E}G_\tau \le \mathbb{E}U_\tau \le U_0$. Finally, on $\{\tau^* > n\}$ one has $U_n = \mathbb{E}[U_{n+1} \mid \mathcal{F}_n]$, so U^{τ^*} is a martingale, whence $U_0 = \mathbb{E}U_{\tau^*} = \mathbb{E}G_{\tau^*}$. □

> **Intuition.** This is the American option algorithm: at each node take the maximum of exercising now and continuing. It also explains the hedge. The seller holds a supermartingale, so the buyer's failure to exercise at the optimal time only leaves money in the seller's hands.

Exercises

2.1 For a walk with up-probability $p > \frac{1}{2}$ started at zero, let τ be the first hit of 1. Using an exponential martingale, compute $\mathbb{E}[s^\tau]$ for $s \in (0,1)$, and from it obtain $\mathbb{E}[\tau]$.

2.2 *(Ballot problem.)* Show that M_n/n is a backward martingale for the symmetric walk conditioned on M_N. Deduce the ballot theorem x/n of the class notes without the reflection principle.

2.3 Show that $W_t^2 - t$ and $W_t^3 - 3tW_t$ are martingales. Find the next one in the sequence (Hermite polynomials).

2.4 Let τ be the first exit of W from $(-a, a)$. Using $W_t^4 - 6tW_t^2 + 3t^2$, compute $\mathbb{E}\tau^2$.

2.5 Prove that if M is a continuous local martingale with $\mathbb{E}\sup_{s\le T}|M_s| < \infty$, then M is a true martingale on $[0,T]$.

2.6 A coin is tossed until the pattern HTH appears. Using a fair-casino martingale argument (a new gambler arrives each round and bets on the pattern), find the expected number of tosses. Do the same for HHH, and explain the difference.

Part II

Stochastic Calculus

Chapter 3

Brownian Motion

3.1 Definition and existence

Definition 3.1. A process $(W_t)_{t\geq 0}$ is a *standard Brownian motion* (Wiener process) with respect to a filtration $(\mathcal{F}_t)$ if

1. $W_0 = 0$ a.s. and W is adapted;
2. for $s < t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and has law $\mathcal{N}(0, t-s)$;
3. $t \mapsto W_t$ is continuous a.s.

Equivalently, W is a continuous centred Gaussian process with $\operatorname{Cov}(W_s, W_t) = s \wedge t$.

The class notes obtain W as a limit of scaled walks; making that limit rigorous requires a tightness argument (Donsker's theorem). Here is a direct construction that shows why continuity is not automatic.

Theorem 3.2 (Lévy's construction)**.** *A standard Brownian motion exists on* $[0,1]$.

Proof. Let $D_n = \{k2^{-n}\}$ be the dyadic points in $[0,1]$ and $D = \bigcup_n D_n$. Let $(Z_d)_{d\in D}$ be i.i.d. $\mathcal{N}(0,1)$. Set $W_0 = 0$, $W_1 = Z_1$. Given W on D_{n-1}, define for $d \in D_n \setminus D_{n-1}$ the midpoint rule

$$W_d = \frac{W_{d-2^{-n}} + W_{d+2^{-n}}}{2} + \frac{Z_d}{2^{(n+1)/2}}.$$

Increments. By induction, increments on D_n are independent $\mathcal{N}(0, 2^{-n})$: the half-difference of the parent increment and the new noise are independent $\mathcal{N}(0, 2^{-n-1})$, and their sum and difference are then independent $\mathcal{N}(0, 2^{-n})$, since for independent Gaussians with equal variance the sum and difference are independent.

Continuity. Let F_n be the piecewise-linear function that vanishes on D_{n-1}, equals $Z_d 2^{-(n+1)/2}$ on $D_n \setminus D_{n-1}$, and is linear between. Then $W = \sum_n F_n$ on D. Since $\mathbb{P}(|Z| \geq c\sqrt{n}) \leq e^{-c^2 n/2}$, the union bound gives

$$\sum_n \mathbb{P}\big(\max_{d\in D_n} |Z_d| \geq c\sqrt{n}\big) \leq \sum_n (2^n + 1) e^{-c^2 n/2} < \infty \quad \text{for } c > \sqrt{2\ln 2}.$$

By Borel–Cantelli, a.s. $\|F_n\|_\infty \leq c\sqrt{n}\, 2^{-(n+1)/2}$ for all large n. These bounds are summable, so the series converges uniformly and the limit is continuous. Gaussian increments extend from D to all of $[0,1]$ by continuity. □

Intuition. Build the path top-down. First pick the endpoint. Then pick the midpoint: the average of its neighbours plus independent noise of the right size. Refine forever. Each level adds noise of size about $2^{-n/2}$, so the corrections shrink fast enough to converge uniformly (giving continuity), but too slowly for the slopes to settle (slope corrections of size $2^{n/2}$ blow up, giving non-differentiability). This also gives the fast way to simulate a path given its end value, and it underlies Brownian bridge constructions in quasi-Monte Carlo.

Symmetries. If W is a Brownian motion, so are each of the following:

- $-W_t$ (reflection), and $W_{t+h} - W_h$ (time shift);
- $c^{-1}W_{c^2 t}$ for $c > 0$ (Brownian scaling);
- $tW_{1/t}$ for $t > 0$, with value 0 at $t = 0$ (time inversion).

Each is a one-line check of the covariance. Scaling is why $\sqrt{t}$ appears everywhere: the size of the move over time t is $\sqrt{t}$, and option prices depend on $\sigma\sqrt{T}$. Time inversion turns large-time behaviour into small-time behaviour; for example it gives $W_t/t \to 0$ a.s. from continuity at 0 (or from the strong law).

3.2 Path properties

Theorem 3.3 (Quadratic variation). *For partitions Π of $[0, T]$ with mesh $\|\Pi\| \to 0$, $\sum_j (W_{t_{j+1}} - W_{t_j})^2 \to T$ in L^2. If $\sum_n \|\Pi_n\| < \infty$, convergence also holds almost surely.*

Proof. Let Q_Π be the sum. Then $\mathbb{E}Q_\Pi = T$, and by independence with $\mathrm{Var}[(\Delta W)^2] = 2(\Delta t)^2$,

$$\mathbb{E}(Q_\Pi - T)^2 = \sum_j 2(\Delta t_j)^2 \le 2T\|\Pi\| \to 0.$$

If the meshes are summable, Chebyshev gives $\sum_n \mathbb{P}(|Q_{\Pi_n} - T| > \epsilon) < \infty$. Borel–Cantelli then yields a.s. convergence. □

Pitfall. The class notes' computation for the scaled walk gave $\sum \frac{1}{p} X_i^2 = T$ exactly, since $X_i^2 = 1$. For Brownian motion Q_Π is random for each finite partition; only its limit is deterministic. The remarkable point is that a *random* path has a *deterministic* quadratic variation. Also: if one takes the supremum over all partitions, not a limit along shrinking ones, the result is $+\infty$ a.s. Quadratic variation is defined by a limit along meshes.

Corollary 3.4. *Almost surely, W has infinite first variation on every interval.*

Proof. $\sum_j (\Delta W_j)^2 \le \max_j |\Delta W_j| \cdot \sum_j |\Delta W_j|$. If the first variation were finite then, since $\max_j |\Delta W_j| \to 0$ by continuity, the left side would tend to zero, contradicting Theorem 3.3. □

This corollary is the entire reason for Itô calculus: an integral $\int H \, dW$ cannot be a Stieltjes integral path by path.

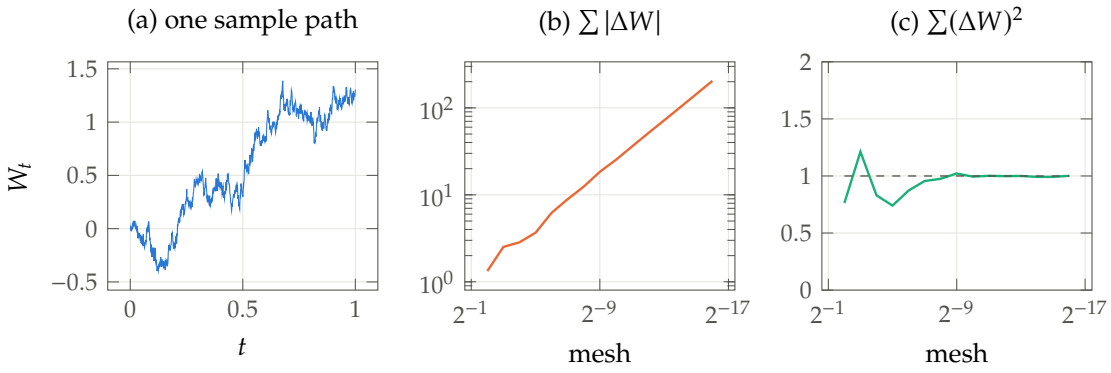


Figure 3.1: One Brownian path on $[0,1]$ sampled at successively finer scales. The first variation (b) grows like $\sqrt{1/\text{mesh}}$ without bound, while the sum of squares (c) settles at $T = 1$ (dashed line).

Theorem 3.5 (Hölder regularity, stated)**.** *Almost surely, W is locally Hölder continuous of every order $\gamma < \frac{1}{2}$, and of no order $\gamma \geq \frac{1}{2}$ at typical points. More precisely, Lévy's modulus of continuity gives*

$$\limsup_{h \downarrow 0} \sup_{0 \leq t \leq 1-h} \frac{|W_{t+h} - W_t|}{\sqrt{2h \ln(1/h)}} = 1 \quad a.s.$$

The Hölder part follows from Kolmogorov's criterion: if $\mathbb{E}|X_t - X_s|^\alpha \leq C|t-s|^{1+\beta}$, then X has a modification Hölder of every order below β/α. For W, $\mathbb{E}|W_t - W_s|^{2m} = c_m|t-s|^m$ gives order $(m-1)/2m \to \frac{1}{2}$.

Theorem 3.6 (Paley–Wiener–Zygmund)**.** *Almost surely, W is differentiable at no point.*

Proof (Dvoretzky–Erdős–Kakutani). Suppose W is differentiable at some $s \in [0,1)$ with derivative bounded by C near s. Then for all large n, with $j = \lfloor ns \rfloor + 1$, the three increments $W_{(j+i)/n} - W_{(j+i-1)/n}$, $i = 1,2,3$, each have absolute value at most MC/n for a fixed numerical constant M, since all these points lie within $4/n$ of s. Fix C, n. Each increment is $\mathcal{N}(0, 1/n)$, so has density at most $\sqrt{n/2\pi}$, so $\mathbb{P}(|\Delta W| \leq MC/n) \leq 2MC\, n^{-1}\sqrt{n/2\pi} = c\, n^{-1/2}$. Three independent increments give $(c\, n^{-1/2})^3$, and there are at most n positions j:

$$\mathbb{P}(\text{such } j \text{ exists}) \leq n \cdot c^3 n^{-3/2} = c^3 n^{-1/2} \to 0.$$

Letting $n \to \infty$ and then taking a union over integer C shows that the event has probability zero. □

The class notes' heuristic $\mathrm{d}W/\mathrm{d}t \sim 1/\sqrt{\mathrm{d}t}$ is correct in spirit; this proof shows why three consecutive small increments, not one, are needed to make the union bound over n positions converge.

Other fine-scale facts worth knowing:

- **Law of the iterated logarithm:** $\limsup_{t\to\infty} W_t/\sqrt{2t \ln\ln t} = 1$ a.s.; by time inversion, the same holds as $t \downarrow 0$ with $\ln\ln(1/t)$.
- **Zero set:** $\{t : W_t = 0\}$ is a.s. closed, has no isolated points, is uncountable, and has Lebesgue measure zero. A path that returns to zero returns infinitely often immediately afterwards.

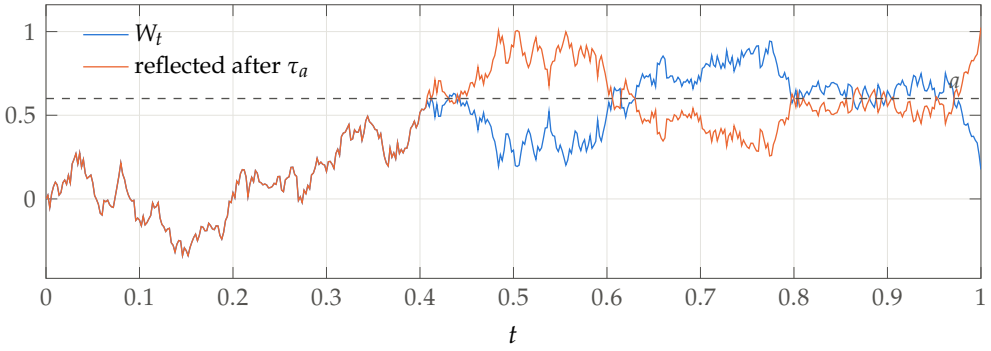


Figure 3.2: The bijection behind Theorem 3.8, on a simulated path. After the first hit of level a, flipping the rest of the path about a yields an equally likely path, turning "hit a and ended below a" into "ended above a".

- **Oscillation:** since $\limsup_{t\downarrow 0} W_t/\sqrt{2t\ln\ln(1/t)} = 1$, W takes both signs in every interval $[0,\epsilon]$.

3.3 Markov properties and the reflection principle

Theorem 3.7 (Strong Markov)**.** *If τ is an a.s. finite stopping time, then $W_{\tau+t} - W_\tau$ is a Brownian motion independent of $\mathcal{F}_\tau$.*

Sketch. For τ taking countably many values, condition on each value and apply the ordinary Markov property. For general τ, approximate from above by $\tau_n = 2^{-n}\lceil 2^n\tau\rceil$ and use path continuity with dominated convergence on bounded functionals. □

Theorem 3.8 (Reflection principle)**.** *Let $a > 0$, $\tau_a = \inf\{t : W_t = a\}$. The process*

$$\widetilde{W}_t = W_t\mathbf{1}_{\{t\le\tau_a\}} + (2a - W_t)\mathbf{1}_{\{t>\tau_a\}}$$

is a Brownian motion. Consequently, with $M_t = \max_{s\le t} W_s$ and $b \le a$,

$$\mathbb{P}(M_t \ge a,\ W_t \le b) = \mathbb{P}(W_t \ge 2a - b).$$

In particular $\mathbb{P}(M_t \ge a) = 2\mathbb{P}(W_t \ge a)$, so $M_t \overset{d}{=} |W_t|$.

Proof. By the strong Markov property, the post-τ_a increment is a Brownian motion independent of the pre-τ_a path, and its negative is too. Gluing either version on gives processes with the same law. The paths that hit a and end below b correspond bijectively, under this reflection, to the paths that end above $2a - b$ (any such path must have hit a). For the last claim, write $\mathbb{P}(M_t \ge a) = \mathbb{P}(M_t \ge a, W_t \ge a) + \mathbb{P}(M_t \ge a, W_t < a) = \mathbb{P}(W_t \ge a) + \mathbb{P}(W_t > a)$. □

Numerical check. With 200,000 paths and 1,000 steps on $[0,1]$: $\mathbb{P}(M_1 \ge 1)$ is estimated as 0.3079 against the exact 0.3173, and $\mathbb{P}(M_1 \ge 0.5)$ as 0.6043 against 0.6171. The estimates are systematically *low*, by far more than sampling error. A discretely observed path misses excursions above the level between grid points. This same bias afflicts barrier option simulation, and Section 12.5 fixes it with a Brownian bridge.

Differentiating the joint probability gives the joint density of (M_t, W_t):

$$f_{M,W}(a,b) = \frac{2(2a-b)}{t\sqrt{2\pi t}} \exp\Big(-\frac{(2a-b)^2}{2t}\Big), \qquad a \ge \max(0,b). \tag{3.1}$$

Corollary 3.9 (First passage time)**.** $\mathbb{P}(\tau_a \le t) = \mathbb{P}(M_t \ge a) = 2\Phi(-a/\sqrt{t})$*. So* τ_a *has the density*

$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi t^3}} e^{-a^2/2t}, \quad t > 0.$$

Hence $\tau_a < \infty$ *a.s. yet* $\mathbb{E}\tau_a = \infty$*, as the density decays like* $t^{-3/2}$*. Its Laplace transform is* $\mathbb{E}e^{-\lambda\tau_a} = e^{-a\sqrt{2\lambda}}$.

Laplace transform. For $\theta = \sqrt{2\lambda} > 0$, $N_t = \exp(\theta W_t - \lambda t)$ is a martingale. The stopped process N^{τ_a} is bounded by $e^{\theta a}$, and on $\{\tau_a = \infty\}$ it tends to zero. Optional stopping gives $1 = e^{\theta a}\mathbb{E}e^{-\lambda\tau_a}$. □

Example 3.10 (Exit from an interval)**.** The exit time τ of $(-a, b)$ satisfies $\mathbb{P}(W_\tau = b) = a/(a+b)$ and $\mathbb{E}\tau = ab$, by exactly the argument of Example 2.7.

Theorem 3.11 (Lévy's arcsine law, stated)**.** *The last zero* $g = \sup\{s \le 1 : W_s = 0\}$ *and the time spent positive* $A = \int_0^1 \mathbf{1}_{\{W_s>0\}}\mathrm{d}s$ *each have the arcsine law,* $\mathbb{P}(\cdot \le x) = \frac{2}{\pi}\arcsin\sqrt{x}$.

This is the continuous limit of the class notes' $\alpha_{2l,2\nu} = \mu_{2l}\mu_{2\nu-2l}$. Using $\mu_{2l} \sim 1/\sqrt{\pi l}$, the mass $\alpha_{2l,2\nu} \approx \frac{1}{\pi\sqrt{l(\nu-l)}}$, whose density on $[0,1]$ is $\frac{1}{\pi\sqrt{x(1-x)}}$. The density is U-shaped: the most likely values for the fraction of time that a fair game keeps you ahead are near 0 and 1, and $\frac{1}{2}$ is the *least* likely. Be careful with this when you read a trader's or a strategy's long winning streak: long stretches on one side happen by pure chance.

3.4 The Brownian bridge

Conditioned on $W_0 = x$ and $W_T = y$, the process is Gaussian with mean $x + (y-x)t/T$ and covariance $s(T-t)/T$ for $s \le t$ (by Example 1.22). The reflection principle gives, for $a > \max(x, y)$,

$$\mathbb{P}\Big(\max_{s\le T} W_s \ge a \;\Big|\; W_0 = x, W_T = y\Big) = \frac{\varphi_T(2a-x-y)}{\varphi_T(y-x)} = \exp\Big(-\frac{2(a-x)(a-y)}{T}\Big), \tag{3.2}$$

where φ_T is the $\mathcal{N}(0,T)$ density. With volatility σ, replace T by $\sigma^2 T$. This gives a closed-form correction for barrier simulation.

3.5 Martingales of Brownian motion

Each of these has a short check by conditional expectation or Itô's formula:

$$W_t, \qquad W_t^2 - t, \qquad W_t^3 - 3tW_t, \qquad W_t^4 - 6tW_t^2 + 3t^2, \qquad \exp(\theta W_t - \tfrac{1}{2}\theta^2 t).$$

The polynomial ones are the Taylor coefficients in θ of the exponential (Hermite polynomials). And by Lévy's characterisation (Theorem 4.10) the first two properties actually characterise Brownian motion among continuous processes. This is a surprisingly strong statement: a continuous martingale whose square minus t is also a martingale must have Gaussian increments.

Heat equation. The transition density $p(t,x,y) = \varphi_t(y-x)$ solves $\partial_t p = \frac{1}{2}\partial_{yy} p$, and $u(t,x) = \mathbb{E}[h(x + W_{T-t})]$ solves $\partial_t u + \frac{1}{2}\partial_{xx} u = 0$. This link between diffusion and PDEs is pursued in Chapter 6. The factor $\frac{1}{2}$ is the same $\frac{1}{2}$ as in Itô's formula.

Exercises

3.1 Show directly that $\int_0^1 W_t \, \mathrm{d}t \sim \mathcal{N}(0, 1/3)$. *Hint:* it is the limit of Gaussian sums; compute the covariance integral.

3.2 Compute $\mathbb{E}[W_s \mid W_t]$ for $s < t$, and $\mathbb{E}[W_t \mid W_s]$. Why are these so different?

3.3 Compute $\mathbb{P}(W_1 > 0, W_2 > 0)$. *Answer:* $3/8$. Use the angle between the two Gaussian vectors.

3.4 Using (3.1), show that $M_t - W_t \overset{d}{=} |W_t|$ (Lévy). What is the expected drawdown at time t of a driftless Brownian motion?

3.5 Let $\tau = \inf\{t : W_t^2 \geq 1 - t\}$. Show that $\tau < 1$ a.s. and compute $\mathbb{E}\tau$ and $\mathbb{E}W_\tau^4$ using the martingales above.

3.6 (Interview classic.) Stock S follows driftless GBM from $S_0 = 1$. What is the probability that it ever reaches 2? And what is the expected maximum of S on $[0, \infty)$? Reconcile the answers using S being a martingale that tends to zero.

Chapter 4

The Itô Integral and Itô's Formula

4.1 Why a new integral, and why the left endpoint

Paths of W have infinite first variation, so $\int_0^T H_t \, \mathrm{d}W_t$ cannot be a Stieltjes integral defined path by path. The second problem is that a Riemann-type limit depends on where in each interval you evaluate. Compare three choices for $\int_0^T W \, \mathrm{d}W$, using a partition with mesh going to zero.

Proposition 4.1. *The following limits hold in L^2:*

$$\sum_j W_{t_j} \Delta W_j \to \tfrac{1}{2}(W_T^2 - T) \qquad \textit{(Itô, left point)},$$

$$\sum_j W_{t_{j+1}} \Delta W_j \to \tfrac{1}{2}(W_T^2 + T) \qquad \textit{(anti-Itô, right point)},$$

$$\sum_j W_{(t_j + t_{j+1})/2} \Delta W_j \to \tfrac{1}{2} W_T^2 \qquad \textit{(Stratonovich, midpoint)}.$$

Proof. From $a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b-a)^2$ and $b(b-a) = \frac{1}{2}(b^2 - a^2) + \frac{1}{2}(b-a)^2$, the first two sums are $\frac{1}{2}W_T^2 \mp \frac{1}{2}\sum(\Delta W_j)^2$, and Theorem 3.3 finishes them. The midpoint sum has mean $\sum_j \frac{1}{2}\Delta t_j = T/2$; a similar computation identifies its limit. □

The class notes mention these choices. The difference between them is exactly the quadratic variation. The Stratonovich answer looks like ordinary calculus. So why choose Itô?

> **Intuition.** Because a trader must decide a position *before* the price moves. In the sum $\sum H_{t_j}(W_{t_{j+1}} - W_{t_j})$, the holding is fixed at the start of the interval. With this choice, Theorem 2.2 says a fair game stays fair, so the integral is a martingale. The left point is the only non-anticipating choice. The right point amounts to trading on tomorrow's price, and it adds a free expected profit of $T/2$ in the calculation above. The cost of Itô's choice is a modified chain rule.

4.2 Construction

Fix $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ and a Brownian motion W. Let L^2_{ad} be the space of progressively measurable H with $\|H\|^2 := \mathbb{E}\int_0^T H_t^2 \,\mathrm{d}t < \infty$. (Progressive measurability is a technical strengthening of adaptedness; adapted and left-continuous is sufficient.)

Step 1: simple processes. A *simple* process is $H_t = \sum_{j=0}^{n-1} \xi_j \mathbf{1}_{(t_j, t_{j+1}]}(t)$ with each ξ_j bounded and $\mathcal{F}_{t_j}$-measurable. Define

$$I_T(H) = \sum_j \xi_j (W_{t_{j+1}} - W_{t_j}).$$

Lemma 4.2 (Itô isometry for simple processes)**.** $\mathbb{E}[I_T(H)] = 0$ *and* $\mathbb{E}[I_T(H)^2] = \mathbb{E}\int_0^T H_t^2 \,\mathrm{d}t$.

Proof. For $j < k$: $\mathbb{E}[\xi_j \Delta W_j \xi_k \Delta W_k] = \mathbb{E}[\xi_j \Delta W_j \xi_k \mathbb{E}[\Delta W_k \mid \mathcal{F}_{t_k}]] = 0$. For $j = k$: $\mathbb{E}[\xi_j^2 (\Delta W_j)^2] = \mathbb{E}[\xi_j^2 \mathbb{E}[(\Delta W_j)^2 \mid \mathcal{F}_{t_j}]] = \mathbb{E}[\xi_j^2]\Delta t_j$. Note how each step uses $\xi_j \in \mathcal{F}_{t_j}$; this is what fails for the other two choices. □

Step 2: density. Simple processes are dense in L^2_{ad}. The approximation proceeds in three stages: a bounded continuous H is approximated by its left-point step functions (dominated convergence); a bounded H by the continuous $\frac{1}{h}\int_{t-h}^t H_s \,\mathrm{d}s$; a general H by truncation.

Step 3: extension. If $H^n \to H$ in L^2_{ad}, then $I_T(H^n)$ is Cauchy in $L^2(\Omega)$ by the isometry, and it has a limit by completeness. The limit is independent of the approximating sequence.

Step 4: a continuous version in t. For simple H, $I_t(H)$ is a continuous martingale. By Doob's inequality (Theorem 2.8),

$$\mathbb{E}\sup_{t \le T} |I_t(H^n) - I_t(H^m)|^2 \le 4\,\mathbb{E}\int_0^T (H_t^n - H_t^m)^2 \,\mathrm{d}t.$$

Along a fast subsequence, Borel–Cantelli gives a.s. uniform convergence. The limit is thus a continuous process.

Theorem 4.3. *For $H, G \in L^2_{\mathrm{ad}}$, the process $I_t = \int_0^t H_s \,\mathrm{d}W_s$ is a continuous square-integrable martingale, linear in H, with*

$$\mathbb{E}[I_t^2] = \mathbb{E}\int_0^t H_s^2 \,\mathrm{d}s, \qquad [I]_t = \int_0^t H_s^2 \,\mathrm{d}s, \qquad \Big[\int H \,\mathrm{d}W, \int G \,\mathrm{d}W\Big]_t = \int_0^t H_s G_s \,\mathrm{d}s.$$

Moreover $I_t^2 - \int_0^t H_s^2 \,\mathrm{d}s$ is a martingale.

Proof. Martingale property and isometry pass to the limit because L^2 limits preserve conditional expectations. For the quadratic variation with a simple H, on each $(t_j, t_{j+1}]$ the integral is ξ_j times a Brownian increment, so Theorem 3.3 gives the quadratic variation $\xi_j^2 \Delta t_j$. A density argument gives the general case. The cross variation follows by polarisation. □

Intuition. The isometry is Pythagoras for orthogonal increments (Proposition 2.4). The quadratic variation statement says: *each infinitesimal bet of size H_t on a noise of variance $\mathrm{d}t$ contributes $H_t^2\mathrm{d}t$*, and those contributions accumulate pathwise. For a deterministic H, I_t is a Gaussian process with independent increments and $\operatorname{Var} I_t = \int_0^t H_s^2 \mathrm{d}s$. With random H the integral is generally *not* Gaussian, even though it is built from Gaussian noise.

Step 5: localisation. Let L^2_{loc} be the space of progressive H with $\int_0^T H_t^2 \,\mathrm{d}t < \infty$ a.s. For such H, the times $\tau_n = \inf\{t : \int_0^t H_s^2 \mathrm{d}s \geq n\}$ increase to infinity, $H\mathbf{1}_{[0,\tau_n]} \in L^2_{\mathrm{ad}}$, and the integrals agree on overlaps, so $\int_0^t H \,\mathrm{d}W$ is well defined. It is a continuous *local* martingale, but it need not be a true martingale.

Pitfall. Every result that uses the martingale property of $\int H \,\mathrm{d}W$ needs $\mathbb{E}\int H^2 \mathrm{d}t < \infty$, or some other argument. Students often write "no $\mathrm{d}t$ term, so it is a martingale" and it is true only for local martingales. For example, take $H_t = 1/\sqrt{T-t}$ on $[0,T)$: then $\int_0^t H^2 \mathrm{d}s = \ln(T/(T-t)) \to \infty$, and the integral (a time-changed Brownian motion) reaches any level before time T almost surely. Stopped at its first hit of $+1$, it has value 1 at time T a.s., starting at zero. It is a local martingale, not a martingale, and it is the mathematical form of the doubling strategy (Section 5.5).

4.3 Itô processes

Definition 4.4. An *Itô process* is $X_t = X_0 + \int_0^t \Delta_s \,\mathrm{d}W_s + \int_0^t \Theta_s \,\mathrm{d}s$ with $\Delta \in L^2_{\mathrm{loc}}$ and $\int_0^T |\Theta_s| \mathrm{d}s < \infty$ a.s. Its quadratic variation is $[X]_t = \int_0^t \Delta_s^2 \mathrm{d}s$, since the finite-variation part contributes nothing.

Theorem 4.5. *The decomposition is unique: a continuous local martingale with finite variation is constant. So Δ and Θ are determined by X, and an Itô process is a local martingale if and only if $\Theta = 0$ a.e.*

Proof. If M is a continuous local martingale of finite variation, then $[M] = 0$ since $\sum(\Delta M)^2 \leq \max|\Delta M| \sum|\Delta M| \to 0$. Stopping to make M bounded, $\mathbb{E}M_t^2 = \mathbb{E}[M]_t = 0$. □

This gives the reasoning behind every "match the $\mathrm{d}W$ terms and the $\mathrm{d}t$ terms" argument in the class notes. Matching coefficients is legitimate exactly because of this uniqueness.

4.4 Itô's formula

Theorem 4.6 (Itô–Doeblin)**.** *Let $f \in C^{1,2}$ and let X be an Itô process. Then a.s., for all t,*

$$f(t, X_t) = f(0, X_0) + \int_0^t \partial_t f(s, X_s)\,\mathrm{d}s + \int_0^t \partial_x f(s, X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t \partial_{xx} f(s, X_s)\,\mathrm{d}[X]_s.$$

Proof for $f = f(x)$, $X = W$. Localisation. Let τ_n be the first exit from $(-n, n)$. It suffices to work on $[0, \tau_n]$, so we can assume f, f' and f'' are bounded and f'' is uniformly continuous.

Taylor expansion. On a partition, there are points ξ_j between W_{t_j} and $W_{t_{j+1}}$ with

$$f(W_t) - f(W_0) = \underbrace{\sum_j f'(W_{t_j})\Delta W_j}_{A_\Pi} + \underbrace{\frac{1}{2}\sum_j f''(W_{t_j})(\Delta W_j)^2}_{B_\Pi} + \underbrace{\frac{1}{2}\sum_j \big(f''(\xi_j) - f''(W_{t_j})\big)(\Delta W_j)^2}_{C_\Pi}.$$

The first term. $A_\Pi \to \int_0^t f'(W_s)\mathrm{d}W_s$ in L^2, as left-point step functions converge to the continuous bounded integrand.

The second term. Compare B_Π with $\frac{1}{2}\sum_j f''(W_{t_j})\Delta t_j \to \frac{1}{2}\int_0^t f''(W_s)\mathrm{d}s$. The difference is $\frac{1}{2}\sum_j f''(W_{t_j})\big[(\Delta W_j)^2 - \Delta t_j\big]$. Its terms have conditional mean zero given earlier terms, so they are orthogonal:

$$\mathbb{E}\Big(\sum_j f''(W_{t_j})[(\Delta W_j)^2 - \Delta t_j]\Big)^2 = \sum_j \mathbb{E}[f''(W_{t_j})^2]\cdot 2\Delta t_j^2 \le 2\|f''\|_\infty^2\, t\,\|\Pi\| \to 0.$$

The third term. $|C_\Pi| \le \frac{1}{2}\max_j |f''(\xi_j) - f''(W_{t_j})| \sum_j (\Delta W_j)^2$. Uniform continuity of f'' and of the path makes the maximum go to zero a.s., while the sum converges to t in probability. So $C_\Pi \to 0$ in probability.

Higher terms. None are needed. Both sides are continuous in t, so the identity holds simultaneously for all t a.s. □

The class notes' derivation replaced $(\Delta W)^2$ by Δt directly; the orthogonality step above is what justifies it.

> **Intuition.** Why does a second-order term survive? $\Delta W \sim \sqrt{\Delta t}$, so $(\Delta W)^2 \sim \Delta t$, the same size as the time step. Each individual $(\Delta W)^2$ is random, but there are n of them with independent fluctuations of size Δt around their mean. So the total fluctuation is of size $\sqrt{n}\,\Delta t \to 0$ while the mean stays fixed. A law of large numbers converts randomness into a deterministic $\mathrm{d}t$. Convexity makes this visible: for convex f, $\frac{1}{2}f''\mathrm{d}t$ is a positive drift, the Jensen gain. For an option holder, this is the gamma, and it reappears as $\frac{1}{2}\Gamma S^2\sigma^2\mathrm{d}t$.

Multi-dimensional form and the product rule. For Itô processes $X^1, \dots, X^n$,

$$\mathrm{d}f(t,X) = \partial_t f\,\mathrm{d}t + \sum_i \partial_i f\,\mathrm{d}X^i + \frac{1}{2}\sum_{i,j}\partial_{ij} f\,\mathrm{d}[X^i, X^j],$$

under the box calculus $\mathrm{d}W^i\mathrm{d}W^j = \delta_{ij}\mathrm{d}t$ (or $\rho_{ij}\mathrm{d}t$ for correlated motions), $\mathrm{d}t\,\mathrm{d}W = \mathrm{d}t\,\mathrm{d}t = 0$. Taking $f(x,y) = xy$,

$$\mathrm{d}(X_tY_t) = X_t\mathrm{d}Y_t + Y_t\mathrm{d}X_t + \mathrm{d}[X,Y]_t.$$

Example 4.7 (Moments by recursion)**.** $\mathrm{d}W^k = kW^{k-1}\mathrm{d}W + \frac{1}{2}k(k-1)W^{k-2}\mathrm{d}t$. Taking expectations, $\mathbb{E}W_t^k = \frac{1}{2}k(k-1)\int_0^t \mathbb{E}W_s^{k-2}\mathrm{d}s$. Hence $\mathbb{E}W_t^4 = 3t^2$ and $\mathbb{E}W_t^6 = 15t^3$. This is a common interview question.

Example 4.8 (Exponential martingale)**.** For $\Theta \in L^2_{\mathrm{loc}}$, set $Z_t = \exp(\int_0^t \Theta\,\mathrm{d}W - \frac{1}{2}\int_0^t \Theta^2\mathrm{d}s)$. Applying Itô's formula to exp gives $\mathrm{d}Z_t = Z_t\Theta_t\mathrm{d}W_t$. So Z is a positive local martingale, and thus a supermartingale. The $-\frac{1}{2}\Theta^2$ term exactly cancels the Itô drift. This is the *Doléans-Dade exponential* $\mathcal{E}(\int \Theta\,\mathrm{d}W)$, the stochastic analogue of e^x: it solves $\mathrm{d}Z = Z\,\mathrm{d}M$.

Example 4.9 (Stochastic integral of a deterministic integrand). $\int_0^t f(s)\mathrm{d}W_s \sim \mathcal{N}(0, \int_0^t f^2)$, and integration by parts gives $\int_0^t W_s \mathrm{d}s = \int_0^t (t-s)\mathrm{d}W_s \sim \mathcal{N}(0, t^3/3)$. This is how one prices Bachelier-model Asians, and how the Vasicek model's integrated short rate is Gaussian.

4.5 Lévy's characterisation

Theorem 4.10. *A continuous local martingale M with $M_0 = 0$ and $[M]_t = t$ is a Brownian motion. The d-dimensional version requires $[M^i, M^j]_t = \delta_{ij} t$.*

Proof. Fix $u \in \mathbb{R}$. Applying Itô's formula to $Y_t = \exp(iuM_t + \frac{1}{2}u^2 t)$, the drift terms cancel: $\mathrm{d}Y_t = iuY_t\,\mathrm{d}M_t$. So Y is a local martingale that is bounded on $[0, T]$, hence a true martingale. Therefore, for $s < t$,

$$\mathbb{E}[e^{iu(M_t - M_s)} \mid \mathcal{F}_s] = e^{-u^2(t-s)/2},$$

which says the increment is $\mathcal{N}(0, t-s)$ and independent of $\mathcal{F}_s$. □

> **Intuition.** This is an extraordinary rigidity result. Continuity plus the martingale property plus a clock running at unit speed *force* Gaussianity. It is why Girsanov's theorem has such a short proof, and it implies the following generalisation.

Theorem 4.11 (Dambis–Dubins–Schwarz, stated). *Every continuous local martingale with $[M]_\infty = \infty$ is a time-changed Brownian motion: $M_t = B_{[M]_t}$ for a Brownian motion B.*

So $[M]_t$ is the intrinsic clock of M. In finance: if $\mathrm{d}\ln F_t = \sigma_t \mathrm{d}W_t - \frac{1}{2}\sigma_t^2 \mathrm{d}t$ then log-returns are Brownian on the clock of integrated variance $\int \sigma^2 \mathrm{d}t$. Fat tails in calendar time can be pure normal noise in "business time".

Theorem 4.12 (Burkholder–Davis–Gundy, stated). *For $p > 0$ there exist constants with $c_p \mathbb{E}[M]_\tau^{p/2} \le \mathbb{E}\sup_{t \le \tau} |M_t|^p \le C_p \mathbb{E}[M]_\tau^{p/2}$ for any continuous local martingale M with $M_0 = 0$ and any stopping time τ.*

4.6 Stratonovich calculus

The Stratonovich integral is $\int H \circ \mathrm{d}X = \int H\,\mathrm{d}X + \frac{1}{2}[H, X]$. It obeys the ordinary chain rule, $\mathrm{d}f(X) = f'(X) \circ \mathrm{d}X$. It has a place in physics and on manifolds, where coordinate changes are essential. Since it is not a martingale transform, in finance it is mainly a computational device: the SDE $\mathrm{d}X = \sigma(X)\mathrm{d}W$ (Itô) has Stratonovich form $\mathrm{d}X = -\frac{1}{2}\sigma\sigma'(X)\mathrm{d}t + \sigma(X) \circ \mathrm{d}W$. The Wong–Zakai theorem says that if you smooth the noise in an ODE and take the limit, you get a Stratonovich SDE, not an Itô SDE.

4.7 Tanaka's formula and local time

Itô's formula needs $f \in C^2$, but option payoffs have kinks. What is $\mathrm{d}|W_t|$?

Theorem 4.13 (Tanaka). *There is a continuous increasing process L^a, the* local time *of W at a, such that*

$$|W_t - a| = |W_0 - a| + \int_0^t \operatorname{sgn}(W_s - a)\,\mathrm{d}W_s + L_t^a, \qquad L_t^a = \lim_{\epsilon \downarrow 0} \frac{1}{2\epsilon} \int_0^t \mathbf{1}_{\{|W_s - a| < \epsilon\}} \mathrm{d}s,$$

where L^a grows only on the set $\{t : W_t = a\}$. Likewise,

$$(W_t - a)^+ = (W_0 - a)^+ + \int_0^t \mathbf{1}_{\{W_s > a\}} \mathrm{d}W_s + \tfrac{1}{2} L_t^a.$$

Idea. Take f_ϵ smooth and convex with $f_\epsilon \to |x - a|$, $f_\epsilon'' = \epsilon^{-1}\mathbf{1}_{(a-\epsilon, a+\epsilon)}$ (smoothed). Itô's formula gives $f_\epsilon(W_t) = f_\epsilon(W_0) + \int f_\epsilon'(W)\mathrm{d}W + \frac{1}{2\epsilon}\int \mathbf{1}_{\{|W-a|<\epsilon\}}\mathrm{d}s$. Pass to the limit. □

Formally, $\frac{1}{2} f'' = \delta_a$ and $L_t^a = \int_0^t \delta_a(W_s)\mathrm{d}s$. The *occupation times formula* generalises this: $\int_0^t g(W_s)\mathrm{d}s = \int_{\mathbb{R}} g(a) L_t^a \,\mathrm{d}a$. So $a \mapsto L_t^a$ is the density of time spent at level a, and it exists precisely because Brownian motion is so rough. A smooth path has no such density at a typical level of a stationary point.

The stochastic integral $\beta_t = \int_0^t \operatorname{sgn}(W_s)\mathrm{d}W_s$ is itself a Brownian motion, by Lévy's characterisation, since $[\beta]_t = t$. So $|W| = \beta + L^0$: reflected Brownian motion is Brownian motion plus an increasing push exerted only at zero.

Example 4.14 (The stop-loss start-gain paradox)**.** Consider this strategy for a call struck at K, with zero rates: hold one share when $S > K$ and none when $S < K$. Payoff replicated: $(S_T - K)^+$. It seems to cost just the intrinsic value $(S_0 - K)^+$ and to be free of model dependence! Yet the call costs more. The resolution is Tanaka–Meyer applied to S:

$$(S_T - K)^+ = (S_0 - K)^+ + \int_0^T \mathbf{1}_{\{S_t > K\}} \mathrm{d}S_t + \tfrac{1}{2}\int_0^T \delta_K(S_t)\,\mathrm{d}[S]_t.$$

The last term is a local time, and it is the *loss* you take by buying slightly above K and selling slightly below on each of the infinitely many crossings. Its expected value is the time value of the option. Discretising the strategy makes it look fine until you notice the crossings get denser as you refine. In this picture, an option's time value is exactly the expected local time at the strike. The same idea gives Dupire's formula (Chapter 9).

Exercises

4.1 Compute $\mathbb{E}\big[\big(\int_0^T W_t\,\mathrm{d}W_t\big)^2\big]$ two ways: by isometry and from Proposition 4.1.

4.2 Find $\operatorname{Var}\big(\int_0^T W_t^2 \mathrm{d}W_t\big)$.

4.3 Solve for the distribution of $\int_0^T e^{-\kappa(T-t)}\mathrm{d}W_t$. Why is it Gaussian?

4.4 Let $X_t = W_t^3 - 3tW_t$. Show by Itô's formula that it is a martingale and identify its integrand. What is $[X]_t$?

4.5 Compute $\mathrm{d}(W_t/\sqrt{1+t})$ and $\mathrm{d}(\ln S_t)$ for GBM. Why is the $\frac{1}{2}\sigma^2$ in the latter sometimes called the *volatility drag*? Show that the median of S_T is below its mean.

4.6 Let W^1, W^2 be independent Brownian motions. Show that $B_t = \int_0^t \cos(W_s^2)\mathrm{d}W_s^1 + \int_0^t \sin(W_s^2)\mathrm{d}W_s^2$ is a Brownian motion. Is it Gaussian jointly with W^2?

4.7 Show that $\mathrm{d}W^3 = 3W^2 \circ \mathrm{d}W$ in the Stratonovich sense, and convert it to an Itô statement to check it.

4.8 Explain why the anti-Itô integral of W fails to be a martingale; compute its drift.

Chapter 5

Girsanov, Representation and the Fundamental Theorems

The class notes state Girsanov's theorem and the martingale representation theorem without proof. Both have short proofs given the tools of Chapter 4, and the proofs clarify the hypotheses.

5.1 Girsanov's theorem

Let Θ be adapted with $\int_0^T \Theta_s^2 \mathrm{d}s < \infty$ a.s., and let

$$Z_t = \exp\Big(-\int_0^t \Theta_s \mathrm{d}W_s - \frac{1}{2}\int_0^t \Theta_s^2 \mathrm{d}s\Big), \qquad \mathrm{d}Z_t = -\Theta_t Z_t \mathrm{d}W_t.$$

Z is a positive local martingale, hence a supermartingale. If $\mathbb{E}Z_T = 1$ it is a true martingale on $[0,T]$ (Proposition 2.12), and $\mathbb{Q}(A) = \mathbb{E}[Z_T \mathbf{1}_A]$ defines a probability measure equivalent to $\mathbb{P}$ on $\mathcal{F}_T$.

Lemma 5.1. *Let Z be a true martingale as above. An adapted continuous Y is a $\mathbb{Q}$-(local) martingale if and only if YZ is a $\mathbb{P}$-(local) martingale.*

Proof. For the martingale case: if YZ is a $\mathbb{P}$-martingale, the Bayes rule (Theorem 1.21) gives, for $s \le t \le T$, using $\mathbb{E}[Z_T \mid \mathcal{F}_t] = Z_t$,

$$\mathbb{E}^{\mathbb{Q}}[Y_t \mid \mathcal{F}_s] = \frac{\mathbb{E}^{\mathbb{P}}[Y_t Z_T \mid \mathcal{F}_s]}{Z_s} = \frac{\mathbb{E}^{\mathbb{P}}[Y_t Z_t \mid \mathcal{F}_s]}{Z_s} = \frac{Y_s Z_s}{Z_s} = Y_s.$$

The converse is the same computation in reverse, and the local case follows by stopping. □

Theorem 5.2 (Girsanov)**.** *Assume $\mathbb{E}Z_T = 1$. Under $\mathbb{Q}$, the process $W_t^{\mathbb{Q}} = W_t + \int_0^t \Theta_s \mathrm{d}s$ is a Brownian motion on $[0,T]$.*

Proof. By the product rule,

$$\mathrm{d}(ZW^{\mathbb{Q}}) = Z\,\mathrm{d}W^{\mathbb{Q}} + W^{\mathbb{Q}}\mathrm{d}Z + \mathrm{d}Z\,\mathrm{d}W^{\mathbb{Q}} = Z(\mathrm{d}W + \Theta \mathrm{d}t) - W^{\mathbb{Q}} Z\Theta\,\mathrm{d}W - Z\Theta\,\mathrm{d}t = Z(1 - \Theta W^{\mathbb{Q}})\mathrm{d}W.$$

The $\mathrm{d}t$ terms cancel, which is the whole purpose of the $-\Theta$ in Z. So $ZW^{\mathbb{Q}}$ is a $\mathbb{P}$-local martingale, and by Lemma 5.1 $W^{\mathbb{Q}}$ is a $\mathbb{Q}$-local martingale. Quadratic variation is an a.s.

limit, so it does not change under an equivalent measure, and adding a finite-variation term does not change it either: $[W^{\mathbb{Q}}]_t = [W]_t = t$. Lévy's characterisation (Theorem 4.10) applied under $\mathbb{Q}$ finishes the proof. □

> **Intuition.** Think of paths as outcomes. Under $\mathbb{Q}$, we reweight each path by a factor Z_T favouring those that went in the direction $-\Theta$. For a constant Θ, $\mathbb{Q}$ tilts the terminal Gaussian exactly as in the class notes' example, moving the mean by $-\Theta t$. Paths are not moved; they are merely given different weights. The same path that looks like driftless noise with a lucky run under one measure looks like drift plus ordinary noise under the other. Because the fine structure of a path (quadratic variation) has probability one under both, only the slow component can change. Volatility is a property of the paths themselves, and drift is a property of the weights.

Theorem 5.3 (Novikov's condition, stated)**.** *If* $\mathbb{E}\exp\left(\frac{1}{2}\int_0^T \Theta_s^2 \mathrm{d}s\right) < \infty$, *then* $\mathbb{E}Z_T = 1$. *In particular this holds for bounded* Θ.

> **Pitfall.** Novikov is sufficient, not necessary, and the condition truly matters. Sin (1998) shows that for a stock with lognormal stochastic volatility, $\mathrm{d}S = \sigma_t S\,\mathrm{d}W$, $\mathrm{d}\sigma_t = \xi\sigma_t \mathrm{d}B$, positive correlation $\rho > 0$ between W and B makes S a strict local martingale: $\mathbb{E}S_T < S_0$. A calibrated model that fails this would misprice a forward. It is one reason for caution with positive spot–vol correlation, which occurs in some FX and commodity markets.

The d-dimensional version. With $\vec{\Theta}$ d-dimensional and $Z_t = \exp(-\int \vec{\Theta}\cdot \mathrm{d}\vec{W} - \frac{1}{2}\int |\vec{\Theta}|^2 \mathrm{d}s)$, each $W^{\mathbb{Q},i} = W^i + \int \Theta^i \mathrm{d}s$ is a $\mathbb{Q}$-Brownian motion, and they are independent, using the d-dimensional Lévy theorem. It is the result used in Chapter 5 of the class notes.

Girsanov on a general semimartingale. If $\mathrm{d}X = \Delta \mathrm{d}W + \mu \mathrm{d}t$, then under $\mathbb{Q}$, $\mathrm{d}X = \Delta \mathrm{d}W^{\mathbb{Q}} + (\mu - \Delta\Theta)\mathrm{d}t$. For a general local martingale M and density process Z, $M - \int \frac{\mathrm{d}[M,Z]}{Z}$ is a $\mathbb{Q}$-local martingale. The drift adjustment equals the instantaneous covariance of M with $\ln Z$. This form makes change of numeraire computations nearly automatic: *drift change equals covariance with the log of the density*.

5.2 Application: Brownian motion with drift

To simulate $\mathbb{E}[f(\text{path of } W_t + \nu t)]$ under $\mathbb{P}$, apply Girsanov with constant $\Theta = -\nu$ (the Cameron–Martin theorem):

$$\mathbb{E}\left[f\left((W_s + \nu s)_{s\le T}\right)\right] = \mathbb{E}\left[f\left((W_s)_{s\le T}\right) e^{\nu W_T - \frac{1}{2}\nu^2 T}\right].$$

Proposition 5.4 (Maximum with drift)**.** *Let* $X_t = W_t + \nu t$ *and* $a > 0$. *Then*

$$\mathbb{P}\Big(\max_{s\le T} X_s \ge a\Big) = \Phi\Big(\frac{\nu T - a}{\sqrt{T}}\Big) + e^{2\nu a}\Phi\Big(\frac{-\nu T - a}{\sqrt{T}}\Big).$$

Proof. By Cameron–Martin, the probability is $\mathbb{E}[\mathbf{1}_{\{M_T \ge a\}} e^{\nu W_T - \nu^2 T/2}]$, where M is the maximum of W. On $\{W_T \ge a\}$ the event holds automatically, and $\mathbb{E}[\mathbf{1}_{\{W_T \ge a\}} e^{\nu W_T - \nu^2 T/2}] = \mathbb{P}(W_T + \nu T \ge a)$ gives the first term. On $\{M_T \ge a, W_T < a\}$, reflect: the path is replaced by one ending at $2a - W_T > a$. The term becomes $\mathbb{E}[\mathbf{1}_{\{W_T > a\}} e^{\nu(2a - W_T) - \nu^2 T/2}] = e^{2\nu a}\mathbb{P}(W_T - \nu T > a)$. □

5.3 Application: barrier options

The class notes reach up-and-out options at the very end. Here is the cleanest closed form.

Theorem 5.5 (Down-and-out call)**.** *In Black–Scholes, let $C(S, K)$ be the vanilla call. For a barrier $B < S_0$ with $B \le K$, continuously monitored,*

$$C_{\mathrm{do}}(S_0) = C(S_0, K) - \left(\frac{S_0}{B}\right)^{1-2r/\sigma^2} C\left(\frac{B^2}{S_0}, K\right).$$

Proof. Under $\mathbb{Q}$, $\ln(S_t/S_0) = \sigma(W_t^{\mathbb{Q}} + \nu t)$, with $\nu = (r - \frac{1}{2}\sigma^2)/\sigma$. Put $b = \ln(B/S_0)/\sigma < 0$. By Cameron–Martin and the reflection principle for the minimum, the sub-density of $y = W_T$ on the event that the path stays above b is

$$e^{\nu y - \frac{1}{2}\nu^2 T}\left[\varphi_T(y) - \varphi_T(2b - y)\right], \qquad y > b.$$

The call pays only when $S_T > K \ge B$, which means $y > b$ automatically. The $\varphi_T(y)$ term gives the vanilla price. In the other term substitute $z = y - 2b$. Then $\varphi_T(2b - y) = \varphi_T(z)$, $S_T = S_0 e^{\sigma(2b+z)} = (B^2/S_0)e^{\sigma z}$ and $e^{\nu y} = e^{2\nu b}e^{\nu z}$. Therefore that term equals $e^{2\nu b}$ times a vanilla call on an underlying starting from B^2/S_0, where $e^{2\nu b} = (B/S_0)^{2\nu/\sigma} = (S_0/B)^{1-2r/\sigma^2}$. □

> **Intuition.** This is the method of images from electrostatics. The price must vanish at $S = B$. Place a virtual, negatively weighted option at the mirror image of S_0 across the barrier (in log space, B^2/S_0). Its weight is set so that the two cancel on $S = B$. Check it: at $S_0 = B$, $(S_0/B)^{\cdot} = 1$ and $B^2/S_0 = S_0$, so the two terms cancel.

> **Numerical check.** For $S_0 = K = 100$, $B = 90$, $r = 5\%$, $\sigma = 20\%$, $T = 1$: vanilla 10.4506 and down-and-out 8.6655. A Monte Carlo with 400,000 paths and 250 steps, testing the barrier only at grid points, gives 8.9347 ± 0.0232, which is biased high by about 12 standard errors because paths can cross and return unobserved. Using the Brownian bridge crossing probability (3.2) on each step gives 8.6817 ± 0.0231, consistent with the formula.

The difference between the two estimates is also real money: a contract monitored daily is a different contract from one monitored continuously. The Broadie–Glasserman–Kou correction shifts the barrier by a factor $\exp(\pm 0.5826\,\sigma\sqrt{\Delta t})$ to convert one to the other.

Remark 5.6. In-out parity: a down-and-in plus a down-and-out gives a vanilla, because exactly one of them is alive at maturity. Up-and-out calls have a more complicated formula, as the payoff is cut off above; the integrand is restricted to $K < S_T < B$, and the result needs digital-type terms. Lookbacks follow from the joint density (3.1) with drift.

5.4 Martingale representation

Theorem 5.7. *Let $(\mathcal{F}_t)$ be the (augmented) filtration generated by W. Every $F \in L^2(\mathcal{F}_T)$ has a unique representation*

$$F = \mathbb{E}F + \int_0^T H_t \, \mathrm{d}W_t, \qquad H \in L^2_{\mathrm{ad}}.$$

Consequently every local martingale of this filtration has a continuous version and is a stochastic integral.

Proof (main ideas). Uniqueness follows from the isometry. *Existence:* let $\mathcal{S} \subseteq L^2(\mathcal{F}_T)$ be the set of F that can be represented. By the isometry $\mathcal{S}$ is a closed linear subspace. It contains the exponentials

$$\mathcal{E}_f = \exp\Big(\int_0^T f\,\mathrm{d}W - \frac{1}{2}\int_0^T f^2\mathrm{d}t\Big) = 1 + \int_0^T \mathcal{E}_f(t) f(t)\,\mathrm{d}W_t,$$

for deterministic step functions f. So it remains to show these are dense. If $G \in L^2(\mathcal{F}_T)$ is orthogonal to every $\mathcal{E}_f$, then

$$\mathbb{E}\big[G \exp\big(\textstyle\sum_j u_j (W_{t_{j+1}} - W_{t_j})\big)\big] = 0 \quad \text{for all real } u_j.$$

This is an entire function of u, so it vanishes also for imaginary u. Hence the signed measure $G\,\mathrm{d}\mathbb{P}$ has zero Fourier transform on every finite collection of increments, and therefore $\mathbb{E}[G \mid W_{t_1}, \dots, W_{t_n}] = 0$. Such σ-algebras generate $\mathcal{F}_T$, and so $G = 0$. □

> **Pitfall.** The filtration must be generated by W. If $\mathcal{F}_t$ contains additional randomness (an independent coin, a second Brownian motion, a jump process), then there are martingales orthogonal to W, and the representation fails. That is the mathematical form of market incompleteness.

Clark–Ocone formula. When F is Malliavin differentiable, the integrand can be named: $H_t = \mathbb{E}[D_t F \mid \mathcal{F}_t]$, where $D_t F$ measures the sensitivity of F to a small bump in the noise at time t. For example, under $\mathbb{Q}$ with $r = 0$, let $F = (S_T - K)^+$. Then $D_t S_T = \sigma S_T$, and $D_t F = \sigma S_T \mathbf{1}_{\{S_T > K\}}$. Hence $H_t = \sigma \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T>K\}} \mid \mathcal{F}_t] = \sigma S_t N(d_+)$. Comparing with $H_t = \sigma S_t \Delta_t$ gives $\Delta_t = N(d_+)$: the Black–Scholes delta, without a PDE.

> **Intuition.** The representation theorem says that *with one source of noise, all uncertainty can be hedged by trading that noise.* Each new piece of information arrives through a $\mathrm{d}W$; a martingale moves only when information arrives, and its move must be proportional to $\mathrm{d}W$, since there is nothing else. A binomial tree has the same property: at each node there are two branches and two assets. With two assets (stock and bond), every one-step claim can be replicated.

5.5 The fundamental theorems, carefully

Consider the market with m stocks driven by d Brownian motions from Chapter 5 of the class notes. There the first fundamental theorem was proved by saying that discounted wealth $D_t X_t$ is a $\mathbb{Q}$-martingale. That hides a real problem: it is only a $\mathbb{Q}$-*local* martingale.

Example 5.8 (Arbitrage from an unbounded strategy)**.** Take one stock with $r = 0$ and $\mathrm{d}S = S\,\mathrm{d}W^{\mathbb{Q}}$. Let $\Delta_t S_t = 1/\sqrt{T-t}$. The wealth from $X_0 = 0$ is $X_t = \int_0^t (T-s)^{-1/2}\mathrm{d}W_s^{\mathbb{Q}}$, and by the pitfall in Chapter 4, $\tau = \inf\{t : X_t = 1\} < T$ almost surely. Stop trading at τ. The strategy turns 0 into 1 almost surely, with an equivalent martingale measure present! The catch is that X can go arbitrarily negative on the way.

Definition 5.9. A self-financing strategy is *admissible* if its discounted wealth is bounded below by a constant: there is some c with $D_t X_t \geq -c$ for all $t \leq T$ a.s.

Theorem 5.10 (First fundamental theorem, sufficient direction)**.** *If an equivalent local martingale measure $\mathbb{Q}$ exists, there is no admissible arbitrage.*

Proof. $D_t X_t + c$ is a nonnegative $\mathbb{Q}$-local martingale, thus a supermartingale by Proposition 2.12. Hence $\mathbb{E}^{\mathbb{Q}}[D_T X_T] \le X_0$. If $X_0 = 0$ and $\mathbb{P}(X_T \ge 0) = 1$, then by equivalence $\mathbb{Q}(X_T \ge 0) = 1$. Combined with $\mathbb{E}^{\mathbb{Q}}[D_T X_T] \le 0$, it follows that $X_T = 0$ $\mathbb{Q}$-a.s., hence $\mathbb{P}$-a.s. □

The class notes' argument becomes correct with "supermartingale" in place of "martingale". A full converse needs more care: *no arbitrage* alone does not imply the existence of a $\mathbb{Q}$ in continuous time. Delbaen and Schachermayer (1994) proved the sharp statement: a suitable condition called *No Free Lunch with Vanishing Risk*, a strengthening of no-arbitrage that also rules out sequences of nearly-arbitrages, is equivalent to the existence of an equivalent local martingale measure.

Theorem 5.11 (Second fundamental theorem)**.** *Assume $\mathcal{F}_t$ is generated by $\vec{W}$ and that a market price of risk exists. The market is complete if and only if the equivalent local martingale measure is unique, if and only if σ_t has rank d for a.e. t, a.s.*

Sketch. If $\operatorname{rank}\sigma = d$: given a claim V_T, the martingale $\mathbb{E}^{\mathbb{Q}}[D_T V_T \mid \mathcal{F}_t]$ is by d-dimensional representation $\int \sum_j H^j \mathrm{d}W^{\mathbb{Q},j}$. Solve $H = D\,\sigma^{\top}(\Delta S)$ for Δ, possible since $\sigma^{\top}$ is onto $\mathbb{R}^d$. If $\operatorname{rank}\sigma < d$, there is a non-zero adapted $\eta \in \ker\sigma$. Then $\vec{\Theta} + \epsilon\eta$ gives a second market price of risk and thus a second measure (Novikov for small bounded η), and the claim $\mathcal{E}(\int \eta \cdot \mathrm{d}W^{\mathbb{Q}})_T$, which has different prices under the two measures, cannot be hedged. □

> **Intuition.** *Prices come from martingale measures, hedges come from completeness.* When σ has rank $< d$ there are more shocks than ways to trade them. Then many $\mathbb{Q}$ are consistent with the traded assets. Each gives a different arbitrage-free price of a non-traded claim, and these prices fill an interval. Stochastic volatility and jump models live here. We calibrate $\mathbb{Q}$ to option prices because no-arbitrage alone cannot pin it down.

Exercises

5.1 Use Proposition 5.4 with $T \to \infty$ to show that for $\nu < 0$ and $a > 0$, $\mathbb{P}(\sup_t X_t \ge a) = e^{2\nu a}$. So the overall maximum is exponential. Deduce the probability that a stock with a negative log-drift ever doubles.

5.2 (First-passage density with drift.) Derive that τ_a for X has density $\frac{a}{\sqrt{2\pi t^3}} e^{-(a-\nu t)^2/2t}$, and find $\mathbb{E}\tau_a$ for $\nu > 0$.

5.3 Derive the price of a one-touch paying 1 at expiry, in Black–Scholes, if S ever hits $B > S_0$.

5.4 Find $F = W_T^2$'s martingale representation by hand, and the integrand for $F = W_T^3$. Compare with Clark–Ocone.

5.5 Show that if $\mathbb{Q} \sim \mathbb{P}$ on $\mathcal{F}_T$, $\mathbb{Q}$ has a density process $Z_t = \mathbb{E}[\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} \mid \mathcal{F}_t]$, and by the representation theorem $Z_t = \mathcal{E}(-\int \Theta\,\mathrm{d}W)_t$ for some adapted Θ. So in a Brownian filtration, *every* equivalent measure change is a Girsanov change.

5.6 Compute $\mathbb{E}^{\mathbb{Q}}$ of the realised variance of $\ln S$ under $\mathbb{Q}$ and under $\mathbb{P}$. Which of the two is the volatility of a Black–Scholes model?

5.7 In a market with $m = 1$, $d = 2$, $\mathrm{d}S = S(\mu \mathrm{d}t + \sigma_1 \mathrm{d}W^1 + \sigma_2 \mathrm{d}W^2)$, describe all market prices of risk. Find a change of basis showing the market has only one effective Brownian motion. Is it complete with respect to the filtration of S? With respect to that of (W^1, W^2)?

Chapter 6

Stochastic Differential Equations and PDEs

6.1 Strong solutions

Consider

$$\mathrm{d}X_t = b(t, X_t)\mathrm{d}t + \sigma(t, X_t)\mathrm{d}W_t, \qquad X_0 = x. \tag{6.1}$$

A *strong solution* is a continuous adapted process (adapted to the filtration generated by W) satisfying the integrated equation a.s. It is a machine: feed in the noise, get out the path.

Theorem 6.1 (Existence and uniqueness)**.** *Suppose b, σ are Lipschitz in x uniformly in t, and $|b(t,x)| + |\sigma(t,x)| \le K(1+|x|)$. Then* (6.1) *has a unique strong solution, with $\mathbb{E}\sup_{t\le T}|X_t|^2 < \infty$.*

Proof. Picard iteration. Let $X^0 \equiv x$ and $X_t^{n+1} = x + \int_0^t b(s, X_s^n)\mathrm{d}s + \int_0^t \sigma(s, X_s^n)\mathrm{d}W_s$. Using $(a+b)^2 \le 2a^2 + 2b^2$, Cauchy–Schwarz for the $\mathrm{d}s$ integral, Doob and the isometry for the $\mathrm{d}W$ integral, and Lipschitz continuity with constant L,

$$\mathbb{E}\sup_{s\le t}|X_s^{n+1} - X_s^n|^2 \le C\int_0^t \mathbb{E}\sup_{r\le s}|X_r^n - X_r^{n-1}|^2\mathrm{d}s, \qquad C = 2L^2(T+4).$$

Iterating gives a bound $A(Ct)^n/n!$. Chebyshev and Borel–Cantelli give a.s. uniform convergence to a limit, which solves the equation.

Uniqueness. If X, Y are two solutions, let $v(t) = \mathbb{E}\sup_{s\le t\wedge\tau}|X_s - Y_s|^2$, stopped so as to be finite. The same estimate gives $v(t) \le C\int_0^t v(s)\mathrm{d}s$. Gronwall's lemma then yields $v \equiv 0$. □

Lemma 6.2 (Gronwall)**.** *If v is bounded, nonnegative and $v(t) \le a + C\int_0^t v(s)\mathrm{d}s$, then $v(t) \le ae^{Ct}$.*

What goes wrong without the conditions.

- *Explosion.* $\mathrm{d}X = X^2\mathrm{d}t$, $X_0 = 1$ has solution $1/(1-t)$, which blows up at $t = 1$. Superlinear growth lets the solution reach infinity. (Superlinear σ does not always explode; it can be saved by negative drift, but the moments may be infinite.)
- *Non-uniqueness.* $\mathrm{d}X = 3X^{2/3}\mathrm{d}t$, $X_0 = 0$ has both $X \equiv 0$ and $X_t = t^3$.

- *No strong solution.* Tanaka's equation $\mathrm{d}X = \operatorname{sgn}(X)\mathrm{d}W$ has solutions (take X a Brownian motion and define $W = \int \operatorname{sgn}(X)\mathrm{d}X$, again Brownian). This is a *weak* solution: the pair (X, W) exists, but X is not a function of W. Since $W = \int \operatorname{sgn}(X)\mathrm{d}X$ can be recovered from $|X|$ by Tanaka's formula, W loses the sign information, so no strong solution exists.

Theorem 6.3 (Yamada–Watanabe, one dimension, stated)**.** *Pathwise uniqueness holds if b is Lipschitz and $|\sigma(x) - \sigma(y)| \le c|x - y|^{1/2}$. Weak existence together with pathwise uniqueness implies strong existence.*

This matters for CIR and Heston, where $\sigma(x) = \xi\sqrt{x}$ is not Lipschitz at zero.

6.2 Solvable models

Linear SDEs. For $\mathrm{d}X = (a_t + b_t X)\mathrm{d}t + (c_t + e_t X)\mathrm{d}W$, let U_t be the homogeneous solution, $U_t = \exp(\int_0^t (b_s - \tfrac{1}{2}e_s^2)\mathrm{d}s + \int_0^t e_s \mathrm{d}W_s)$. Then

$$X_t = U_t\Big(x + \int_0^t \frac{a_s - e_s c_s}{U_s}\mathrm{d}s + \int_0^t \frac{c_s}{U_s}\mathrm{d}W_s\Big),$$

which one checks by applying the product rule to the claimed solution.

Ornstein–Uhlenbeck. $\mathrm{d}X = \kappa(\theta - X)\mathrm{d}t + \sigma\mathrm{d}W$. Multiply by $e^{\kappa t}$:

$$X_t = \theta + (x - \theta)e^{-\kappa t} + \sigma\int_0^t e^{-\kappa(t-s)}\mathrm{d}W_s. \tag{6.2}$$

So X_t is Gaussian with mean $\theta + (x-\theta)e^{-\kappa t}$ and variance $\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$. Its stationary law is $\mathcal{N}(\theta, \sigma^2/2\kappa)$, and the half-life of a shock is $\ln 2/\kappa$. Sampled discretely with step h, OU is an AR(1) with coefficient $e^{-\kappa h}$. This is the model behind Vasicek rates and pairs trading.

Cox–Ingersoll–Ross. $\mathrm{d}X = \kappa(\theta - X)\mathrm{d}t + \sigma\sqrt{X}\mathrm{d}W$. Taking expectations in the integrated equation gives $\mathbb{E}X_t = \theta + (x-\theta)e^{-\kappa t}$, and Itô on X^2 gives

$$\operatorname{Var} X_t = x\frac{\sigma^2}{\kappa}\big(e^{-\kappa t} - e^{-2\kappa t}\big) + \frac{\theta\sigma^2}{2\kappa}\big(1 - e^{-\kappa t}\big)^2.$$

The law is a scaled noncentral chi-squared: $X_t \sim c\,\chi'^2(\delta, \lambda)$, with $c = \sigma^2(1 - e^{-\kappa t})/4\kappa$, degrees of freedom $\delta = 4\kappa\theta/\sigma^2$, and noncentrality $\lambda = xe^{-\kappa t}/c$. The stationary law is Gamma$(2\kappa\theta/\sigma^2,\ 2\kappa/\sigma^2)$ (shape, rate).

Proposition 6.4 (Feller condition)**.** *If $2\kappa\theta \ge \sigma^2$, X never reaches 0 (from $x > 0$). If $2\kappa\theta < \sigma^2$, it reaches 0 a.s. and is instantaneously reflected.*

This follows from the scale function criterion below. The dimension $\delta = 4\kappa\theta/\sigma^2$ is the analogue of the dimension of a squared Bessel process, for which zero is hit exactly when $\delta < 2$. In equity-calibrated Heston models, the Feller condition is often violated.

6.3 The generator, Dynkin and exit problems

Let X solve a time-homogeneous version of (6.1). Define the *generator*

$$\mathcal{A}f(x) = b(x)f'(x) + \tfrac{1}{2}\sigma^2(x)f''(x).$$

By Itô's formula, $f(X_t) - \int_0^t \mathcal{A}f(X_s)\mathrm{d}s$ is a local martingale. In particular, for a stopping time τ with enough integrability, we have *Dynkin's formula*:

$$\mathbb{E}_x f(X_\tau) = f(x) + \mathbb{E}_x \int_0^\tau \mathcal{A}f(X_s)\mathrm{d}s.$$

> **Intuition.** $\mathcal{A}f(x)$ is the expected rate of change of $f(X)$ when X sits at x: $\lim_{h\downarrow 0} \frac{\mathbb{E}_x f(X_h) - f(x)}{h}$. It is the drift of $f(X_t)$. Every PDE in this chapter says one thing: *the drift of something is zero (or equals a known running cost).*

Exit problems. Let τ be the exit time from (a, b). Solving $\mathcal{A}s = 0$ gives the *scale function* $s'(x) = \exp(-\int^x 2b/\sigma^2)$, and $s(X_t)$ is a local martingale. Therefore

$$\mathbb{P}_x(X_\tau = b) = \frac{s(x) - s(a)}{s(b) - s(a)}.$$

Solving $\mathcal{A}m = -1$ with $m(a) = m(b) = 0$ gives the expected exit time $m(x) = \mathbb{E}_x\tau$. Boundary behaviour is read off s: a point at 0 is not reached from $x > 0$ if $s(0+) = -\infty$. For CIR, $s'(x) \propto x^{-2\kappa\theta/\sigma^2} e^{2\kappa x/\sigma^2}$, whose integral diverges at 0 exactly when $2\kappa\theta/\sigma^2 \geq 1$. This proves Proposition 6.4 (up to behaviour at ∞).

6.4 Kolmogorov equations

Theorem 6.5 (Backward equation)**.** *If $u \in C^{1,2}$ with suitable growth solves $\partial_t u + \mathcal{A}u = 0$, $u(T, x) = h(x)$, then $u(t, x) = \mathbb{E}[h(X_T) \mid X_t = x]$.*

Proof. By Itô's formula, $u(s, X_s)$ has no drift; if it is a true martingale, then $u(t, X_t) = \mathbb{E}[u(T, X_T) \mid \mathcal{F}_t] = \mathbb{E}[h(X_T) \mid \mathcal{F}_t]$. □

Forward (Fokker–Planck) equation. Let $p(T, y)$ be the density of X_T given $X_t = x$. For any test function f, Dynkin's formula gives $\partial_T \mathbb{E}f(X_T) = \mathbb{E}\mathcal{A}f(X_T)$, that is, $\partial_T \int fp = \int (\mathcal{A}f)p$. Integrating by parts in y,

$$\partial_T p = -\partial_y(bp) + \tfrac{1}{2}\partial_{yy}(\sigma^2 p). \tag{6.3}$$

The forward equation evolves *probabilities* forward in time, and the backward equation evolves *payoffs* backward. They are adjoint. Dupire's formula is an immediate consequence of the forward equation (Chapter 9).

Setting $\partial_T p = 0$ gives the stationary density, for one-dimensional diffusion, $p_\infty(y) \propto \frac{1}{\sigma^2(y)} \exp(\int^y 2b/\sigma^2)$ (speed density). Check for OU: this gives $\mathcal{N}(\theta, \sigma^2/2\kappa)$. For CIR it gives the stated Gamma.

6.5 Feynman–Kac

Theorem 6.6 (Feynman–Kac). *Let $u \in C^{1,2}([0,T)\times\mathbb{R}) \cap C([0,T]\times\mathbb{R})$ solve*

$$\partial_t u + \mathcal{A}u - r(t,x)u + g(t,x) = 0, \qquad u(T,x) = h(x),$$

with r bounded below. Suppose in addition that the process $Y_s = e^{-\int_t^s r}u(s,X_s) + \int_t^s e^{-\int_t^v r} g(v,X_v)\mathrm{d}v$ is a true martingale (for instance $\partial_x u\,\sigma$ with polynomial growth and polynomial moments). Then

$$u(t,x) = \mathbb{E}_{t,x}\left[e^{-\int_t^T r(s,X_s)\mathrm{d}s}h(X_T) + \int_t^T e^{-\int_t^s r(v,X_v)\mathrm{d}v} g(s,X_s)\mathrm{d}s\right].$$

Proof. By Itô's formula and the PDE,

$$\mathrm{d}Y_s = e^{-\int_t^s r}\big(\partial_t u + \mathcal{A}u - ru + g\big)\mathrm{d}s + e^{-\int_t^s r}\partial_x u\,\sigma\,\mathrm{d}W_s = e^{-\int_t^s r}\partial_x u\,\sigma\,\mathrm{d}W_s.$$

So Y is a local martingale, and under the assumption a martingale. Taking $\mathbb{E}_{t,x}$ of $Y_t = \mathbb{E}[Y_T \mid \mathcal{F}_t]$ gives the formula. □

> **Pitfall.** The PDE alone does not determine the solution; you need growth conditions. The heat equation has non-zero solutions with zero initial data (Tychonoff's examples) that grow like $e^{x^{2+\epsilon}}$. Feynman–Kac selects the solution with controlled growth. In finance the same issue appears when the stock is a strict local martingale: the Black–Scholes-type PDE then has several solutions, and the price (the expectation) may not be the one with the "expected" boundary behaviour.

The converse. The class notes argue the other way: define u by the expectation; then $u(s,X_s)$ is a martingale by the Markov property and the tower law, and if u is smooth enough, its $\mathrm{d}t$ term must vanish, which gives the PDE. Smoothness is the delicate part. It holds when σ is uniformly elliptic and coefficients are Hölder.

Example 6.7 (Black–Scholes both ways)**.** Under $\mathbb{Q}$, $\mathcal{A} = rx\partial_x + \frac{1}{2}\sigma^2x^2\partial_{xx}$, so with $g = 0$ the PDE is the BSM equation and the expectation is the risk-neutral price. The equation is thus the *infinitesimal* form of $e^{-rt}V_t$ having zero drift.

Example 6.8 (Bond prices)**.** In a short-rate model, $P(t,T) = \mathbb{E}_t[e^{-\int_t^T r_s\mathrm{d}s}]$ is given by Feynman–Kac with potential $r(x) = x$, $h = 1$. This is the term-structure PDE of Chapter 11.

Example 6.9 (Occupation time / running payoffs)**.** $u(t,x) = \mathbb{E}_{t,x}[\exp(-\lambda\int_t^T \mathbf{1}_{\{W_s>0\}}\mathrm{d}s)]$ solves a PDE with a discontinuous potential. Solving it (Kac's original use) gives the arcsine law. In the same way, Parisian options and corridor products are priced with a discontinuous killing rate.

6.6 Optimal stopping and American options

An American option with payoff g has value, under $\mathbb{Q}$,

$$V(t,x) = \sup_{\tau\in[t,T]} \mathbb{E}^{\mathbb{Q}}_{t,x}\left[e^{-r(\tau-t)}g(S_\tau)\right].$$

This is the Snell envelope (Theorem 2.14) in continuous time. The discounted value is the smallest $\mathbb{Q}$-supermartingale above the discounted payoff. Therefore the *drift of* $e^{-rt}V$ *is* ≤ 0, and it is 0 wherever it is optimal to continue. In formulas, with $\mathcal{L} = \partial_t + rx\partial_x + \frac{1}{2}\sigma^2x^2\partial_{xx} - r$,

$$\min\big(-\mathcal{L}V,\ V - g\big) = 0, \qquad V(T,x) = g(x). \tag{6.4}$$

This *variational inequality* splits space into a continuation region $\{V > g\}$, where the BSM equation holds, and an exercise region $\{V = g\}$. The boundary between them is free, part of the solution.

Theorem 6.10 (Early exercise of a call)**.** *If $r \ge 0$ and the stock pays no dividends, an American call is worth the European call. It is never optimal to exercise early.*

Proof. Let $\tau \le T$ be a stopping time. By conditional Jensen, the martingale property of $e^{-rt}S_t$, and $e^{-rT} \le e^{-r\tau}$,

$$\mathbb{E}^{\mathbb{Q}}[e^{-rT}(S_T - K)^+ \mid \mathcal{F}_\tau] \ge (e^{-r\tau}S_\tau - Ke^{-rT})^+ \ge e^{-r\tau}(S_\tau - K)^+.$$

Taking expectations shows that holding to maturity dominates stopping at any τ. □

> **Intuition.** Exercising a call early pays the strike sooner (losing interest) and gives up the insurance against the stock falling below K. Both effects favour waiting. A put is different: exercising early *receives* K sooner, so with $r > 0$ a deep in-the-money put should be exercised once the interest on K outweighs the remaining insurance value.

Theorem 6.11 (Perpetual American put)**.** *With $T = \infty$, $r > 0$, the optimal rule is to exercise on first reaching*

$$L^* = \frac{2r}{2r + \sigma^2}K,$$

and for $x \ge L^*$ the value is $V(x) = (K - L^*)\,(x/L^*)^{-2r/\sigma^2}$. For $x \le L^*$, $V(x) = K - x$.

Proof. Candidate. In continuation, time does not matter, so $\frac{1}{2}\sigma^2x^2V'' + rxV' - rV = 0$. Trying x^p gives $p = 1$ or $p = -\gamma$, where $\gamma = 2r/\sigma^2$. As V must stay bounded as $x \to \infty$, $V = Ax^{-\gamma}$. Value matching at L gives $AL^{-\gamma} = K - L$. So for each threshold L the value of the rule "exercise at L" is $V_L(x) = (K - L)(x/L)^{-\gamma}$. (It is also $(K - L)\mathbb{E}_x^{\mathbb{Q}}[e^{-r\tau_L}]$, and the Laplace transform of the hitting time gives this.) Maximising over L gives $L^* = \gamma K/(1 + \gamma)$.

Smooth pasting. At L^*, $V'(L^*) = -\gamma(K - L^*)/L^* = -1$, matching the slope of $K - x$.

Verification. $V \ge (K - x)^+$; $\mathcal{L}V = 0$ on $x > L^*$; and for $x < L^*$, $\mathcal{L}V = -rK < 0$. Hence $e^{-rt}V(S_t)$ is a supermartingale (V being C^1 with a piecewise C^2 form, Itô's formula still holds), and it dominates the discounted payoff, giving $V \ge$ every rule's value. Stopped at τ_{L^*} it is a martingale, so V is attained. □

> **Numerical check.** For $K = 100$, $r = 5\%$, $\sigma = 20\%$: $L^* = 71.43$ and $V(100) = 12.3200$. A CRR tree at $T = 100$ years with 20,000 steps gives 12.3179. For a $T = 1$ put with $S_0 = 100$, a tree with 5,000 steps gives 6.0902, against the European 5.5731: an early exercise premium of 0.5171.

Exercises

6.1 Solve $\mathrm{d}X = -\frac{X}{T-t}\mathrm{d}t + \mathrm{d}W$ on $[0, T)$ with $X_0 = 0$, and show it is a Brownian bridge from 0 to 0.

6.2 For OU, compute $\operatorname{Cov}(X_s, X_t)$, and then $\operatorname{Var}\int_0^t X_s \mathrm{d}s$. This variance appears in Vasicek bond prices.

6.3 For $\mathrm{d}X = \mu X \mathrm{d}t + \sigma X \mathrm{d}W$, $\mu \neq \frac{1}{2}\sigma^2$, find the scale function. What is the probability that it reaches b before a?

6.4 Use Fokker–Planck to verify that the lognormal density solves the forward equation for GBM.

6.5 Show that a perpetual American call on a dividend-paying stock with yield $q > 0$ has threshold $L^* = \frac{p}{p-1}K$, where $p > 1$ solves $\frac{1}{2}\sigma^2 p(p-1) + (r-q)p - r = 0$.

6.6 Let $h = (x - K)^+$. Compute ∂_{xx} of the BSM solution near T, and show the PDE's smoothing: gamma blows up like $(T-t)^{-1/2}$ at $x = K$.

6.7 (Interview) Let $u(x) = \mathbb{E}_x[\int_0^\tau e^{-\lambda t}\mathrm{d}t]$ where τ is the exit of Brownian motion from $(-1, 1)$. Set up and solve the ODE.

Chapter 7

Jump Processes

Continuous models cannot produce overnight gaps or the steepness of very short-dated option smiles. Allowing jumps also breaks completeness.

7.1 The Poisson process

Definition 7.1. A *Poisson process* of intensity $\lambda > 0$ is an adapted counting process N with $N_0 = 0$, with right-continuous paths, and increments independent of the past with $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$.

Equivalently, the waiting times between jumps are i.i.d. exponential with mean $1/\lambda$. Given $N_T = n$, the jump times are distributed like the order statistics of n uniform points on $[0, T]$; this gives an efficient simulation method.

Proposition 7.2. *The following are martingales: the compensated process* $\tilde{N}_t = N_t - \lambda t$; $\tilde{N}_t^2 - \lambda t$; $\tilde{N}_t^2 - N_t$; *and* $\exp(uN_t - \lambda t(e^u - 1))$.

Here is the first contrast with Brownian motion: $[\tilde{N}]_t = \sum_{s \le t} (\Delta N_s)^2 = N_t$ (random) whereas the predictable variation is $\langle \tilde{N} \rangle_t = \lambda t$ (deterministic). Both make the square of $\tilde{N}$ a martingale after subtraction.

> **Pitfall.** *Predictability matters more with jumps.* For a càdlàg integrand, $\int_0^t H_{s-} \mathrm{d}\tilde{N}_s$ is a martingale, but $\int_0^t H_s \mathrm{d}\tilde{N}_s$ need not be. For example,
>
> $$\int_0^t N_{s-} \mathrm{d}N_s = \sum_{k=1}^{N_t} (k-1) = \tfrac{1}{2} N_t(N_t - 1), \quad \text{while} \quad \int_0^t N_s \mathrm{d}N_s = \tfrac{1}{2} N_t(N_t + 1).$$
>
> The difference is N_t, a clearly non-zero expected gain. With Brownian motion, we could get away with being careless in notation because of continuity. With jumps, the left limit H_{s-} is the position taken *before* the jump, which is the only one a trader can have.

Compound Poisson. $J_t = \sum_{i=1}^{N_t} Y_i$ with Y_i i.i.d. with law ν_Y, independent of N. Conditioning on N_t,

$$\mathbb{E} e^{iuJ_t} = \exp\big(\lambda t(\varphi_Y(u) - 1)\big), \quad \mathbb{E} J_t = \lambda t \mathbb{E} Y, \quad \operatorname{Var} J_t = \lambda t \mathbb{E} Y^2.$$

7.2 Itô's formula with jumps

Let X be a process of finite-activity jump-diffusion form:

$$\mathrm{d}X_t = \mu_t \mathrm{d}t + \sigma_t \mathrm{d}W_t + \mathrm{d}J_t.$$

Theorem 7.3. *For $f \in C^2$,*

$$f(X_t) = f(X_0) + \int_0^t f'(X_{s-})(\mu_s \mathrm{d}s + \sigma_s \mathrm{d}W_s) + \frac{1}{2}\int_0^t f''(X_{s-})\sigma_s^2 \mathrm{d}s + \sum_{s \le t} \big[f(X_s) - f(X_{s-})\big].$$

Proof. Between jumps, X is an Itô process, and the usual formula applies. At a jump, add the actual change in f. There are finitely many jumps on any interval. □

> **Intuition.** Taylor expansion fails for big moves: $f(x+y) - f(x)$ is not approximated by $f'y + \frac{1}{2}f''y^2$. So we keep the whole difference. If $\mathrm{d}J$ has intensity λ and sizes Y, the drift contributed by the jumps is $\lambda \mathbb{E}[f(x+Y) - f(x)]$, and the generator becomes non-local:
>
> $$\mathcal{A}f(x) = bf' + \tfrac{1}{2}\sigma^2 f'' + \lambda \int \big(f(x+y) - f(x)\big)\nu_Y(\mathrm{d}y).$$
>
> Pricing PDEs become partial integro-differential equations (PIDEs).

Stock with jumps. Consider $\mathrm{d}S_t = S_{t-}(\mu \mathrm{d}t + \sigma \mathrm{d}W_t + \mathrm{d}J_t)$ with $Y_i > -1$. Theorem 7.3 applied to $\ln S$ gives

$$S_t = S_0 \exp\big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big) \prod_{i=1}^{N_t} (1 + Y_i). \tag{7.1}$$

A jump of relative size Y moves the stock from S_- to $S_-(1+Y)$. The term $\mathbb{E}S_t$ grows at the rate $\mu + \lambda \mathbb{E}Y$, so for $e^{-rt}S_t$ to be a martingale we need $\mu = r - \lambda \bar{m}$, $\bar{m} = \mathbb{E}Y$. The drift is lowered, compensating for expected jumps.

7.3 Change of measure with jumps

Theorem 7.4 (Girsanov for compound Poisson)**.** *Let J have intensity λ and jump density f under $\mathbb{P}$. Let $\tilde{\lambda} > 0$ and let $\tilde{f}$ be a density with the same support as f. Define*

$$Z_t = e^{(\lambda - \tilde{\lambda})t} \prod_{i=1}^{N_t} \frac{\tilde{\lambda}\tilde{f}(Y_i)}{\lambda f(Y_i)}.$$

Then Z is a positive mean-one martingale, and under $\mathrm{d}\mathbb{Q} = Z_T \mathrm{d}\mathbb{P}$, J is compound Poisson with intensity $\tilde{\lambda}$ and jump density $\tilde{f}$.

Proof. Compute the joint likelihood of the data. The density of seeing n jumps at times $t_1 < \cdots < t_n$ with sizes y_i on $[0,T]$ under (λ, f) is $\lambda^n e^{-\lambda T} \prod f(y_i)$, and Z_T is the ratio. Multiplicativity over disjoint intervals gives the martingale property. □

Intuition. Unlike Brownian motion, where only the drift could change, here intensities and jump-size laws can change, so long as the set of possible jump sizes stays the same. An equivalent measure cannot create or remove jumps of a given size; it can only reweight them. Note that the measure cannot remove the diffusion either. This much freedom is why the risk-neutral measure is highly non-unique in jump models.

Combining with Brownian Girsanov, Θ and $(\tilde{\lambda}, \tilde{f})$ must satisfy one equation to make $e^{-rt}S_t$ a martingale:

$$\mu - r - \sigma\Theta + \tilde{\lambda}\mathbb{E}^{\tilde{f}}[Y] = 0.$$

One equation, infinitely many unknowns. The market price of diffusion risk and of jump risk can be traded against one another.

7.4 Incompleteness: a one-period view

Suppose in one period the stock can go to uS_0, mS_0 or dS_0, with $u > 1 + r > d$. Risk-neutral probabilities (q_u, q_m, q_d) must satisfy two linear equations, so there is a one-parameter family. Two assets cannot replicate an arbitrary claim on three states. The set of arbitrage-free prices of a claim V is an interval

$$\Big[\inf_{\mathbb{Q}} \mathbb{E}^{\mathbb{Q}}[\tfrac{V}{1+r}],\ \sup_{\mathbb{Q}} \mathbb{E}^{\mathbb{Q}}[\tfrac{V}{1+r}]\Big],$$

and the top of the interval is the *super-replication price* (linear programming duality): the cheapest portfolio that is at least V in all states. In the Merton model, the super-replication interval for a call is essentially the whole no-arbitrage range. Super-replication is thus too expensive to use in practice. Instead one fixes a $\mathbb{Q}$ by calibrating to traded options.

7.5 Merton's jump-diffusion model

Let $\ln(1 + Y) \sim \mathcal{N}(\mu_J, \delta^2)$, so $\bar{m} = e^{\mu_J + \delta^2/2} - 1$. Merton (1976) argued that jump risk is diversifiable and so carries no risk premium. Under this assumption, $\mathbb{Q}$ leaves λ and the law of Y unchanged. (Later models calibrate to options and drop this assumption, in which case λ and μ_J in this section are risk-neutral parameters.)

Theorem 7.5 (Merton's formula)**.** *With $\lambda' = \lambda(1 + \bar{m})$,*

$$C = \sum_{n=0}^{\infty} e^{-\lambda' T} \frac{(\lambda' T)^n}{n!}\, C_{\mathrm{BS}}\big(S_0, K, T; r_n, \sigma_n\big),$$

where $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda\bar{m} + n\ln(1 + \bar{m})/T$.

Proof. Condition on $N_T = n$. Under $\mathbb{Q}$, from (7.1), $\ln S_T$ is normal with variance $\sigma^2 T + n\delta^2$ and mean such that $\mathbb{E}[S_T \mid N_T = n] = S_0 e^{(r-\lambda\bar{m})T}(1 + \bar{m})^n$. So the conditional price is $e^{-rT}\mathbb{E}[(S_T - K)^+ \mid n]$, a Black–Scholes value with forward $F_n = S_0 e^{(r-\lambda\bar{m})T}(1 + \bar{m})^n$. Rewrite $e^{-rT}\Phi$-terms using $e^{-rT} = e^{-r_n T} \cdot e^{(r_n - r)T}$, with $e^{(r_n - r)T} = e^{-\lambda\bar{m}T}(1 + \bar{m})^n$, and combine the factor with the Poisson weight $e^{-\lambda T}(\lambda T)^n/n!$. The product is exactly the stated weight with λ'. □

Intuition. A jump-diffusion is a Black–Scholes model whose volatility is random: with probability w_n you get a variance of $\sigma^2 T + n\delta^2$ (and an adjusted forward). A *mixture of normals* has fat tails, and that produces a smile. With negative mean jump, $\mu_J < 0$, you get a negative skew.

Numerical check. For $S_0 = 100$, $r = 0$, $T = 1$, $\sigma = 15\%$, $\lambda = 0.5$, $\mu_J = -15\%$, $\delta = 15\%$: the series gives 21.6958, 7.8826, 1.7042 for $K = 80, 100, 120$, agreeing to four decimals with independent Fourier inversion (Section 10.4).

7.6 Lévy processes

Definition 7.6. A *Lévy process* is an adapted càdlàg process with $X_0 = 0$, with stationary increments independent of the past.

Brownian motion with drift, Poisson and compound Poisson are all examples.

Theorem 7.7 (Lévy–Khintchine, stated). $\mathbb{E}e^{iuX_t} = e^{t\psi(u)}$, *where*

$$\psi(u) = ibu - \tfrac{1}{2}\sigma^2 u^2 + \int_{\mathbb{R}} \left(e^{iux} - 1 - iux\mathbf{1}_{\{|x|<1\}}\right)\nu(\mathrm{d}x),$$

for $b \in \mathbb{R}$, $\sigma \geq 0$, and a measure ν with $\nu\{0\} = 0$ and $\int \min(1, x^2)\nu(\mathrm{d}x) < \infty$.

Theorem 7.8 (Lévy–Itô decomposition, stated). *X splits into independent pieces: drift, Brownian motion with variance σ^2, a compound Poisson process of big jumps (with Lévy measure ν on $|x| \geq 1$), and a compensated limit of small jumps.*

Intuition. $\nu(A)$ is the expected number of jumps per unit time with size in A. If $\nu(\mathbb{R}) < \infty$, the process has finitely many jumps: *finite activity*. If $\nu(\mathbb{R}) = \infty$, there are infinitely many tiny jumps: *infinite activity*. The condition $\int_{|x|<1} x^2\nu(\mathrm{d}x) < \infty$ is what makes the sum of squared jumps finite, hence the compensated sum converges in L^2. The compensator $iux\mathbf{1}_{\{|x|<1\}}$ is what keeps the small-jump sum from blowing up. A Lévy process has no "free parameter" other than (b, σ, ν), which is the infinitely-divisible distribution theorem in disguise.

Examples of pure-jump models.

- **Variance gamma (Madan–Seneta):** Brownian motion with drift evaluated at a Gamma time, $X_t = \theta G_t + \sigma W_{G_t}$. It models a random business clock. Its Lévy density is $\nu(\mathrm{d}x) = \frac{e^{-\lambda_\pm|x|}}{\kappa|x|}\mathrm{d}x$. It has infinite activity but finite variation.
- **CGMY/KoBoL:** Lévy density $C\, e^{-\lambda_\pm|x|}/|x|^{1+Y}$, with $Y < 2$. The parameter Y governs small-jump activity: finite activity for $Y < 0$, infinite with finite variation for $0 \leq Y < 1$, infinite variation for $1 \leq Y < 2$.
- **Normal inverse Gaussian:** Brownian motion evaluated at an inverse Gaussian time.

All have a known ψ, so options are priced by Fourier methods. The risk-neutral stock is $S_t = S_0 e^{(r-\psi(-i))t + X_t}$, the *mean-correcting martingale*.

Why stationarity fails in practice. Since Lévy increments are i.i.d., cumulants of $\ln S_T$ add up linearly in T. The skewness of $\ln S_T$ therefore decays like $T^{-1/2}$ and the excess kurtosis like T^{-1} (a central limit effect), and the at-the-money slope of implied volatility in k decays like $1/T$. Index skews decay more slowly than this. Additive (Sato) processes, or stochastic volatility with jumps (Bates), address the gap.

Why jumps matter for hedging. If Δ hedged, the P&L at a jump of size y in S is $V(S+y) - V(S) - \Delta y$. For small y, it is $\frac{1}{2}\Gamma y^2$, the usual form, but for big y the error is not controlled by rebalancing frequency. *Hedging more often does not help against a jump.* Only trading other options does. This is a clean illustration of incompleteness: gap risk cannot be delta-hedged away.

Exercises

7.1 Show that $\exp(uN_t - \lambda t(e^u - 1))$ is a martingale. Use it to find the distribution of the first time $N_t - ct$ exceeds a positive level, when $\lambda < c$ (ruin theory).

7.2 Show that for a Lévy process with $\mathbb{E}|X_1| < \infty$, $X_t - t\mathbb{E}X_1$ is a martingale. If X_t has an exponential moment, find the martingale form of e^{X_t}.

7.3 Compute the variance, skewness and excess kurtosis of $\ln S_T$ in Merton's model. How do skewness and kurtosis scale in T? Why does this imply a smile that flattens with maturity?

7.4 In the three-state one-period market with $S_0 = 100$, $u = 1.2$, $m = 1$, $d = 0.8$, $r = 0$, find the interval of arbitrage-free call prices at $K = 100$. Find the cheapest super-replicating portfolio.

7.5 Let $\mathrm{d}S = S_-\mathrm{d}\tilde{J}$ with $\tilde{J}$ a compensated Poisson process of jump size a, $a > -1$, λ intensity. Find S_t. Is this a complete market? Solve for the hedging ratio replicating a claim $V(S_T)$ with the stock and a bond.

7.6 For $H_t = \mathbf{1}_{\{t \le \tau_1\}}$ with τ_1 the first jump of N, compute $\int_0^t H_s \mathrm{d}\tilde{N}_s$. Show it is a martingale with mean zero. This is the basic payoff of a credit default swap; find the *hazard rate* formulation of default probability.

Part III

Pricing Beyond Black–Scholes

Chapter 8

The Volatility Smile

8.1 Implied volatility

Black–Scholes has one unknown input, σ. The call price is strictly increasing in σ (vega $= S\varphi(d_+)\sqrt{T} > 0$), moving from intrinsic value at $\sigma \to 0$ to S at $\sigma \to \infty$. So every price between these bounds has a unique *implied volatility* $\sigma_{\text{imp}}(K, T)$.

> **Intuition.** Implied volatility is not a model belief; it is a quoting convention, like quoting a bond by yield. It is the *wrong number to put in the wrong formula to get the right price*. Yet it is a useful unit. It strips out the first-order dependence on spot, strike and maturity, so that options on different underlyings can be compared.

If Black–Scholes were right, σ_{imp} would be flat across K and T. In practice, equity index options show a downward sloping *skew*: low-strike puts trade at higher vols than high-strike calls. FX options show a more symmetric *smile*. Commodities can tilt either way. After the 1987 crash, the equity skew emerged and never left.

Useful coordinates. Let $F = Se^{(r-q)T}$ be the forward, $k = \ln(K/F)$ log-moneyness, and $w = \sigma_{\text{imp}}^2 T$ the total implied variance. Then Black's formula depends only on (k, w) after dividing by F and discount:

$$\frac{C}{DF} = N(d_+) - e^k N(d_-), \qquad d_\pm = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}.$$

8.2 Model-free constraints

Fix T, let $C(K)$ be the undiscounted call price $\mathbb{E}^{\mathbb{Q}}(S_T - K)^+$, and assume no arbitrage. Then

1. C is decreasing in K, with $-1 \le \partial_K C \le 0$ (a call spread is positive, and at most worth its width);
2. C is convex in K (a butterfly spread has positive payoff, hence positive value);
3. $C(K) \ge (F - K)^+$ and $C(K) \to 0$ as $K \to \infty$;
4. total variance is nondecreasing in T at a fixed forward-moneyness: $\partial_T w(k, T) \ge 0$ (calendar spreads, using that $C_T(Fe^k)/F$ is convex in k and that a convex function of a

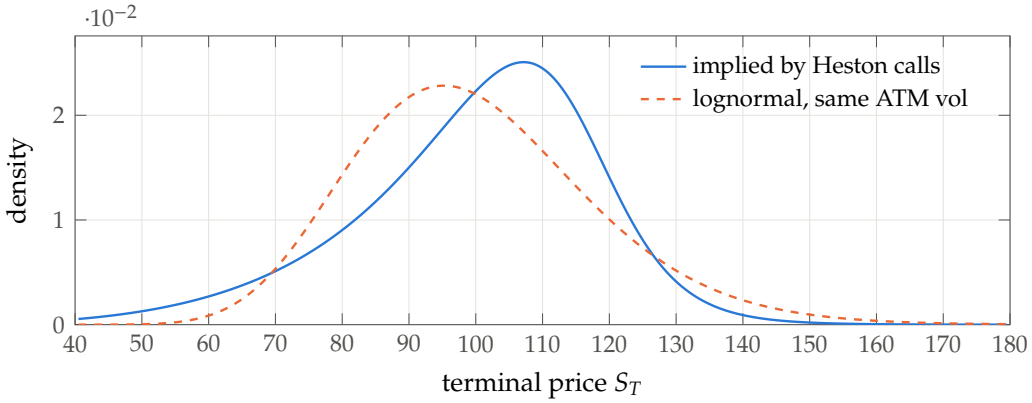


Figure 8.1: Breeden–Litzenberger: density from the second difference of call prices for the Heston parameters of Section 10.3, against the lognormal with the same at-the-money implied volatility (18.05%). Negative spot–vol correlation fattens the left tail and thins the right.

martingale increases).

Each comes from an explicit portfolio, so a violation would be a real arbitrage.

Theorem 8.1 (Breeden–Litzenberger). *Differentiating* $C(K) = \int (s-K)^+ \rho(s)\mathrm{d}s$ *twice gives*

$$\partial_K C = -\mathbb{Q}(S_T > K), \qquad \partial_{KK} C = \rho(K).$$

The risk-neutral density of S_T *is the second strike derivative of undiscounted call prices.*

> **Intuition.** A butterfly at K with width h, scaled by $1/h$, is an approximate Arrow–Debreu claim: a payoff of about 1 if S_T is near K, and 0 otherwise. Its value is thus the state price density. Options reveal the market's whole probability distribution, not just a volatility. Note that it reveals $\mathbb{Q}$, not $\mathbb{P}$. High put prices are a mix of a crash forecast and a premium for insurance.

8.3 Static replication

Theorem 8.2 (Carr–Madan). *For* $f \in C^2$ *and any reference* x_0,

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \int_0^{x_0} f''(K)(K-x)^+ \mathrm{d}K + \int_{x_0}^{\infty} f''(K)(x-K)^+ \mathrm{d}K.$$

Proof. It is Taylor's theorem with integral remainder, written with payoff kinks. For $x > x_0$, $\int_{x_0}^{x} f''(K)(x-K)\mathrm{d}K = f(x) - f(x_0) - f'(x_0)(x-x_0)$ by parts; for $x < x_0$ use the put term. □

So any European payoff is a bond, a forward and a strip of options, with the strip weighted by $f''(K)$. Taking risk-neutral expectations, with F as x_0 and D the discount factor,

$$V_0 = D f(F) + \int_0^{F} f''(K) P(K) \mathrm{d}K + \int_F^{\infty} f''(K) C(K) \mathrm{d}K.$$

8.4 Variance swaps

A *variance swap* pays at T the realised variance $\frac{1}{T}\sum_i (\ln S_{t_{i+1}}/S_{t_i})^2$ minus a fixed strike K_{var}. In the continuous limit, the realised variance is $\frac{1}{T}\int_0^T \sigma_t^2 \mathrm{d}t$ for an Itô stock with any stochastic σ (no jumps).

Theorem 8.3 (Log-contract replication)**.** *Let* $\mathrm{d}S = S(\mu_t \mathrm{d}t + \sigma_t \mathrm{d}W_t)$*, with* S *continuous, and* $r = 0$ *for simplicity. Then*

$$\frac{1}{2}\int_0^T \sigma_t^2 \mathrm{d}t = \int_0^T \frac{\mathrm{d}S_t}{S_t} - \ln\frac{S_T}{S_0},$$

and the fair strike is

$$K_{\text{var}} = \frac{1}{T}\mathbb{E}^{\mathbb{Q}}\int_0^T \sigma_t^2 \mathrm{d}t = \frac{2}{T}\Big[\int_0^F \frac{P(K)}{K^2}\mathrm{d}K + \int_F^\infty \frac{C(K)}{K^2}\mathrm{d}K\Big].$$

Proof. Itô's formula for $\ln S$: $\mathrm{d}\ln S = \mathrm{d}S/S - \frac{1}{2}\sigma^2 \mathrm{d}t$. This is a pathwise identity. Since $\int \mathrm{d}S/S$ is a $\mathbb{Q}$-martingale (under integrability), $\mathbb{E}^{\mathbb{Q}}\int_0^T \sigma_t^2 \mathrm{d}t = -2\mathbb{E}^{\mathbb{Q}}\ln(S_T/S_0)$. Apply Carr–Madan to $f(x) = -\ln x$ with $f'' = 1/x^2$ and $x_0 = F = S_0$. □

Intuition. The result is striking: *whatever* σ_t *does*, realised variance is replicated by holding a static log contract and delta hedging it, which means keeping a constant dollar amount in the stock (that is $1/S_t$ shares per unit of notional). The weight $1/K^2$ means low strikes matter a lot, so this is expensive when the skew is steep. It needs no volatility model, but it does assume no jumps, a limit on the range of strikes, and continuous rebalancing. The VIX index is exactly this strip computed from listed S&P options, at 30 days.

Numerical check. For the Heston parameters of Section 10.3 with $v_0 = \theta = 0.04$, $T = 1$, the strip gives fair variance 0.04000, against $\mathbb{E}^{\mathbb{Q}}\int v_t \mathrm{d}t/T = 0.04000$. With $v_0 = 0.09$, the strip gives 0.06162, matching the theoretical $\theta + (v_0 - \theta)\frac{1-e^{-\kappa T}}{\kappa T} = 0.06162$, whereas the ATM implied variance is 0.05189. The variance swap strike lies *above* ATM implied variance when the skew is negative: it pays for the wings.

8.5 What delta hedging actually earns

This is the most important practical identity in the theory of volatility.

Theorem 8.4 (Hedging P&L)**.** *Let* S *have realised volatility* σ_t *(any adapted process). Buy an option at implied volatility* σ_i*, and delta hedge using the Black–Scholes delta at* σ_i*, with* $V(t,S)$ *denoting the Black–Scholes price at* σ_i*. With all money financed at rate* r*, the terminal P&L is*

$$P\&L = \frac{1}{2}\int_0^T e^{r(T-t)}\,\Gamma_t\,S_t^2\left(\sigma_t^2 - \sigma_i^2\right)\mathrm{d}t,$$

where $\Gamma_t = \partial_{SS}V(t,S_t)$*. A short position earns the negative of this.*

Proof. Take $r = 0$ for readability. By Itô's formula, $\mathrm{d}V = \partial_t V \mathrm{d}t + \partial_S V \mathrm{d}S + \frac{1}{2}\Gamma S^2 \sigma_t^2 \mathrm{d}t$. The Black–Scholes PDE with σ_i gives $\partial_t V = -\frac{1}{2}\Gamma S^2 \sigma_i^2$. So

$$\mathrm{d}V - \partial_S V \, \mathrm{d}S = \tfrac{1}{2}\Gamma S^2 (\sigma_t^2 - \sigma_i^2)\mathrm{d}t.$$

The left side is the increment in the hedged portfolio's value. Integrating, and using $V(T, \cdot) =$ payoff, we get the result. □

Intuition. Over a small step, long gamma earns $\frac{1}{2}\Gamma(\Delta S)^2$ and pays theta $\frac{1}{2}\Gamma S^2 \sigma_i^2 \Delta t$. You break even when the stock's squared return equals $\sigma_i^2 \Delta t$, i.e. when it moves by one "implied" standard deviation. Note what this does and does not say. You earn $\sigma^2 - \sigma_i^2$ weighted by *dollar gamma*, which is large near the money and near expiry. Two paths with the same realised volatility but with different timing of moves relative to strike give different P&Ls. A big move while you are far from the strike earns nothing. This path-dependence is the reason a variance swap, whose weighting is uniform, exists.

Numerical check. Selling a 1-year ATM call at 20%, hedging daily at $\sigma_i = 20\%$, when the stock realises 25%, over 50,000 paths, gives a mean P&L of −1.990 with standard deviation 1.003 (see `results.txt`). The Black–Scholes value difference at 25% and 20% is 1.982. So the average is the expected model difference, while the standard deviation shows how much realised path-dependence remains.

8.6 Why smiles exist

The Black–Scholes assumptions fail in three ways, each giving a smile or skew.

1. **Fat tails.** Returns are leptokurtic, so OTM options are more valuable than a lognormal suggests. Mixtures of normals give a symmetric smile (Hull–White, Theorem 10.1).
2. **Leverage and spot–vol correlation.** When equity falls, volatility rises. So large down moves occur with high volatility, fattening the left tail; large up moves occur at low volatility, thinning the right. This gives a skew.
3. **Jumps and crash risk.** A negative jump premium lifts short-dated OTM puts. Demand from portfolio insurers, who structurally buy puts, adds to this.

Term structure. At-the-money skew $\partial_k \sigma_{\text{imp}}$ decays with maturity, in equity indices roughly like $T^{-\alpha}$ with α around 0.3 to 0.5. Diffusive stochastic volatility with mean reversion predicts a roughly flat short end, then decay like $1/T$. Lévy models also decay like $1/T$ at long maturities, and they are much steeper at short ones. Rough volatility (Section 10.7) reproduces the power law.

8.7 Asymptotic constraints

Theorem 8.5 (Lee's moment formula, stated)**.** *Let* $\tilde{p} = \sup\{p : \mathbb{E}S_T^{1+p} < \infty\}$. *Then*

$$\limsup_{k \to \infty} \frac{w(k)}{k} = \beta_R \in [0, 2], \quad \textit{where} \quad \tilde{p} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2},$$

and similarly for the left wing, with negative moments. So implied total variance is at most linear *in log-strike at extreme strikes, with slope at most* 2.

The bound $\beta_R \le 2$ comes from no arbitrage: a steeper wing implies call prices increasing in K. Therefore any parametrisation should go linearly in the wings. The SVI parametrisation (Gatheral) has this by construction:

$$w(k) = a + b\big(\rho(k-m) + \sqrt{(k-m)^2 + s^2}\big).$$

It has five parameters: level a, wing slope b, rotation ρ, shift m and curvature s. With constraints, it fits most equity slices well and can be made free of static arbitrage.

8.8 How smiles are quoted and how they move

FX conventions. Volatility is quoted by *delta*, not strike, using ATM vol, risk reversal $\mathrm{RR}_{25} = \sigma_{25C} - \sigma_{25P}$ (the skew) and butterfly $\mathrm{BF}_{25} = \frac{1}{2}(\sigma_{25C} + \sigma_{25P}) - \sigma_{\mathrm{ATM}}$ (the convexity).

Sticky rules. When spot moves, how does the smile move?

- *Sticky strike:* $\sigma(K)$ stays put. The delta is then Black–Scholes delta.
- *Sticky delta (moneyness):* $\sigma(K/S)$ stays put. The smile moves with spot. Then $\partial_S \sigma$ contributes to the hedge: with $\sigma_{\mathrm{imp}} = g(K/S)$, the true delta is $\Delta_{\mathrm{BS}} + \mathrm{vega} \cdot \partial_S \sigma$.
- A true model gives its own rule, and for the skew it matters: a local vol model predicts the smile moves opposite to spot, and stochastic vol moves with it (Chapter 9).

Choice of the smile dynamics affects delta even when all vanilla prices are fit, so it directly affects the hedge. It is a key risk in practice.

Exercises

8.1 Show that if $C(K_1) - C(K_2) > K_2 - K_1$ with $K_1 < K_2$, there is an arbitrage; write down the trades.

8.2 Show that for a given maturity, a smile implies digital prices differ from their naive BS values: $D(K) = -\partial_K C = \Phi(d_-) - \mathrm{vega} \cdot \partial_K \sigma_{\mathrm{imp}}$. With a negative skew, is the digital call worth more or less than its flat-vol value?

8.3 Use Carr–Madan to replicate x^2 and $\sqrt{x}$, and price them in Black–Scholes as a check.

8.4 For ATM forward options, $\sigma_{\mathrm{imp}} \approx C\sqrt{2\pi}/(S\sqrt{T})$ (Brenner–Subrahmanyam). Derive it from $C \approx 0.4 S\sigma\sqrt{T}$.

8.5 Show the fair volatility swap strike is below the square root of the fair variance strike, by Jensen. What is the size of the convexity gap in terms of the variance of realised variance?

8.6 In the theorem on hedging P&L, set $\sigma_i = \sigma$ constant, and consider hedging at $\sigma_h \neq \sigma_i$. How does the P&L formula change? Discuss which hedge volatility gives a deterministic P&L, if any.

Chapter 9

Local Volatility

The simplest way to produce a smile is to let volatility depend on the spot price and time:

$$\mathrm{d}S_t = (r-q)S_t\mathrm{d}t + \sigma_{\mathrm{loc}}(t,S_t)S_t\mathrm{d}W_t.$$

There remains one Brownian motion, so the market is complete, and vanilla options are hedged with the stock alone. Dupire (1994) and Derman–Kani (1994) showed there is exactly one such function matching a full, arbitrage-free surface of call prices.

9.1 Dupire's formula

Theorem 9.1 (Dupire)**.** *Let $C(K,T)$ be the (discounted) call price with spot S_0 and let r, q be constants. If C is smooth, with $\partial_{KK}C > 0$, then the local volatility reproducing all call prices is*

$$\sigma^2_{\mathrm{loc}}(T,K) = \frac{\partial_T C + (r-q)K\partial_K C + qC}{\frac{1}{2}K^2\partial_{KK}C}.$$

Proof. Take $r = q = 0$ for clarity. Let $\rho(T,y) = \partial_{KK}C(K,T)|_{K=y}$ be the density (Theorem 8.1). The forward equation (6.3) says

$$\partial_T \rho = \tfrac{1}{2}\partial_{yy}\big(\sigma^2_{\mathrm{loc}}y^2\rho\big).$$

Write $\partial_T C = \int (y-K)^+ \partial_T\rho\,\mathrm{d}y = \frac{1}{2}\int (y-K)^+\partial_{yy}(\sigma^2_{\mathrm{loc}}y^2\rho)\mathrm{d}y$. Integrate by parts twice: $\partial_{yy}(y-K)^+ = \delta_K(y)$, and boundary terms vanish. So

$$\partial_T C = \tfrac{1}{2}\sigma^2_{\mathrm{loc}}(T,K)K^2\rho(T,K) = \tfrac{1}{2}\sigma^2_{\mathrm{loc}}K^2\partial_{KK}C.$$

□

Intuition. The Dupire equation is a *forward* PDE in (T,K), whereas Black–Scholes is backward in (t,S). Numerator: the calendar spread $\partial_T C$, the cost of extending maturity by a moment. Denominator: the butterfly $\partial_{KK}C$, the probability density of being at K. By Tanaka (Example 4.14), extending a call from T to $T+\mathrm{d}T$ pays off exactly $\frac{1}{2}\sigma^2S^2\delta_K(S)\mathrm{d}T$: gamma P&L at the strike. Thus local variance is the ratio *cost of a calendar spread* over *cost of a butterfly*.
So no-arbitrage on the surface (Chapter 8) is exactly what makes σ^2_{loc} positive and finite.

In implied-variance coordinates. Prices are hard to differentiate numerically from noisy data. Let $w(k,T) = \sigma_{\text{imp}}^2 T$ with $k = \ln(K/F_T)$. Gatheral shows, by the chain rule on Black's formula,

$$\sigma_{\text{loc}}^2(T,k) = \frac{\partial_T w}{1 - \frac{k}{w}\partial_k w + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2}\right)(\partial_k w)^2 + \frac{1}{2}\partial_{kk} w}. \tag{9.1}$$

Numerator and denominator are precisely the calendar and butterfly no-arbitrage conditions. The surface is free of static arbitrage if and only if both are nonnegative.

> **Numerical check.** For the toy surface $\sigma_{\text{imp}}(k) = 0.20 - 0.10k + 0.15k^2$ at $T = 0.5$, price-based Dupire (by finite differences) and formula (9.1) agree to four decimals: 22.30%, 19.71% and 18.28% at $K = 90, 100, 110$, against implied 21.22%, 20.00% and 19.18%. Local vol is steeper than implied, as the next section explains.

9.2 Local vol as a conditional expectation

Theorem 9.2 (Gyöngy; Dupire). *Let $\mathrm{d}S_t = \sigma_t S_t \mathrm{d}W_t$ for an arbitrary adapted σ_t. Then the local vol model with*

$$\sigma_{\text{loc}}^2(T,K) = \mathbb{E}^{\mathbb{Q}}\left[\sigma_T^2 \mid S_T = K\right]$$

has the same one-dimensional marginal laws of S_T as the original model, for every T.

Proof. By Tanaka–Meyer applied to $(S_T - K)^+$,

$$\mathbb{E}(S_{T+\mathrm{d}T} - K)^+ - \mathbb{E}(S_T - K)^+ = \tfrac{1}{2}\mathbb{E}[\delta_K(S_T) S_T^2 \sigma_T^2]\mathrm{d}T = \tfrac{1}{2}K^2 \mathbb{E}[\sigma_T^2 \mid S_T = K]\rho(T,K)\mathrm{d}T.$$

Comparing with the Dupire equation gives the result. □

> **Intuition.** Local variance is the market's consensus forecast of future instantaneous variance, conditional on the stock being at that level at that date. It is to instantaneous variance what the forward rate is to the short rate. A local vol model is the unique Markov model with these marginals. But marginals are not joint distributions: Stochastic vol and its local vol "mimic" price all European options identically, yet disagree on everything path-dependent.

9.3 Relating implied to local volatility

For $r = 0$ the Berestycki–Busca–Florent (BBF) short-maturity result says that implied volatility at K is a harmonic-mean-type average of local vol along the straight line from spot to strike:

$$\frac{1}{\sigma_{\text{imp}}(K)} = \frac{1}{\ln(K/S)} \int_S^K \frac{\mathrm{d}s}{s\,\sigma_{\text{loc}}(s)} \quad (T \to 0).$$

Taylor-expanding, it gives the *one-half rule*.

Proposition 9.3 (Skew rule). *For short maturities, near the money, implied volatility is the average of local volatility between spot and strike. So*

$$\sigma_{\text{imp}}(K) \approx \frac{\sigma_{\text{loc}}(S) + \sigma_{\text{loc}}(K)}{2}, \qquad \partial_K \sigma_{\text{loc}} \approx 2\,\partial_K \sigma_{\text{imp}}.$$

Numerical check. For the toy surface (slope -0.10 in k), the local vol slope at the money computed from (9.1) is -0.1997 at $T = 0.02$ (ratio 1.997) and -0.1928 at $T = 0.5$ (ratio 1.93). The factor of 2 holds well, particularly at short maturity.

9.4 The flaw: smile dynamics

Suppose today's smile is fitted. What does the model say about tomorrow's smile if spot rises? Roughly, the implied vol at strike K averages local vol over the path from S to K. If S rises, the average shifts to include higher strikes, where local vol is lower on a negative skew. By the rule above, the local vol model predicts that the at-the-money vol falls as spot rises at about twice the slope of the skew:

$$\frac{\mathrm{d}\sigma_{\mathrm{ATM}}}{\mathrm{d}S} \approx 2\,\partial_K \sigma_{\mathrm{imp}} \quad \text{(local vol)}, \qquad \text{but} \quad \sigma_{\mathrm{imp}}(K;S) \approx g\Big(\frac{K+S}{2}\Big).$$

Thus the local vol smile moves *opposite* to spot: a rise in spot moves the smile left. Hagan et al. (2002) highlighted that this is contrary to what is observed and to what stochastic vol models give, and showed that delta hedges computed from local vol can be worse than plain Black–Scholes.

Pitfall. Fitting today's surface perfectly does not make a model right. Local vol has zero residual on vanilla options by construction, which proves nothing about their *dynamics*. It also predicts that forward smiles flatten as the local vol surface flattens with time. So forward-starting options and cliquets, which depend on forward smiles, are systematically mispriced by local vol.

9.5 Practice

Local vol remains in wide use: it gives a consistent price for any barrier or path-dependent option that matches the vanilla hedges, and a PDE exists in one spatial dimension. To compute it stably, fit a smooth arbitrage-free parametrisation (say, SVI by slices) first, and then use (9.1). Applying the price formula to raw quotes amplifies noise: second derivatives of noisy data are unusable. Combining with stochastic volatility yields stochastic-local volatility (SLV):

$$\mathrm{d}S = \sqrt{v_t}\, L(t,S)\, S\mathrm{d}W, \quad L^2(t,K) = \frac{\sigma^2_{\mathrm{Dup}}(t,K)}{\mathbb{E}[v_t \mid S_t = K]},$$

where the leverage function L is chosen by Theorem 9.2 to refit vanillas exactly while retaining stochastic-vol dynamics.

Exercises

9.1 Suppose σ_{imp} is flat. Show that Dupire returns that constant.

9.2 Show that the CEV model $\sigma_{\mathrm{loc}}(S) = \sigma S^{\beta-1}$ gives an implied skew of $\approx \frac{1}{2}(\beta-1)\sigma$ in k, with $\sigma \equiv \sigma_{\mathrm{loc}}(S)$.

9.3 Derive Dupire's formula with $r, q \neq 0$. *Hint:* start from the discounted forward equation for $e^{-rT}\rho$.

9.4 Why might a butterfly price zero at some strike imply an infinite local vol? Interpret with a discrete distribution for S_T.

9.5 Consider a two-state stochastic vol model with $\sigma_T \in \{\sigma_L, \sigma_H\}$ fixed at time zero, probabilities $\frac{1}{2}$. Write the local vol from Theorem 9.2 using Bayes. Sketch it for $\sigma_L = 10\%$, $\sigma_H = 30\%$, $T = 1$. Compare path-dependent price of a one-touch in the two models qualitatively.

Chapter 10

Stochastic Volatility and Fourier Pricing

10.1 Pricing when volatility is a second risk

Let, under $\mathbb{P}$,

$$\mathrm{d}S = \mu S \mathrm{d}t + \sqrt{v} S \mathrm{d}W^1, \qquad \mathrm{d}v = a(t,v)\mathrm{d}t + b(t,v)\mathrm{d}W^2, \qquad \mathrm{d}W^1 \mathrm{d}W^2 = \rho \mathrm{d}t.$$

The variance v is not traded, so the stock alone cannot hedge an option. The usual argument uses a *second option*. Let V and U be two options. Form $\Pi = V - \Delta S - \phi U$. Applying Itô's formula and choosing $\phi = \partial_v V / \partial_v U$ and $\Delta = \partial_S V - \phi \partial_S U$ eliminates both $\mathrm{d}W^1$ and $\mathrm{d}W^2$. So Π is riskless and must earn r. Rearranging gives

$$\frac{\mathcal{G} V + r S \partial_S V - r V}{\partial_v V} = \frac{\mathcal{G} U + r S \partial_S U - r U}{\partial_v U}, \qquad \mathcal{G} = \partial_t + \tfrac{1}{2} v S^2 \partial_{SS} + \rho \sqrt{v}\, b S \partial_{Sv} + \tfrac{1}{2} b^2 \partial_{vv}.$$

The left side depends only on V and the right side only on U, so both equal a function $-(a - \lambda b)$ of (t, S, v). So every derivative solves

$$\partial_t V + \tfrac{1}{2} v S^2 \partial_{SS} V + \rho \sqrt{v}\, b S \partial_{Sv} V + \tfrac{1}{2} b^2 \partial_{vv} V + r S \partial_S V + (a - \lambda b)\partial_v V - r V = 0. \tag{10.1}$$

Here $\lambda(t, S, v)$ is the *market price of volatility risk*. By Feynman–Kac, it is a pricing equation under the measure $\mathbb{Q}$ for which $\mathrm{d}v = (a - \lambda b)\mathrm{d}t + b \mathrm{d}W^{\mathbb{Q},2}$. No-arbitrage alone cannot fix λ. This is the second fundamental theorem at work (Theorem 5.11): two shocks, one traded asset, infinitely many $\mathbb{Q}$. In practice one takes the model under $\mathbb{Q}$ directly and calibrates to option prices. What a calibrated θ means under $\mathbb{Q}$ need not be the actual long-run variance; the difference is the variance risk premium, which is typically negative for the buyer.

10.2 The mixing theorem

Theorem 10.1 (Hull–White)**.** *If $\rho = 0$ and v is independent of W^1, then*

$$C = \mathbb{E}^{\mathbb{Q}}\big[C_{\mathrm{BS}}(S_0, K, T, r, \bar{\sigma})\big], \qquad \bar{\sigma}^2 = \frac{1}{T}\int_0^T v_t \mathrm{d}t.$$

Proof. Conditional on the whole path of v, $\ln S_T$ is normal with variance $\int_0^T v_t \mathrm{d}t$, because W^1 is independent of v. So the conditional price is Black–Scholes with volatility $\bar{\sigma}$. Apply the tower law. □

Intuition. The price is an average of Black–Scholes prices over total variance. Black–Scholes is roughly linear in σ near the money and strongly convex in σ in the wings (*volga* is positive there). By Jensen, randomness in volatility lifts wing prices more than ATM prices. This gives a smile that is *symmetric* in log-moneyness. Non-zero ρ tilts it. If $\rho < 0$, down-moves come with high vol. Then the left tail fattens, and the smile turns into a skew.

10.3 The Heston model

Heston (1993) takes the CIR process for variance, under $\mathbb{Q}$ with zero dividends:

$$\mathrm{d}S = rS\mathrm{d}t + \sqrt{v}S\mathrm{d}W^1, \qquad \mathrm{d}v = \kappa(\theta - v)\mathrm{d}t + \xi\sqrt{v}\mathrm{d}W^2, \qquad \mathrm{d}W^1\mathrm{d}W^2 = \rho\mathrm{d}t.$$

Parameters: v_0 (spot variance), θ (long-run variance), κ (speed of mean reversion), ξ (vol of vol), ρ (spot–vol correlation).

The characteristic function. Let $X_t = \ln(S_t/F_t)$, with $F_t = S_0e^{rt}$. Then $\mathrm{d}X = -\frac{1}{2}v\mathrm{d}t + \sqrt{v}\mathrm{d}W^1$. Define $\phi(\tau; u, x, v) = \mathbb{E}[e^{iuX_T} \mid X_{T-\tau} = x, v_{T-\tau} = v]$. By Feynman–Kac (with τ time to go),

$$\partial_\tau\phi = \tfrac{1}{2}v(\partial_{xx}\phi - \partial_x\phi) + \rho\xi v\partial_{xv}\phi + \tfrac{1}{2}\xi^2 v\partial_{vv}\phi + \kappa(\theta - v)\partial_v\phi, \qquad \phi(0) = e^{iux}.$$

All coefficients are affine in the state (x, v), so we guess an exponential-affine solution $\phi = \exp(C(\tau) + D(\tau)v + iux)$. Substituting and matching the coefficient of v and the constant term gives

$$\begin{aligned} D' &= \tfrac{1}{2}\xi^2 D^2 - (\kappa - \rho\xi iu)D - \tfrac{1}{2}(u^2 + iu), & D(0) &= 0, \\ C' &= \kappa\theta D, & C(0) &= 0. \end{aligned}$$

The first is a scalar Riccati equation with constant coefficients. With $a = \kappa - \rho\xi iu$, $d = \sqrt{a^2 + \xi^2(iu + u^2)}$ and $g = (a - d)/(a + d)$, its roots are $(a \pm d)/\xi^2$, and the solution is

$$D(\tau) = \frac{a - d}{\xi^2} \cdot \frac{1 - e^{-d\tau}}{1 - ge^{-d\tau}}, \qquad C(\tau) = \frac{\kappa\theta}{\xi^2}\left[(a - d)\tau - 2\ln\frac{1 - ge^{-d\tau}}{1 - g}\right]. \tag{10.2}$$

Pitfall. *The little Heston trap.* The original paper's form used g^{-1} and $e^{d\tau}$. The complex logarithm in that form crosses its branch cut for long maturities, giving wrong prices without warning. The form above (Albrecher et al. 2007) stays on the principal branch. Always plot the integrand when coding a Fourier pricer.

10.4 Fourier pricing

Theorem 10.2 (Lewis's formula). *Let $\phi(u) = \mathbb{E}^{\mathbb{Q}}e^{iuX}$ be the characteristic function of $X = \ln(S_T/F)$, with $\mathbb{E}^{\mathbb{Q}}e^X = 1$, and let $k = \ln(F/K)$. Then the undiscounted call price is*

$$\mathbb{E}^{\mathbb{Q}}(S_T - K)^+ = F - \frac{\sqrt{FK}}{\pi}\int_0^\infty \frac{\mathrm{Re}\left[e^{iuk}\phi(u - \frac{i}{2})\right]}{u^2 + \frac{1}{4}}\mathrm{d}u.$$

Proof. Since $(s-K)^+ = s - \min(s,K)$ and $\mathbb{E}S_T = F$, we need $\mathbb{E}\min(S_T,K)$. With $y = \ln(S_T/K) = X + k$,

$$\min(S_T,K) = \sqrt{S_T K}\, e^{-|y|/2} = \sqrt{FK}\, e^{X/2} e^{-|X+k|/2}.$$

The function $e^{-|y|/2}$ has Fourier transform $\int e^{-|y|/2} e^{-iuy}\,\mathrm{d}y = \frac{1}{u^2+1/4}$, so $e^{-|y|/2} = \frac{1}{\pi}\int_0^\infty \frac{\operatorname{Re} e^{iuy}}{u^2+1/4}\mathrm{d}u$. Insert this, exchange order (legitimate because $\mathbb{E}e^{X/2} < \infty$), and use $\mathbb{E}[e^{X/2}e^{iuX}] = \phi(u - \frac{i}{2})$. □

Intuition. Characteristic functions are often known in closed form when densities are not. Taking a Fourier transform turns a convolution (payoff against density) into a product. The damping factor $e^{X/2}$ shifts the integration contour so that the payoff's transform exists; that shift is the $-i/2$. The Carr–Madan method does the same, with a dampening parameter and an FFT, computing all strikes at once on a grid. COS methods use a cosine expansion of the density.

Any model with known ϕ can use this: Heston, Merton, variance gamma, Bates. For Merton's model,

$$\phi(u) = \exp\Big(T\big[-\tfrac{1}{2}\sigma^2(iu+u^2) - iu\lambda\bar{m} + \lambda\big(e^{iu\mu_J - \frac{1}{2}\delta^2 u^2} - 1\big)\big]\Big).$$

Numerical check. With $v_0 = \theta = 0.04$, $\kappa = 2$, $\xi = 0.5$, $\rho = -0.7$, $S_0 = 100$, $r = 0$, $T = 1$. (Note that $2\kappa\theta - \xi^2 = -0.09 < 0$: the Feller condition fails.)

K	Formula (10.2) + Lewis	Monte Carlo (400k paths, 400 steps)	Std error
80	21.8133	21.8205	0.0237
100	7.1926	7.1931	0.0149
120	0.7981	0.7968	0.0051

The Lewis pricer applied to a pure Black–Scholes characteristic function reproduces the closed form to at least eight significant figures, which verifies the integrator.

10.5 What each parameter does

Parameter	Main effect on the implied surface
v_0	level of short-dated volatility
θ	level of long-dated volatility
κ	speed of move from v_0 to θ; larger values flatten the long-dated smile
ξ	curvature of the smile (kurtosis of returns); larger means more convex
ρ	slope of the smile (skewness of returns); more negative means steeper skew

To first order in small T the ATM skew is $\partial_k \sigma_{\text{imp}} \to \rho\xi/(4\sqrt{v_0})$: *constant* as $T \to 0$. The Heston model does not produce the steep short-dated skew observed in indices. At large T, it decays like $1/(\kappa T)$. The expected variance has the simple form

$$\mathbb{E}^{\mathbb{Q}}\Big[\frac{1}{T}\int_0^T v_t\,\mathrm{d}t\Big] = \theta + (v_0 - \theta)\frac{1 - e^{-\kappa T}}{\kappa T}.$$

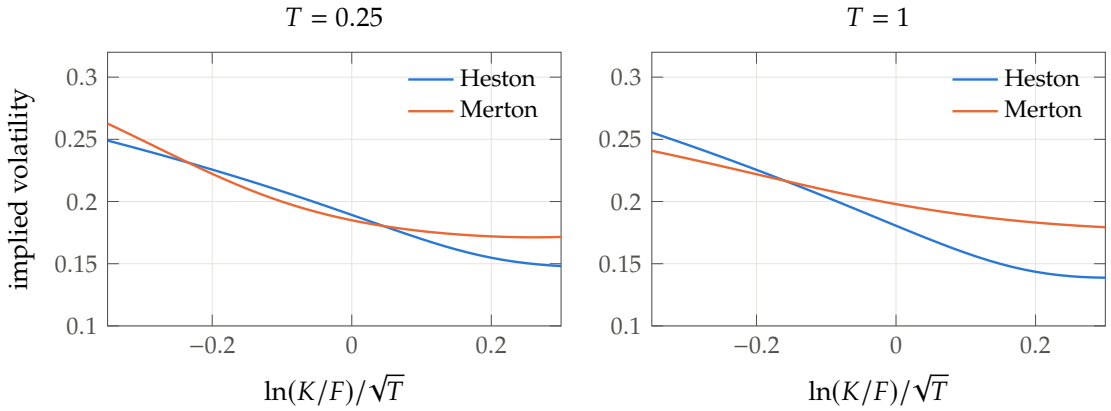


Figure 10.1: Implied volatility from Fourier prices in Heston (parameters above) and Merton ($\sigma = 15\%$, $\lambda = 0.5$, $\mu_J = -15\%$, $\delta = 15\%$). The parameters are for illustration and not fitted to each other. Merton's jumps give the sharper short-dated skew.

Calibration in practice. One minimises the weighted squared differences of model and market prices or vols. There are five parameters and a surface of quotes. κ and ξ are usually poorly identified separately (they trade off), and parameters can be unstable day to day, even when the fit is good. So a stable hedge may need regularisation or a fixed κ.

10.6 The SABR model

Hagan, Kumar, Lesniewski and Woodward (2002) proposed, for a forward F under its own forward measure,

$$\mathrm{d}F = \alpha F^{\beta}\mathrm{d}W^{1}, \qquad \mathrm{d}\alpha = \nu\alpha\mathrm{d}W^{2}, \qquad \mathrm{d}W^{1}\mathrm{d}W^{2} = \rho\mathrm{d}t.$$

Singular perturbation in small $\nu^2 T$ gives the implied volatility approximation

$$\sigma_{\mathrm{imp}}(K) = \frac{\alpha}{(FK)^{\frac{1-\beta}{2}}\left[1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}\right]} \cdot \frac{z}{x(z)} \cdot \left\{1 + \left[\frac{(1-\beta)^2\alpha^2}{24(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4(FK)^{\frac{1-\beta}{2}}} + \frac{2-3\rho^2}{24}\nu^2\right]T\right\},$$

with $z = \frac{\nu}{\alpha}(FK)^{\frac{1-\beta}{2}}\ln\frac{F}{K}$ and $x(z) = \ln\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}$.

Numerical check. For $F = 1$, $\alpha = 0.3$, $\beta = 0.5$, $\rho = -0.3$, $\nu = 0.4$, $T = 1$:

K	0.70	0.85	1.00	1.20	1.40
Hagan	35.90%	32.64%	30.24%	28.08%	26.82%
Monte Carlo	36.02%	32.70%	30.29%	28.11%	26.81%

The simulation uses 400,000 paths and 500 Euler steps with absorption at zero. The expansion is within about 12 basis points at the lowest strike, and closer elsewhere.

Interpretation.

- α sets the ATM level. The *backbone* of how ATM vol moves with forward is governed by β: $\sigma_{\text{ATM}} \approx \alpha F^{\beta-1}$.
- Skew comes from both β and ρ. Since they are close to redundant in fit, β is usually fixed, from a view on the backbone. $\beta = 1$ is lognormal and $\beta = 0$ normal.
- ν gives curvature.

Since SABR puts its smile dynamics in the right direction (the smile follows the forward), it remedies the flaw of local vol, and it became the standard for interest-rate smiles.

> **Pitfall.** The Hagan formula is an asymptotic expansion, not an exact price. For small F and long T (low strikes) it implies a negative density, a real arbitrage. Normal SABR variants and arbitrage-free PDE reformulations fix this.

10.7 Rough volatility

Gatheral, Jaisson and Rosenbaum (2018) found that realised log-volatility behaves like fractional Brownian motion with very small Hurst exponent, $H \approx 0.1$, far from the Markov case $H = \frac{1}{2}$. In a rough Bergomi-type model,

$$v_t = \xi_0(t)\,\mathcal{E}\Big(\eta\sqrt{2H}\int_0^t (t-s)^{H-\frac{1}{2}}\mathrm{d}W_s\Big),$$

and the implied skew is predicted to blow up as $T^{H-\frac{1}{2}}$. This matches the power law for short-dated index skew with few parameters. The cost: the process is not Markov and not a semimartingale in its driver, and simulation is expensive. It is an active research area.

10.8 How to choose?

Model	Fits	Weaknesses
Black–Scholes	one vol	no smile; still the language of quotes
Local vol	all vanillas	wrong smile dynamics, flat forward smile
Heston	long-dated smile	short-dated skew, unstable calibration
SABR	one maturity at a time	not a joint model across maturities, wings
Merton/Bates	short-dated skew	jump parameters unstable, incomplete hedge
SLV	all vanillas, tunable dynamics	costly, non-trivial numerics
Rough vol	skew term structure	costly, non-Markov

> **Intuition.** Models in this field are *interpolators between hedging instruments*. No model is correct. The questions for a desk are: Does it reprice the hedges? Does it give stable Greeks? Does it make the right prediction for what you are exposed to? For a vanilla book, SABR or SVI may be enough. For a barrier product, the smile dynamics matter. For a cliquet, the forward skew does.

Exercises

10.1 Show that in Theorem 10.1 with v taking two values σ_L^2, σ_H^2, the implied smile has a minimum at $k = 0$. Using Taylor expansion to the second order, estimate its curvature.

10.2 Derive the Riccati system for $\mathbb{E}[e^{iuX_T}]$ in the affine model with $\mathrm{d}X = -\frac{1}{2}v\mathrm{d}t + \sqrt{v}\mathrm{d}W$. Verify D in (10.2) by differentiation.

10.3 In Heston, compute $\mathbb{E}^{\mathbb{Q}}[\ln S_T]$ and thus the fair strike of a variance swap, and check against Theorem 8.3.

10.4 Show $\mathbb{E}[v_T]$ in closed form and derive the Heston forward variance curve $\xi_0(T) = \mathbb{E}^{\mathbb{Q}}[v_T]$.

10.5 Set $\beta = 1$ in SABR and show $\sigma_{\mathrm{imp}}(K) \approx \alpha \frac{z}{x(z)}$ with $z = \frac{\nu}{\alpha}\ln(F/K)$. Expand to first order in $\ln(K/F)$: show the skew is $\frac{1}{2}\rho\nu$.

10.6 In Heston, find the moment explosion time $T^*(p)$ such that $\mathbb{E}S_T^p = \infty$ for $T > T^*$. Using Lee's formula, explain how it determines the wing of the smile. *Hint:* the Riccati blow-up time.

Chapter 11

Interest Rates and Change of Numeraire

11.1 Instruments

A *zero-coupon bond* $P(t,T)$ pays 1 at T. Everything in the rates world is defined from it.

- **Zero yield:** $y(t,T) = -\ln P(t,T)/(T-t)$.
- **Instantaneous forward rate:** $f(t,T) = -\partial_T \ln P(t,T)$, so $P(t,T) = \exp(-\int_t^T f(t,u)\mathrm{d}u)$.
- **Short rate:** $r_t = f(t,t)$. Under the risk-neutral measure, with money market account $B_t = \exp(\int_0^t r_s \mathrm{d}s)$,
$$P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s \mathrm{d}s}\right].$$
- **Simple forward rate:** for accrual $\tau = T_2 - T_1$, $L(t;T_1,T_2) = \frac{1}{\tau}\left(\frac{P(t,T_1)}{P(t,T_2)} - 1\right)$. This is the rate fixed today that makes a forward rate agreement worth zero: it is a pure no-arbitrage statement, since $P(t,T_1)/P(t,T_2)$ is the replicating ratio.
- **Caplet:** pays $\tau(L(T_1;T_1,T_2) - K)^+$ at T_2. A *cap* is a strip of caplets.
- **Swap:** exchanges fixed payments at rate K for floating. Its value is zero at the *par swap rate*
$$S(t) = \frac{P(t,T_0) - P(t,T_n)}{A(t)}, \qquad A(t) = \sum_{i=1}^{n} \tau_i P(t,T_i) \quad \text{(the annuity)}.$$
- **Swaption:** an option at T_0 to enter a swap. A payer swaption pays $A(T_0)(S(T_0) - K)^+$ at T_0.

Intuition. A term structure is not one random variable but a whole *curve* moving through time. Short-rate models say one number (a low-dimensional state) drives everything. Forward-rate models (HJM, LMM) treat the whole curve as primitive. Also, the first principal component of yield curve moves (parallel shift) explains most variance, with *slope* and *curvature* explaining most of the rest. That is why low-dimensional models work at all.

11.2 Short-rate models

Term-structure equation. Let $\mathrm{d}r = a(t,r)\mathrm{d}t + b(t,r)\mathrm{d}W$ under $\mathbb{P}$. Bonds of two maturities can hedge one another, and the argument of Chapter 10 gives a market price of risk $\lambda(t,r)$, the same for all maturities. Under $\mathbb{Q}$, r has drift $a - \lambda b$, and $P(t,T) = u(t,r_t)$ satisfies, by Feynman–Kac,

$$\partial_t u + (a - \lambda b)\partial_r u + \tfrac{1}{2} b^2 \partial_{rr} u - ru = 0, \qquad u(T,r) = 1.$$

Affine models. Suppose under $\mathbb{Q}$ the drift $\kappa(\theta - r)$ and the variance $b^2 = \beta_0 + \beta_1 r$ are affine in r. Try $P(t,T) = \exp(A(\tau) - B(\tau)r)$, with $\tau = T - t$. Matching powers of r gives the Riccati system

$$B' = 1 - \kappa B - \tfrac{1}{2}\beta_1 B^2, \qquad A' = -\kappa\theta B + \tfrac{1}{2}\beta_0 B^2, \qquad A(0) = B(0) = 0.$$

It is the same structure as Heston's Riccati equation. The affine class (Duffie–Kan) is where most tractable models live.

Theorem 11.1 (Vasicek)**.** *If* $\mathrm{d}r = \kappa(\theta - r)\mathrm{d}t + \sigma\mathrm{d}W^{\mathbb{Q}}$*, then*

$$P(t,T) = \exp(A(\tau) - B(\tau)r_t), \quad B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}, \quad A(\tau) = \Big(\theta - \frac{\sigma^2}{2\kappa^2}\Big)(B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}.$$

Proof. Take $\beta_0 = \sigma^2, \beta_1 = 0$ in the Riccati system. The equation $B' = 1 - \kappa B$ gives B. Integrate for A. Alternatively, from (6.2), $\int_t^T r_s \mathrm{d}s$ is Gaussian given r_t; apply $\mathbb{E}e^{-G} = e^{-\mathbb{E}G + \frac{1}{2}\operatorname{Var} G}$. □

Theorem 11.2 (Cox–Ingersoll–Ross)**.** *If* $\mathrm{d}r = \kappa(\theta - r)\mathrm{d}t + \sigma\sqrt{r}\mathrm{d}W^{\mathbb{Q}}$*, with* $\gamma = \sqrt{\kappa^2 + 2\sigma^2}$*,* $P(t,T) = \mathcal{A}(\tau)e^{-B(\tau)r_t}$*, where*

$$B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}, \quad \mathcal{A}(\tau) = \Big[\frac{2\gamma e^{(\kappa+\gamma)\tau/2}}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}\Big]^{2\kappa\theta/\sigma^2}.$$

Numerical check. For Vasicek with $\kappa = 0.5$, $\theta = 4\%$, $\sigma = 1\%$, $r_0 = 2\%$: $P(0,5) = 0.84974$ from the formula, against Monte Carlo 0.84976 ± 0.00006. CIR with $\sigma = 0.1$ (and the same other parameters): 0.85051 versus 0.85058 ± 0.00010.

Comparison. Vasicek's rates are Gaussian, may go negative (once a criticism, now sometimes a feature), and give closed forms for bond options. CIR keeps $r > 0$, with rate-dependent volatility. Neither can fit today's whole curve with a few constants.

Hull–White: fitting today's curve. Let θ depend on time: $\mathrm{d}r = (\theta(t) - \kappa r)\mathrm{d}t + \sigma\mathrm{d}W^{\mathbb{Q}}$. Choosing

$$\theta(t) = \partial_T f(0,t) + \kappa f(0,t) + \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa t}\big)$$

makes the model reproduce all initial bond prices. The theorems above extend, and the model retains Gaussian tractability. This is a simple working model for rate-sensitive exotics such as Bermudan swaptions, usually in a two-factor version to decorrelate forward rates.

11.3 Change of numeraire

Theorem 11.3. *Let N be a strictly positive traded asset (with no dividends), so $D_t N_t$ is a $\mathbb{Q}$-martingale, where $D_t = 1/B_t$. Define $\mathbb{Q}^N$ on $\mathcal{F}_T$ by*

$$\frac{\mathrm{d}\mathbb{Q}^N}{\mathrm{d}\mathbb{Q}} = \frac{D_T N_T}{N_0}.$$

For every traded asset V, the ratio V_t/N_t is a $\mathbb{Q}^N$-martingale, and so

$$V_t = N_t \, \mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{V_T}{N_T}\right].$$

Proof. $Z_t = D_t N_t / N_0$ is a positive mean-one martingale, so $\mathbb{Q}^N$ is a probability. By Bayes (Theorem 1.21),

$$\mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{V_T}{N_T}\right] = \frac{\mathbb{E}_t^{\mathbb{Q}}[Z_T V_T/N_T]}{Z_t} = \frac{\mathbb{E}_t^{\mathbb{Q}}[D_T V_T]}{D_t N_t} = \frac{D_t V_t}{D_t N_t} = \frac{V_t}{N_t}. \qquad \square$$

> **Intuition.** Prices have units. Quote every asset in units of the numeraire: dollars, apples, or shares of IBM. With each numeraire there is a probability measure in which every relative price is a fair game. The risk-neutral measure is the one for the money market account. Choosing a clever numeraire removes a random discount factor from the expectation, turning a hard joint problem into a marginal one. The rule of thumb: *pick the numeraire that makes the payoff simplest*.

Effect on Brownian motion. If $\mathrm{d}N/N = r\mathrm{d}t + \sigma^N \mathrm{d}W^{\mathbb{Q}}$, then $\mathrm{d}Z/Z = \sigma^N \mathrm{d}W^{\mathbb{Q}}$, and Girsanov gives that

$$\mathrm{d}W^N = \mathrm{d}W^{\mathbb{Q}} - \sigma^N \mathrm{d}t$$

is a $\mathbb{Q}^N$-Brownian motion. So the drift change equals the numeraire's volatility, or, for a correlated asset X, the instantaneous covariance of $\ln N$ with X.

Example 11.4 (Stock numeraire and the $N(d_+)$ term)**.** Take $N = S$. Then

$$C = \mathbb{E}^{\mathbb{Q}}[D_T S_T \mathbf{1}_{\{S_T > K\}}] - K\mathbb{E}^{\mathbb{Q}}[D_T \mathbf{1}_{\{S_T > K\}}] = S_0 \mathbb{Q}^S(S_T > K) - Ke^{-rT}\mathbb{Q}(S_T > K).$$

So $N(d_+)$ is the probability of finishing in the money under the stock measure, and $N(d_-)$ is the same probability under the risk-neutral measure. Under $\mathbb{Q}^S$, $\ln S$ has drift $r + \frac{1}{2}\sigma^2$ rather than $r - \frac{1}{2}\sigma^2$, whence the difference between d_+ and d_-.

Example 11.5 (Margrabe's exchange option)**.** A payoff $(S_T^1 - S_T^2)^+$. Under $\mathbb{Q}^{S^2}$, the price is $V_0 = S_0^2 \mathbb{E}^{S^2}[(S_T^1/S_T^2 - 1)^+]$. The ratio $R = S^1/S^2$ is a martingale, with volatility $\sigma_m^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$. It is thus Black's formula with zero rate, no matter what the rate r is:

$$V_0 = S_0^1 N(d_+) - S_0^2 N(d_-), \qquad d_\pm = \frac{\ln(S_0^1/S_0^2) \pm \frac{1}{2}\sigma_m^2 T}{\sigma_m \sqrt{T}}.$$

Numerical check. For $S_0^1 = 100$, $S_0^2 = 95$, $\sigma_1 = 30\%$, $\sigma_2 = 20\%$, $\rho = 0.5$, $T = 1$, Margrabe gives 12.9523. Monte Carlo at $r = 3\%$ (2m paths) gives 12.9838 ± 0.0137, which is 2.3 standard errors high. A larger independent run (8m paths) gives 12.9585 ± 0.0068, within one standard error.

Example 11.6 (Quanto adjustment)**.** Let S be a foreign stock with volatility σ_S, and let X be the FX rate (domestic per foreign) with volatility σ_X and correlation ρ. Under the foreign risk-neutral measure, S drifts at r_f. Using the numeraire covariance rule, the drift of S under the domestic risk-neutral measure is $r_f - \rho\sigma_S\sigma_X$. So a quanto call on S paid in domestic currency at a fixed FX rate is a Black–Scholes call with a dividend-yield-like adjustment.

11.4 The forward measure and Black's formula

Let $N = P(t,T)$. This defines the *T-forward measure* $\mathbb{Q}^T$. Then $P(T,T) = 1$ removes the stochastic discount factor:

$$V_t = P(t,T)\mathbb{E}_t^T[V_T].$$

Three consequences:

1. The forward price of a traded asset, $S_t/P(t,T)$, is a $\mathbb{Q}^T$-martingale.
2. Taking $V_t = P(t,T+\epsilon)$ and $\epsilon \to 0$ gives $f(t,T) = \mathbb{E}_t^T[r_T]$. The forward rate is the $\mathbb{Q}^T$-expectation of the future short rate. It is *not* the $\mathbb{Q}$-expectation, and it is not a real-world forecast.
3. $L(t;T_1,T_2)$ is a $\mathbb{Q}^{T_2}$-martingale, because it is $\frac{1}{\tau}(P(t,T_1)-P(t,T_2))/P(t,T_2)$, a traded asset divided by the numeraire.

Theorem 11.7 (Black's caplet formula)**.** *If L is lognormal under $\mathbb{Q}^{T_2}$ with volatility σ, then*

$$\textit{Caplet} = \tau P(0,T_2)\big[L_0 N(d_+) - K N(d_-)\big], \qquad d_\pm = \frac{\ln(L_0/K) \pm \frac{1}{2}\sigma^2 T_1}{\sigma\sqrt{T_1}}.$$

Proof. $V_0 = P(0,T_2)\mathbb{E}^{T_2}[\tau(L_{T_1} - K)^+]$, and the expectation is the zero-rate Black–Scholes formula applied to the martingale L. □

Market quotes for caplets are Black vols (or, with low or negative rates, Bachelier normal vols, with L an arithmetic Brownian motion).

Example 11.8 (Option on a bond)**.** Under Vasicek, the forward bond price $P(t,S)/P(t,T)$ is a lognormal $\mathbb{Q}^T$-martingale with deterministic volatility $\sigma(B(S-t) - B(T-t))$. So the call with strike K on a bond maturing at S, expiring at T, has Jamshidian's formula

$$\text{ZBC} = P(0,S)N(h) - KP(0,T)N(h-\sigma_p),$$

where

$$\sigma_p = \sigma\sqrt{\frac{1-e^{-2\kappa T}}{2\kappa}}\,B(S-T), \qquad h = \frac{1}{\sigma_p}\ln\frac{P(0,S)}{KP(0,T)} + \frac{\sigma_p}{2}.$$

Numerical check. Vasicek with the earlier parameters, a one-year call on a five-year bond at strike $P(1,5)|_{r=r_0}$: the formula gives 0.001020, and Monte Carlo under $\mathbb{Q}$, with discounting along each path, gives 0.001020 ± 0.000007.

Swaptions. Take $N = A$, the annuity. Then $S(t)$ is a $\mathbb{Q}^A$-martingale, as a difference of bonds divided by the numeraire. The swaption price is

$$V_0 = A(0)\mathbb{E}^A[(S(T_0) - K)^+],$$

so Black (or SABR) on the swap rate applies. Each swap rate has its own measure, and rates that are lognormal under one measure generally are not under another.

11.5 The HJM framework

Heath, Jarrow and Morton (1992) model the whole forward curve. Under $\mathbb{Q}$:

$$\mathrm{d}f(t,T) = \alpha(t,T)\mathrm{d}t + \sigma(t,T)\mathrm{d}W_t, \qquad T \geq t.$$

Theorem 11.9 (HJM drift condition)**.** *If the discounted bonds are $\mathbb{Q}$-martingales, then*

$$\alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,u)\mathrm{d}u.$$

Proof. Let $X_t = \ln P(t,T) = -\int_t^T f(t,u)\mathrm{d}u$. By stochastic Fubini,

$$\mathrm{d}X_t = \big(r_t - \alpha^*(t,T)\big)\mathrm{d}t + \Sigma(t,T)\mathrm{d}W_t, \quad \alpha^* = \int_t^T \alpha(t,u)\mathrm{d}u, \quad \Sigma = -\int_t^T \sigma(t,u)\mathrm{d}u.$$

By Itô, $\mathrm{d}P/P = (r_t - \alpha^* + \frac{1}{2}\Sigma^2)\mathrm{d}t + \Sigma\mathrm{d}W$. The discounted bond has zero drift if and only if $\alpha^* = \frac{1}{2}\Sigma^2$ for all T. Differentiating in T gives $\alpha = \Sigma\,\partial_T\Sigma = \sigma\int_t^T \sigma(t,u)\mathrm{d}u$. □

Intuition. Under $\mathbb{Q}$, the drift of forward rates is not a free choice. *Specify volatilities only*; no-arbitrage determines drifts. It is the rates analogue of the fact that the stock's drift under $\mathbb{Q}$ is r, independent of its real-world drift. Why the positive convexity term? A bond price is a convex function of rates. Without the drift, bonds would earn more than r on average.

Examples.

- Ho–Lee: $\sigma(t,T) = \sigma$. Then $\alpha = \sigma^2(T-t)$, and $r_t = f(0,t) + \frac{1}{2}\sigma^2 t^2 + \sigma W_t$.
- Hull–White: $\sigma(t,T) = \sigma e^{-\kappa(T-t)}$. The short rate is Markov.

For general $\sigma(t,T)$, r is not Markov and the state is infinite-dimensional. A separable volatility $\sigma(t,T) = g(t)h(T)$ gives Markov representations (Cheyette models).

11.6 The LIBOR market model

Brace–Gątarek–Musiela (1997) and Jamshidian model the discrete forwards that are actually quoted, $L_k(t) = L(t; T_k, T_{k+1})$, with $\tau_k = T_{k+1} - T_k$. Under its own forward measure $\mathbb{Q}^{k+1}$, each is a lognormal martingale:

$$\mathrm{d}L_k = L_k \sigma_k(t) \mathrm{d}W_k^{k+1}(t).$$

That reproduces Black's caplet formula by construction. But to price anything depending on several L_k, all dynamics must be in one measure.

Proposition 11.10 (Drift under the terminal and spot measures)**.** *Let ρ_{ki} be the correlation between W_k and W_i. Moving from $\mathbb{Q}^{i+1}$ to $\mathbb{Q}^i$ changes the driver of L_k by*

$$\mathrm{d}W_k^i = \mathrm{d}W_k^{i+1} - \rho_{ki} \frac{\tau_i L_i \sigma_i}{1 + \tau_i L_i} \mathrm{d}t.$$

Therefore, under the spot measure (numeraire: rolled-over short bonds, with $q(t)$ the index of the next reset),

$$\mathrm{d}L_k = L_k \sigma_k \sum_{i=q(t)}^{k} \rho_{ki} \frac{\tau_i L_i \sigma_i}{1 + \tau_i L_i} \mathrm{d}t + L_k \sigma_k \mathrm{d}W_k.$$

Proof. The density of $\mathbb{Q}^i$ relative to $\mathbb{Q}^{i+1}$ is proportional to the ratio of numeraires $P(t, T_i)/P(t, T_{i+1}) = 1 + \tau_i L_i$, a $\mathbb{Q}^{i+1}$-martingale with $\mathrm{d}\ln(1+\tau_i L_i)$ having volatility $\tau_i L_i \sigma_i/(1+\tau_i L_i)$. Girsanov gives the first statement (with correlation ρ_{ki} for the k-th driver), and telescoping gives the second. □

The drift is state-dependent, so there are no closed-form path distributions; the LMM is simulated by Monte Carlo. Calibrating the correlation structure and the swaption surface is an art of its own.

After LIBOR. USD LIBOR ceased in 2023, replaced by the overnight secured rate SOFR, a *backward-looking* compounded rate. A compounded-in-arrears caplet depends on the path over the accrual period, and its volatility decays during the period. Lyashenko and Mercurio's generalised forward market model extends the LMM to these rates. The change-of-numeraire logic is the same.

Futures versus forwards. A futures price is a $\mathbb{Q}$-martingale (it is marked to market daily, and has zero value at resets), while a forward price is a $\mathbb{Q}^T$-martingale. They differ by a covariance: $\mathrm{Fut}_t - \mathrm{For}_t \approx \mathrm{Cov}^{\mathbb{Q}}(\text{asset}, \int r)$. In Ho–Lee, the Eurodollar futures convexity adjustment is approximately $\frac{1}{2}\sigma^2 T_1 T_2$. For a typical bank desk, this was a real and well-known source of P&L.

Exercises

11.1 Show that Vasicek's $P(t,T)$ has bond volatility $\sigma B(T-t)$. Find the model's long-run yield, $y_\infty = \lim_{T\to\infty} y(t,T)$. Can you find an arbitrage in a model where the long yield moves?

11.2 In Hull–White, derive $\theta(t)$ from the condition that the model matches $f(0,T)$.

11.3 Show that the $\mathbb{Q}^T$-dynamics in Vasicek are $\mathrm{d}r = \kappa(\theta - \frac{\sigma^2}{\kappa}B(T-t) - r)\mathrm{d}t + \sigma \mathrm{d}W^T$. Use this to find $\mathbb{E}^T[r_T]$ and hence verify $f(0,T)$.

11.4 Price a caplet under Vasicek by writing it as a put on a bond. Why is this exact?

11.5 Use a change of numeraire to price an option paying $S_T \mathbf{1}_{\{S_T > K\}}$ (asset-or-nothing).

11.6 Derive the Ho–Lee convexity adjustment for a futures rate, and calculate its size for $T_1 = 5$, $T_2 = 5.25$, $\sigma = 1\%$.

11.7 Taking S as numeraire, show that $1/S$ is a $\mathbb{Q}^S$-martingale with the same volatility as S. Use this to relate a call on S to a put on $1/S$ (put–call symmetry). Which FX fact does it explain?

Part IV

Numerical Methods

Chapter 12

Monte Carlo Methods

12.1 The basic estimator

To compute $\mu = \mathbb{E}[Y]$ we draw i.i.d. copies and form $\hat{\mu}_n = \frac{1}{n}\sum_i Y_i$. The strong law gives consistency, and the CLT gives

$$\sqrt{n}(\hat{\mu}_n - \mu) \Rightarrow \mathcal{N}(0, \sigma_Y^2),$$

so a 95% confidence interval is $\hat{\mu}_n \pm 1.96\,\hat{\sigma}_Y/\sqrt{n}$. The error decays as $n^{-1/2}$ *whatever the dimension*.

> **Intuition.** This is the only feature of Monte Carlo that matters. Quadrature in d dimensions with n points has error about $n^{-p/d}$, which degrades with dimension (the curse of dimensionality). A path-dependent option with 250 dates is a 250-dimensional integral. Monte Carlo pays a slow but dimension-free $n^{-1/2}$, and to gain a decimal of accuracy you need 100 times the samples. So every clever idea either reduces σ_Y or controls bias.

Efficiency. If one sample costs c and has variance σ^2, then the variance for budget T is $\sigma^2 c/T$. Comparison between methods should use $\sigma^2 c$, not σ^2 alone (Hammersley–Handscomb).

Generating randomness. Normals are produced by inverse transform $\Phi^{-1}(U)$ or by Box–Muller. A vector with covariance Σ is LZ, where $LL^\top = \Sigma$ (Cholesky). A Gaussian path W_{t_i} is built by increments, by Brownian bridge, or by PCA of the covariance. A good pseudorandom generator (for example PCG64) has a long period and good equidistribution, and one must seed parallel streams with care.

12.2 Discretising SDEs

Let $\mathrm{d}X = b(X)\mathrm{d}t + \sigma(X)\mathrm{d}W$ and let $h = T/m$.

$$\begin{aligned} \text{Euler–Maruyama:}\quad & X_{k+1} = X_k + b(X_k)h + \sigma(X_k)\Delta W_k, \\ \text{Milstein:}\quad & X_{k+1} = X_k + b(X_k)h + \sigma(X_k)\Delta W_k + \tfrac{1}{2}\sigma\sigma'(X_k)\big(\Delta W_k^2 - h\big). \end{aligned}$$

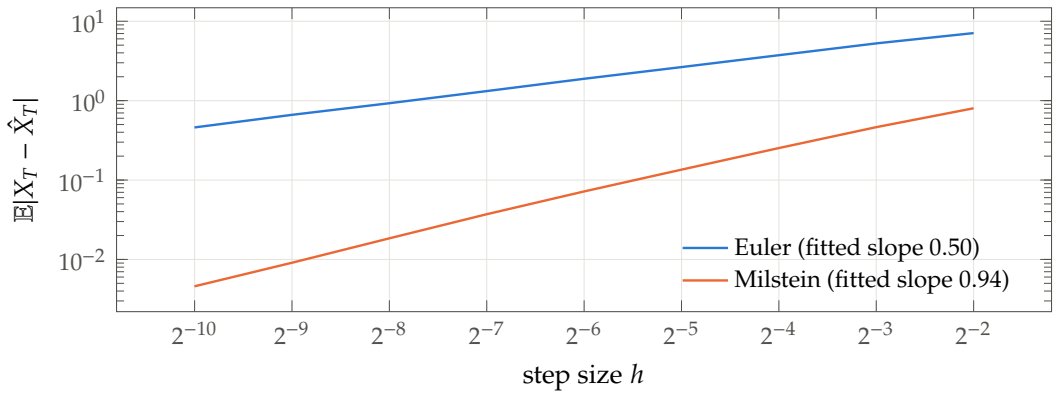


Figure 12.1: Strong error for GBM with $\mu = 5\%$, $\sigma = 50\%$, $S_0 = 100$, $T = 1$, using 5,000 paths and an exact solution driven by the same Brownian increments. The slopes match the theoretical orders $\frac{1}{2}$ and 1.

The extra Milstein term comes from applying Itô's formula to $\sigma(X)$ inside the stochastic integral: $\int_{t_k}^{t_{k+1}} \sigma(X_s)\mathrm{d}W_s \approx \sigma(X_k)\Delta W_k + \sigma\sigma'(X_k)\int_{t_k}^{t_{k+1}}(W_s - W_{t_k})\mathrm{d}W_s$, and that iterated integral equals $\frac{1}{2}(\Delta W_k^2 - h)$ by Proposition 4.1.

Definition 12.1. A scheme has *strong order* γ if $\mathbb{E}|X_T - \hat{X}_T| \le Ch^\gamma$, and *weak order* β if $|\mathbb{E}f(X_T) - \mathbb{E}f(\hat{X}_T)| \le Ch^\beta$ for smooth f.

Theorem 12.2 (Stated)**.** *Under Lipschitz and smoothness conditions, Euler has strong order* $\frac{1}{2}$ *and weak order* 1*; Milstein has strong order* 1 *and weak order* 1*.*

Intuition. Strong error measures path closeness, relevant for hedging and path-dependent payoffs. Weak error measures closeness of laws, relevant for vanilla pricing. The weak order is better because errors average: the leading Euler error per step is the mean-zero term $\frac{1}{2}\sigma\sigma'(\Delta W^2 - h)$, of size h per step. Added over $1/h$ steps, it has root-sum-square size $\sqrt{1/h}\cdot h = \sqrt{h}$. Hence strong order $\frac{1}{2}$. Being mean zero, it cancels in expectation, and only terms of size h^2 per step, h in total, survive.

Numerical check. Weak error of Euler for $\mathbb{E}[X_T^2]$ is analytic here: $S_0^2((1+\mu h)^2 + \sigma^2 h)^{1/h}$ against $S_0^2 e^{(2\mu+\sigma^2)T}$. The errors are 195.9, 101.0, 51.3, 25.8, 13.0 at $h = 2^{-2}, \ldots, 2^{-6}$, halving at every step: order one.

Bias–variance trade-off. With n paths and m steps of a weak-order-1 scheme, the mean squared error is $C_1/m^2 + C_2/n$, and the cost is nm. To achieve root mean squared error ϵ we need $m \sim \epsilon^{-1}$ and $n \sim \epsilon^{-2}$: total cost ϵ^{-3}. So more steps without more paths is wasteful, and the reverse too.

Square-root processes. Euler can produce negative variance in CIR/Heston. *Full truncation* (used in Chapter 10) replaces v by v^+ in the coefficients, and has small bias in practice. Exact simulation from the noncentral chi-squared (Broadie–Kaya) is possible but slow; Andersen's quadratic-exponential scheme matches moments and is the usual compromise.

12.3 Variance reduction

Antithetic variates. Average $\frac{1}{2}(f(Z) + f(-Z))$. If f is monotone, the two terms are negatively correlated. Then the variance per pair is below $\frac{1}{2}\operatorname{Var} f$, and so antithetic beats using independent samples at equal cost. It does nothing for an even function f.

Control variates. Let C have known mean μ_C. The estimator $Y - \beta(C - \mu_C)$ is unbiased, and its variance is minimised by $\beta^* = \operatorname{Cov}(Y, C)/\operatorname{Var} C$, giving

$$\operatorname{Var}(Y - \beta^*(C - \mu_C)) = (1 - \rho_{YC}^2)\operatorname{Var} Y.$$

Everything depends on ρ: reduction by a factor $1/(1-\rho^2)$. Estimating β from the same run introduces a small bias, of order $1/n$.

Stratified sampling. Divide the probability space into cells and sample each with fixed proportion. Then the variance becomes the average *within-cell* variance: by the total variance law (Exercise 1.2) the between-cell variance is removed. In one dimension, $U_i = (i-1+V_i)/n$ removes nearly all the variance. It extends in several dimensions via Latin hypercube (one coordinate at a time) or via stratifying the terminal value with a Brownian bridge.

Importance sampling. For density g and a proposal h, $\mathbb{E}_g[f(X)] = \mathbb{E}_h[f(X)\,g(X)/h(X)]$. The optimal $h \propto |f|g$ has zero variance if $f \geq 0$, but it is uncomputable. A practical choice is a shift of the Gaussian, i.e. Girsanov: sample $Z + \theta$ with weight $e^{-\theta(Z+\theta)+\theta^2/2}$. Choose θ to make the payoff event likely. This is essential for rare events: deep OTM options, credit tail losses.

Numerical check. Call with $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$; exact price 10.4506. In a single run of 100,000 normals:

Method	Estimate	Std error
Plain	10.4981	0.0468
Antithetic (2 evaluations per normal)	10.4951	0.0234
Control variate S_T ($\rho = 0.925$)	10.4842	0.0178
Stratified ($40 \times 5{,}000$ paths)	10.4505	0.00038

Antithetic's standard error is that of 200,000 independent samples (0.0331) reduced by a further factor. Stratification wins hugely because this payoff depends on one normal; on a truly high-dimensional problem the gain would be much lower.
Rare event, $K = 200$**:** exact 0.004799. Plain: 0.004500 ± 0.000907. Importance sampling, shifting the mean so the strike is at the centre ($\theta = 3.316$): 0.004814 ± 0.000020. Standard error falls by about a factor of 45, or a factor of about 2,000 in samples.
Asian call, 12 fixings, 1m paths: plain 6.1536 ± 0.0085. Geometric-average Asian as control, with closed form 5.9402 (correlation 0.9996): 6.1563 ± 0.00024, a factor of about 35 in standard error.

Conditional Monte Carlo. Replace Y by $\mathbb{E}[Y \mid \mathcal{G}]$ when this is known; the variance can only fall. Example: in stochastic volatility with $\rho = 0$, simulate v and use Black–Scholes

for the conditional price (Theorem 10.1). With $\rho \neq 0$, the same works after a shift of the forward.

12.4 Quasi-Monte Carlo

Low-discrepancy sequences (Sobol', Halton) fill the cube more evenly than random points. The Koksma–Hlawka inequality bounds the error by variation times discrepancy: $|\text{error}| \leq V(f)D_n^*$, with $D_n^* = O((\log n)^d/n)$. In practice convergence is near n^{-1} when the *effective dimension* is low. So pair QMC with a Brownian bridge or PCA path construction, which loads most of the variance into the first few coordinates. Randomised (scrambled) QMC gives an error estimate.

12.5 Barrier options and the bridge correction

As in Section 5.3, discrete monitoring misses crossings. Given consecutive log prices x_k, x_{k+1} above a lower barrier b, by (3.2) (with minimum version),

$$\mathbb{P}(\text{crossing in step}) = \exp\Big(-\frac{2(x_k - b)(x_{k+1} - b)}{\sigma^2 h}\Big).$$

So for each step, kill the path with that probability; or better, weight the path by the survival product to reduce variance. Section 5.3 has the numbers: the bias of 12 standard errors disappears.

12.6 Greeks

1. **Finite differences.** Bump and revalue, $\Delta \approx (V(S+\epsilon) - V(S-\epsilon))/2\epsilon$. Use *common random numbers*, or the variance blows up like $1/\epsilon^2$. The bias is $O(\epsilon^2)$. It works with discontinuous payoffs only with large ϵ.

2. **Pathwise.** Differentiate the payoff: $\partial_S \mathbb{E}[f(S_T)] = \mathbb{E}[f'(S_T)\partial_S S_T]$. For a call with GBM, $\partial_S S_T = S_T/S_0$, and so $\Delta = \mathbb{E}[e^{-rT}\mathbf{1}_{\{S_T>K\}}S_T/S_0]$. It is unbiased, and needs a.e. differentiability of the payoff, and Lipschitz continuity. It fails for digitals.

3. **Likelihood ratio.** Differentiate the density instead: $\partial_S \mathbb{E}[f(S_T)] = \mathbb{E}[f(S_T)\partial_S \ln p(S_T; S)]$. For GBM the score is $Z/(S_0\sigma\sqrt{T})$. It works for any payoff, even discontinuous, but the variance is typically higher and grows as $h \to 0$ for path-dependent payoffs. It is equivalent to a Malliavin weight.

4. **Adjoint algorithmic differentiation (AAD).** A single reverse pass gives all sensitivities at a cost about 4 times one price, no matter how many inputs. It made risk reports for large books practical.

> **Numerical check.** ATM call delta (exact 0.6368), 400k paths: pathwise 0.6378 ± 0.0009; likelihood ratio 0.6357 ± 0.0023; bump with common normals 0.6377 ± 0.0009.

12.7 American options: Longstaff–Schwartz

Backward induction needs the continuation value $c_k(x) = \mathbb{E}[e^{-rh}V_{k+1} \mid S_k = x]$, a conditional expectation, and conditional expectation is a projection (Chapter 1). The Longstaff–Schwartz idea is to *regress* it on basis functions of S_k across simulated paths.

1. Simulate n paths forward.
2. At $k = m$, set the cash flow to the payoff. For $k = m-1, \dots, 1$: on in-the-money paths, regress discounted future cash flow on basis functions $\psi_j(S_k)$; exercise if the immediate payoff exceeds the fitted continuation; update the cash flow.
3. Average discounted cash flows.

Only the exercise *decision* uses the regression; the actual cash flow is realised along the path. Any exercise rule is suboptimal, so this gives a *low-biased* estimate. Using a dual formula (Rogers; Haugh–Kogan; Andersen–Broadie), one obtains an upper bound: for any martingale M with $M_0 = 0$,

$$V_0 \le \mathbb{E}\Big[\max_k \big(G_k - M_k\big)\Big].$$

A good approximation to the continuation value gives a tight bound and so a confidence interval. The same regression idea is the basis of nested simulations for exposures.

> **Numerical check.** American put, $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. LSM with 200,000 antithetic paths, 50 exercise dates, cubic polynomials: 6.0754 ± 0.0161. A tree with continuous exercise approximated by 5,000 steps gives 6.0902.

12.8 Multilevel Monte Carlo

Giles (2008) writes the fine-grid estimator as a telescoping sum over grids $h_\ell = 2^{-\ell}T$:

$$\mathbb{E}[\hat{f}_L] = \mathbb{E}[\hat{f}_0] + \sum_{\ell=1}^{L} \mathbb{E}[\hat{f}_\ell - \hat{f}_{\ell-1}].$$

Differences on two levels share Brownian increments, so they have small variance (it is the strong error that matters here). Most samples go on the cheap coarse levels and few on the fine levels. With Euler on Lipschitz payoffs, the cost to reach RMSE ϵ falls from ϵ^{-3} to $\epsilon^{-2}(\log \epsilon)^2$. With Milstein, it is ϵ^{-2}.

Exercises

12.1 Implement antithetic sampling for a straddle. Why does it fail to reduce the variance compared with independent sampling?

12.2 Find the optimal importance-sampling mean shift θ for $\mathbb{P}(Z > a)$. Compute the variance reduction for $a = 4$.

12.3 Prove the formula for the variance of the optimal control variate. How would you use several controls? Show that this is a regression.

12.4 Derive the pathwise and likelihood-ratio estimators of vega for a call. Which one should be used for a digital?

12.5 Why does the Euler scheme for $\mathbb{E}[X_T^2]$ in GBM have error proportional to h? Expand the analytic formula.

12.6 Implement LSM for the American put with several basis sets, then compute the dual upper bound.

12.7 Estimate the number of paths necessary for a standard error of one cent on a one-year ATM call at $S = 100$, $\sigma = 20\%$. Assume no variance reduction.

Chapter 13

Finite Differences and Trees

Monte Carlo is good in high dimensions. For one or two factors, solving the PDE gives prices and all Greeks on a grid, with deterministic accuracy.

13.1 Setting up

With $x = \ln(S/K)$ and time to maturity $\tau = T - t$, the BSM equation has constant coefficients:

$$\partial_\tau V = \tfrac{1}{2}\sigma^2 \partial_{xx} V + \big(r - \tfrac{1}{2}\sigma^2\big)\partial_x V - rV, \qquad V(0,x) = K(e^x - 1)^+. \tag{13.1}$$

Put a grid $x_j = x_{\min} + jh$, $\tau_n = n\Delta$. Use central differences in space:

$$(\mathcal{L}_h V)_j = \tfrac{1}{2}\sigma^2 \frac{V_{j+1} - 2V_j + V_{j-1}}{h^2} + \big(r - \tfrac{1}{2}\sigma^2\big)\frac{V_{j+1} - V_{j-1}}{2h} - rV_j.$$

This is $\ell V_{j-1} + dV_j + uV_{j+1}$ with $\ell = a - b$, $d = -2a - r$, $u = a + b$, where $a = \sigma^2/2h^2$ and $b = (r - \sigma^2/2)/2h$.

The θ-scheme.

$$\frac{V^{n+1} - V^n}{\Delta} = \theta \mathcal{L}_h V^{n+1} + (1-\theta)\mathcal{L}_h V^n.$$

Here $\theta = 0$ is explicit, $\theta = 1$ fully implicit, $\theta = \frac{1}{2}$ Crank–Nicolson. For $\theta > 0$, each step solves a tridiagonal system, which costs $O(J)$ by the Thomas algorithm.

Boundary conditions. Choose $x_{\min}, x_{\max}$ several standard deviations from today's spot, $\pm 5\sigma\sqrt{T}$ being usual. For a call use $V = 0$ at the bottom and $V = Se^{-q\tau} - Ke^{-r\tau}$ at the top. Alternatively use linearity, $\partial_{SS} V = 0$, which needs no knowledge of the product.

13.2 Stability

Theorem 13.1 (Lax equivalence, stated)**.** *A consistent linear scheme for a well-posed problem converges if and only if it is stable.*

Von Neumann analysis. Insert a Fourier mode $V_j^n = A^n e^{i\omega jh}$ into the scheme. For the explicit method,

$$A = 1 + \Delta\big(-2a(1 - \cos\omega h) + 2ib\sin\omega h - r\big).$$

The worst mode has $\omega h = \pi$, where $A \approx 1 - 4a\Delta$. Stability requires $|A| \le 1$, so

$$\sigma^2\Delta/h^2 \le 1. \tag{13.2}$$

The time step must shrink like the *square* of the space step. Double the spatial resolution, and work grows by a factor of 8. For implicit schemes, $A = 1/(1 + \Delta \cdot \text{positive})$, unconditionally stable. For Crank–Nicolson, $|A| \le 1$ also, but $A \to -1$ for the highest modes as Δ/h^2 grows, so these modes neither decay nor blow up: they just oscillate.

Intuition. The explicit scheme is a trinomial tree (Section 13.5), with weights $\Delta\ell$, $1 + \Delta d$, Δu. Condition (13.2) says the probability of staying put, $1 - 2a\Delta$ (ignoring r), must not be negative. Information cannot diffuse faster than one grid cell per time step. Implicit schemes pass information across the grid instantly by solving a linear system.

Numerical check. ATM call, $r = 5\%$, $\sigma = 20\%$, $T = 1$, on the grid $x \in [-4, 4]$ with 200 cells. Then $h = 0.04$ and the limit is $\Delta \le 0.04$, i.e. $N_t \ge 25$. The explicit price errors are:

N_t	20	23	24	25	26	30
$\Delta/\Delta_{\max}$	1.25	1.09	1.04	1.00	0.96	0.83
Error	−334.6	+3.59	−0.580	+0.089	−0.0053	−0.0008

The loss of accuracy is sudden and comes with alternating signs, a typical signature of instability.

13.3 Accuracy: Crank–Nicolson and the kink

Truncation error is $O(h^2) + O(\Delta)$ for the implicit scheme and $O(h^2) + O(\Delta^2)$ for Crank–Nicolson. These are the rates for smooth data; our data has a kink at the strike. With a grid node at the strike, second-order space convergence survives.

Numerical check. Time refinement with a fine space grid ($N_x = 1600$). The price error at the money:

N_t	25	50	100	200	400
Implicit	−0.04244	−0.02153	−0.01105	−0.00581	−0.00319
Crank–Nicolson	+0.03634	−0.00221	−0.00054	−0.00056	−0.00056

Implicit shows first order in Δ. Crank–Nicolson reaches the spatial error (−0.00056) quickly. Its $N_t = 25$ entry shows a large positive error from a different cause, which appears next.

Pitfall. *Crank–Nicolson with non-smooth payoffs.* The kink has high-frequency content, and these modes have $A \approx -1$ under CN when $\Delta \gg h^2$. So they persist and oscillate.

Price error may look acceptable, while Greeks are garbage. With $N_x = 1600$ and $N_t = 10$, the gamma at the strike (exact 0.01876) is:

Implicit	0.01957
Crank–Nicolson	1.25265; neighbouring nodes −0.38228, −0.37191
CN with Rannacher start	0.01881

Rannacher's fix: replace the first few CN steps with two half-steps of fully implicit each. The implicit part damps the kink's high frequencies, and the overall second order is retained. Always use it for digital, kinked or barrier payoffs. And always plot the gamma.

Numerical check. Space and time refined together (error at the money):

(N_x, N_t)	$(100, 25)$	$(200, 100)$	$(400, 400)$	$(800, 1600)$
Implicit	−0.19428	−0.04700	−0.01167	−0.00291
CN	−0.14891	−0.03633	−0.00903	−0.00225
CN–Rannacher	−0.15071	−0.03644	−0.00904	−0.00226

Halving h reduces each error fourfold: clean second order. The grid here is wide, $x \in [-4, 4]$. A narrower grid, or a non-uniform one concentrated at the strike, gives far smaller errors for the same effort.

13.4 Early exercise

For the inequality (6.4), the discrete problem is a *linear complementarity problem*:

$$(I - \theta\Delta\mathcal{L}_h)V^{n+1} \geq b, \quad V^{n+1} \geq g, \quad \left[(I - \theta\Delta\mathcal{L}_h)V^{n+1} - b\right]^\top (V^{n+1} - g) = 0.$$

- **PSOR** (projected successive over-relaxation): Gauss–Seidel sweeps with $V_j \leftarrow \max(V_j, g_j)$ after each update.
- **Brennan–Schwartz:** a one-pass tridiagonal solve, valid for a single exercise boundary.
- **Penalty method:** add a term $\rho \max(g - V, 0)$ to the PDE with large ρ, and solve by Newton iteration.

Convergence near the free boundary is usually lower than second order without local refinement. Delta remains continuous at the boundary (smooth pasting), but gamma jumps.

13.5 Trees

A binomial tree is an explicit finite difference scheme on a recombining grid, with a particular relationship between h and Δ. The Cox–Ross–Rubinstein choice is

$$u = e^{\sigma\sqrt{\Delta}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta} - d}{u - d}.$$

It matches mean and variance to order Δ. By construction $h = \sigma\sqrt{\Delta}$, which is exactly at the stability limit, $\sigma^2\Delta/h^2 = 1$. And that is the reason a two-branch tree has no middle probability.

Numerical check. Convergence for an ATM call (exact 10.4506):

n	10	11	50	51	100	101	500	501
Error	−0.197	+0.161	−0.040	+0.034	−0.020	+0.017	−0.004	+0.0035

Convergence is $O(1/n)$ with an odd–even sawtooth. It comes from the position of the strike relative to the terminal nodes. Averaging consecutive n cancels most of the error. Leisen–Reimer trees choose parameters to give $O(1/n^2)$ on vanillas.

When to use which. Trees are simple and intuitive for American options. Finite differences are more flexible: non-uniform grids, barriers and local vol. In two dimensions (stochastic vol, or a convertible bond) use alternating-direction implicit (ADI) splitting, as a full implicit matrix solve is expensive. With three or more dimensions, go to Monte Carlo. With jumps, the PIDE's integral operator produces a dense matrix, handled using FFT or an implicit-explicit split.

Exercises

13.1 Derive the von Neumann amplification factor for the implicit and CN schemes for the heat equation $\partial_\tau V = \partial_{xx} V$.

13.2 Show that CN's amplification for a mode with eigenvalue $-\Lambda$ is $\frac{1-\Lambda\Delta/2}{1+\Lambda\Delta/2}$, and its limit as $\Lambda\Delta \to \infty$. Why does this explain the oscillation?

13.3 Modify the code in `code/make_data.py` to price a down-and-out call, setting the grid boundary at the barrier. Compare to Theorem 5.5. Why is the grid so much easier here than in Monte Carlo?

13.4 Implement PSOR for an American put. Plot the exercise boundary against time, and check the limit L^* of Theorem 6.11 at long maturity.

13.5 Using (13.1), show that the substitution $V = Ke^{\alpha x+\beta\tau}u(x,\tau)$ with suitable α and β reduces the PDE to the heat equation. Then derive Black–Scholes by the heat kernel.

13.6 Show that a trinomial tree with branching $x \pm h$, x is an explicit scheme. Compute the branch probabilities in terms of σ, Δ, h, and find the range of h for which they are positive.

Part V

Risk, Portfolios and Data

Chapter 14

Risk: Greeks and Risk Measures

14.1 The Greeks as a Taylor expansion

For a position $V(t, S, \sigma, r)$, the change over a short interval is

$$\mathrm{d}V \approx \Theta\,\mathrm{d}t + \Delta\,\mathrm{d}S + \tfrac{1}{2}\Gamma\,\mathrm{d}S^2 + \mathcal{V}\,\mathrm{d}\sigma + \mathrm{Vanna}\,\mathrm{d}S\,\mathrm{d}\sigma + \tfrac{1}{2}\mathrm{Volga}\,\mathrm{d}\sigma^2 + \rho\,\mathrm{d}r.$$

Greek	Definition	Black–Scholes call (no dividends)
Delta Δ	$\partial_S V$	$N(d_+)$
Gamma Γ	$\partial_{SS} V$	$\varphi(d_+)/(S\sigma\sqrt{T})$
Vega $\mathcal{V}$	$\partial_\sigma V$	$S\varphi(d_+)\sqrt{T}$
Theta Θ	$\partial_t V$	$-S\varphi(d_+)\sigma/(2\sqrt{T}) - rKe^{-rT}N(d_-)$
Rho	$\partial_r V$	$KTe^{-rT}N(d_-)$
Vanna	$\partial_{S\sigma} V$	$-\varphi(d_+)d_-/\sigma$
Volga	$\partial_{\sigma\sigma} V$	$\mathcal{V}d_+d_-/\sigma$

Useful identities: $\mathcal{V} = \Gamma S^2\sigma T$ (gamma and vega are one function up to scale, in Black–Scholes), and the BSM equation written as

$$\Theta + \tfrac{1}{2}\sigma^2 S^2\Gamma = r(V - S\Delta).$$

> **Intuition.** With $r = 0$, the BSM equation says theta and gamma are exactly balanced: *time decay is the price of convexity*. A long option position hopes for moves larger than implied; a short position collects the decay and hopes for small moves. Greeks are not separate risks. They are a single budget, spent according to the distribution of future moves. Near expiry at the money, gamma and theta both blow up like $1/\sqrt{T-t}$; gamma risk is concentrated in time and strike. Vega, in contrast, is largest for long-dated ATM options. Hence "gamma short-dated, vega long-dated" is the trader's first map.

Vanna and volga matter because volatility is stochastic: a Black–Scholes book with zero vega can still have vega that changes when spot or vol moves. The widely used *vanna–volga* method in FX adjusts Black–Scholes prices using the cost of hedging these exposures by three market instruments.

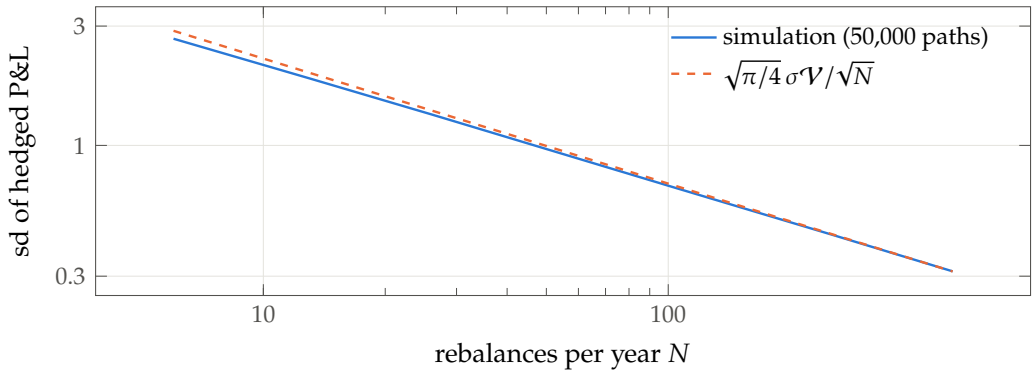


Figure 14.1: Hedging error of a short 1-year ATM call, $S_0 = 100$, $\sigma = 20\%$, $r = 0$. The mean P&L is within 0.01 of zero at each frequency; the spread falls as $1/\sqrt{N}$.

14.2 Discrete hedging error

Theorem 8.4 assumed continuous rebalancing. Hedging at N dates leaves a residual. Over one step, the P&L of the hedged option is about $\frac{1}{2}\Gamma S^2(\Delta R^2 - \sigma^2\Delta t)$, where ΔR is the return.

Theorem 14.1 (Boyle–Emanuel; Derman–Kamal)**.** *For an ATM option hedged at N equally spaced dates, with correct volatility, the standard deviation of final P&L is approximately*

$$\mathrm{sd}(P\&L) \approx \sqrt{\frac{\pi}{4}}\,\frac{\sigma\mathcal{V}}{\sqrt{N}}.$$

Sketch. Each step has an error $\frac{1}{2}\Gamma_t S_t^2\sigma^2\Delta t(Z^2-1)$ with $Z \sim \mathcal{N}(0,1)$ independent of the past, whose variance is $\frac{1}{2}\Gamma^2 S^4\sigma^4\Delta t^2$. Summing and integrating against the distribution of S_t, with Γ_t peaked at the strike, gives $\int_0^T \mathbb{E}[\Gamma_t^2 S_t^4]\,\mathrm{d}t$; this has a closed form that yields $\frac{\pi}{4}\cdot\frac{\mathcal{V}^2\sigma^2}{N}$ at the money. □

> **Numerical check.** Daily rebalancing ($N = 252$) gives 0.441 in simulation and 0.443 in theory, about 5.5% of the option's price of 7.97. Quadrupling N halves the error. At weekly ($N = 52$) it is 0.949 against 0.976. The approximation improves with N.

So there is no perfect hedge. An option seller holding many independent positions diversifies this; one big position does not.

Transaction costs. Rebalancing costs money, with a cost that grows with N, while the hedging error shrinks with N. The optimum balances them. Leland (1985) showed that with proportional cost k (round-trip) and interval δt, the option should be priced at the modified volatility

$$\sigma_L^2 = \sigma^2\Big(1 + \sqrt{\frac{2}{\pi}}\,\frac{k}{\sigma\sqrt{\delta t}}\Big)$$

for a short position (and lower for a long position). Whalley–Wilmott and Hodges–Neuberger derive no-trade bands around the delta. Practitioners rebalance when the delta drifts out of a band, not on a schedule.

14.3 Value at Risk and Expected Shortfall

Let L be the loss over a horizon (say 1 or 10 days).

Definition 14.2. At confidence α (for instance 99%), $\mathrm{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \le \ell) \ge \alpha\}$, the α-quantile of loss. The *Expected Shortfall* is

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\mathrm{d}u,$$

which equals $\mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha]$ when L has a continuous law.

For $L \sim \mathcal{N}(\mu, \sigma^2)$, $\mathrm{VaR}_\alpha = \mu + \sigma\Phi^{-1}(\alpha)$ and $\mathrm{ES}_\alpha = \mu + \sigma\varphi(\Phi^{-1}(\alpha))/(1-\alpha)$.

> **Numerical check.** At 99%, a unit variance normal has VaR = 2.326 and ES = 2.666. A unit-variance Student t_4 (5 million simulated draws) has VaR = 2.650 and ES = 3.688. So fat tails make a modest difference at the 99% quantile but a big one in shortfall. This is one reason regulators moved from 99% VaR to 97.5% ES in the Fundamental Review of the Trading Book.

Coherence. Artzner et al. (1999) ask for four axioms for a risk measure ϱ: monotonicity, translation invariance ($\varrho(L + c) = \varrho(L) + c$), positive homogeneity, and subadditivity ($\varrho(L_1 + L_2) \le \varrho(L_1) + \varrho(L_2)$). Subadditivity says diversification should not raise measured risk.

Example 14.3 (VaR is not subadditive)**.** Two independent bonds each lose 100 with probability 4% and 0 otherwise. At 95%, each has VaR = 0. But the sum loses at least 100 with probability $1 - 0.96^2 = 7.84\% > 5\%$, so $\mathrm{VaR}_{95\%}(L_1 + L_2) = 100 > 0 + 0$. The VaR rule says a concentrated position in one bond is safer than a diversified one. This is exactly the type of position (small probability, large loss: writing deep OTM options) that VaR hides.

Theorem 14.4 (Rockafellar–Uryasev)**.** $\mathrm{ES}_\alpha(L) = \min_{c\in\mathbb{R}}\left\{c + \frac{1}{1-\alpha}\mathbb{E}(L-c)^+\right\}$*, and the minimum is attained at* $c = \mathrm{VaR}_\alpha(L)$*.*

Proof. For continuous L: the function is convex in c, with derivative $1 - \frac{1}{1-\alpha}\mathbb{P}(L > c)$, zero at VaR_α. Plugging in gives $\mathrm{VaR}_\alpha + \frac{1}{1-\alpha}\mathbb{E}(L - \mathrm{VaR}_\alpha)^+ = \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha]$. □

This form shows that ES is subadditive, since $(x + y)^+ \le x^+ + y^+$ and the minimisation can be split. It also makes ES optimisation a linear program, over scenarios.

14.4 Computing and validating risk

Methods.

- **Parametric (delta-normal).** $L \approx -\delta^\top \Delta X$, with $\Delta X \sim \mathcal{N}(0, \Sigma)$: fast, poor for options.
- **Delta-gamma.** Adds $\frac{1}{2}\Delta X^\top \Gamma \Delta X$, a quadratic form in Gaussians, computed with moment expansions or Fourier methods.
- **Historical simulation.** Apply past factor returns to today's positions. There is no distribution assumption, but it knows only past scenarios and is slow to respond.

- **Monte Carlo.** Full revaluation under a fitted risk-factor model. The most flexible and costly.

The scaling rule $\mathrm{VaR}_{10d} = \sqrt{10}\ \mathrm{VaR}_{1d}$ assumes i.i.d. normal returns. Volatility clustering, autocorrelation and fat tails all make it inaccurate.

Backtesting. Count exceedances x in T days. Under correct coverage, $x \sim \mathrm{Binomial}(T, p)$, with $p = 1 - \alpha$. Kupiec's likelihood ratio test uses

$$\mathrm{LR} = -2 \ln \frac{(1-p)^{T-x} p^x}{(1 - x/T)^{T-x} (x/T)^x} \sim \chi^2_1.$$

Christoffersen's test adds independence: exceedances should not cluster. A correct-coverage model that fails the second is under-reacting to volatility.

Extreme value theory. Beyond a high threshold u, excess losses are approximately Generalised Pareto, $\mathbb{P}(L - u > y \mid L > u) \approx (1 + \xi y/\beta)^{-1/\xi}$, by the Pickands–Balkema–de Haan theorem. The tail index $\xi > 0$ gives a power-law tail, and moments of order $\geq 1/\xi$ are infinite. Equity returns have $\xi \approx 0.2$ to 0.4 (finite variance, possibly infinite fourth moment). The practical use is to extrapolate quantiles beyond the data.

Stress testing. A statistical model is calibrated to a past regime. Stress tests revalue the book under named scenarios (1987, 2008, March 2020) and under hypothetical joint shocks. Liquidity risk adds that a position's liquidation horizon lengthens exactly when it matters.

> **Pitfall.** Correlation is not a constant. A portfolio's measured VaR falls as correlations fall in calm periods, and then correlations jump towards one in a crisis. Gaussian copulas have zero tail dependence, and this mattered in pricing CDOs before 2008. A Student-t copula, or scenario-based thinking, addresses the point. Always ask: *what position is this number hiding?*

Exercises

14.1 Verify the Black–Scholes Greek table, using $S\varphi(d_+) = Ke^{-rT}\varphi(d_-)$.

14.2 Show that vega and gamma peak at slightly different strikes, and find them. What is the delta of a straddle at inception?

14.3 Use delta–gamma to find the break-even daily move for a long ATM option in terms of implied vol.

14.4 Compute the 95% and 99% ES for a Student t_ν (with unit variance), in closed form. Compare with the simulations above.

14.5 Construct a portfolio of an option and a stock whose 1-day VaR is zero by parametric delta-normal, but with very large actual tail losses.

14.6 A risk manager wants a 10-day 99% VaR estimate from 250 days of history. How many independent 10-day windows are there? What does that imply about the standard error of this quantile? Derive it from the asymptotic quantile variance $\alpha(1-\alpha)/(nf^2)$.

Chapter 15

Portfolio Theory and Asset Pricing

15.1 Mean–variance optimisation

There are n risky assets with mean returns μ and covariance $\Sigma > 0$. A portfolio w with $\mathbf{1}^\top w = 1$ has mean $w^\top \mu$ and variance $w^\top \Sigma w$. Markowitz (1952) asks for the least variance at each given mean:

$$\min_w \tfrac{1}{2} w^\top \Sigma w \quad \text{subject to} \quad \mu^\top w = m,\ \mathbf{1}^\top w = 1.$$

Theorem 15.1 (Frontier). *With $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$ and $D = AC - B^2 > 0$ (assets not all the same mean), the solution is $w = \Sigma^{-1}(\lambda \mu + \gamma \mathbf{1})$ with*

$$\lambda = \frac{Am - B}{D}, \quad \gamma = \frac{C - Bm}{D}, \qquad \text{and} \qquad \sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}.$$

Proof. The Lagrangian condition $\Sigma w = \lambda \mu + \gamma \mathbf{1}$ gives w. Substituting into the constraints gives the linear system $C\lambda + B\gamma = m$, $B\lambda + A\gamma = 1$. Then $\sigma^2 = w^\top \Sigma w = \lambda m + \gamma$. □

The frontier is a parabola in (m, σ^2), a hyperbola in (σ, m). The vertex is the *global minimum variance* portfolio $w_g = \Sigma^{-1}\mathbf{1}/A$, of mean B/A and variance $1/A$. Since w is linear in m, any two frontier portfolios span the entire frontier (*two-fund separation*).

Adding a riskless asset. With rate r, invest x in risky assets and $1 - \mathbf{1}^\top x$ in cash. Minimising the variance at a target excess return gives $x \propto \Sigma^{-1}(\mu - r\mathbf{1})$. So the risky proportions are the same for everyone. The *tangency portfolio* is

$$w_T = \frac{\Sigma^{-1}(\mu - r\mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\mu - r\mathbf{1})},$$

which has the highest Sharpe ratio, with $\mathrm{SR}^2 = (\mu - r\mathbf{1})^\top \Sigma^{-1}(\mu - r\mathbf{1})$. The frontier becomes a straight line, the *capital market line*.

> **Intuition.** Diversification works because variance adds with correlation, while mean adds without it. Note the structure of the optimal weights, $\Sigma^{-1}(\mu - r)$. The matrix inverse is the danger. It finds the combinations with the smallest apparent risk and bets heavily on them, and these are exactly the eigenvectors of Σ with the smallest,

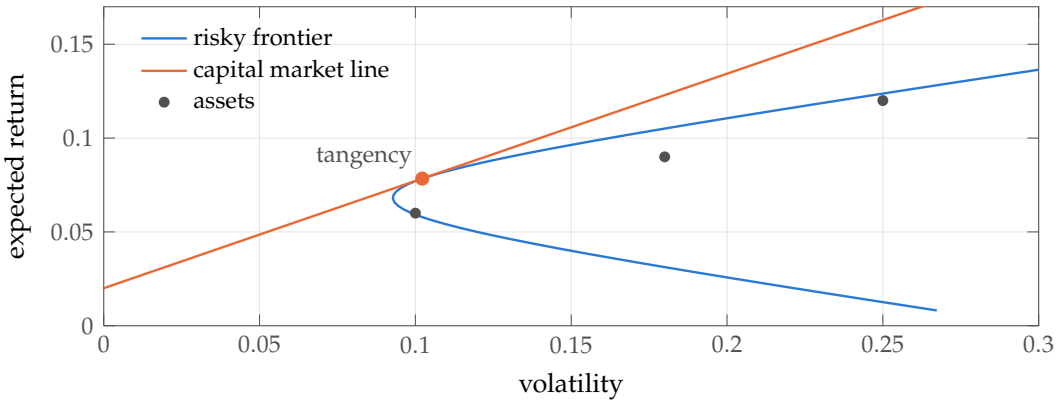


Figure 15.1: Frontier for three assets (means $6, 9, 12\%$; volatilities $10, 18, 25\%$; correlations $0.2, 0.1, 0.4$) with $r = 2\%$. The tangency portfolio $(0.589, 0.208, 0.202)$ has mean 7.84%, volatility 10.22%, and Sharpe ratio 0.5716, which equals the quantity $\sqrt{(\mu - r\mathbf{1})^\top \Sigma^{-1} (\mu - r\mathbf{1})}$.

worst-estimated eigenvalues. An optimiser is an *error maximiser*.

15.2 Estimation error

The sampling error in means is huge. The standard error of a mean annual return over T years is $\sigma/\sqrt{T}$, whatever the sampling frequency. With $\sigma = 20\%$, even 25 years leaves $\pm 4\%$. This is often as large as the premium itself. (Volatility, in contrast, is estimated much better by sampling more often.)

Numerical check. For the three assets above, with the true covariance known and only means estimated from five years of monthly returns, 5,000 simulations of the tangency weights give these 5%–95% intervals: first asset $[-1.05, 2.42]$, second $[-1.53, 1.99]$, third $[-1.19, 1.45]$. The true weights are 0.589, 0.208, 0.202. So the data are almost silent, even in this idealised experiment.

Remedies:

- **Shrinkage of the covariance** (Ledoit–Wolf): $\hat{\Sigma} = (1 - \delta) S + \delta F$, with target F being a multiple of the identity or a factor model, and an optimal δ estimated from the data. With n comparable to T, the sample covariance's eigenvalues are badly spread (Marčenko–Pastur), and shrinking them helps.
- **Constraints:** no-short limits act as implicit shrinkage (Jagannathan–Ma).
- **Bayesian views (Black–Litterman).** Start with the equilibrium returns $\pi = \delta \Sigma w_{\text{mkt}}$ implied by market-cap weights. Express views as $P\mu = q + \epsilon$, with $\epsilon \sim \mathcal{N}(0, \Omega)$. The posterior mean is

$$\mu_{\text{BL}} = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P \right]^{-1} \left[(\tau\Sigma)^{-1} \pi + P^\top \Omega^{-1} q \right].$$

  It is Gaussian conditioning (Example 1.22) again, and yields sensible portfolios close to market weights unless views are strong.

- **Risk parity:** ignore means, and equalise each asset's contribution to risk, $w_i(\Sigma w)_i = \text{const}$. It avoids the worst estimation problem at the price of ignoring return views.

15.3 CAPM and factor models

Theorem 15.2 (CAPM)**.** *Suppose all investors are mean–variance optimisers with the same beliefs and riskless borrowing. Then the market portfolio is the tangency portfolio, and for every asset,*

$$\mathbb{E}R_i - r = \beta_i(\mathbb{E}R_m - r), \qquad \beta_i = \frac{\operatorname{Cov}(R_i, R_m)}{\operatorname{Var} R_m}.$$

Proof. Each investor holds a combination of cash and w_T. Summing over investors, supply equals demand, so $w_T = w_m$. From $\Sigma w_m \propto \mu - r\mathbf{1}$, the i-th row reads $\operatorname{Cov}(R_i, R_m) = c(\mathbb{E}R_i - r)$. Multiply by $w_m^\top$ to find $c = \operatorname{Var} R_m / (\mathbb{E}R_m - r)$. □

> **Intuition.** The only risk that earns a premium is the part you cannot diversify. An asset's idiosyncratic variance is irrelevant to its expected return. What counts is covariance with what you already hold. An asset that pays off in bad times (low beta, or an insurance asset) can have a low or negative premium.

Arbitrage pricing and factor models. Suppose $R_i = \mathbb{E}R_i + \sum_k \beta_{ik} F_k + \epsilon_i$, with ϵ_i diversifiable. Ross's APT says that if there is no asymptotic arbitrage, $\mathbb{E}R_i - r \approx \sum_k \beta_{ik}\lambda_k$. Unlike CAPM, it needs no equilibrium, just a law of large numbers. Empirical factors include value, size, momentum, quality and low-vol (Fama–French, Carhart). Factor models also give a low-rank covariance $\Sigma = BFB^\top + D$, which has far fewer parameters than $n^2/2$, and an easy inverse (Woodbury).

15.4 Stochastic discount factors

No arbitrage implies (in a finite world, by separation of hyperplanes; generally, by Theorem 5.10) a positive *stochastic discount factor* M such that for every asset

$$P_t = \mathbb{E}_t[M_T P_T / M_t].$$

Comparing with the change of measure, $M_T/M_t = D_{t,T}\, Z_T/Z_t$: this is the state price density ζ of the class notes. So *a single object unifies pricing*: real-world expectation times a kernel, or risk-neutral expectation with discounting. From $\mathbb{E}[M(R_i - r)] = 0$,

$$\mathbb{E}R_i - r = -(1 + r)\operatorname{Cov}(M, R_i).$$

Assets with a payoff low when M is high (bad states, when wealth is scarce) must offer a higher return. CAPM is the special case where M is linear in R_m. Cauchy–Schwarz gives the *Hansen–Jagannathan bound*:

$$\frac{\sigma(M)}{\mathbb{E}M} \geq \frac{|\mathbb{E}R_i - r|}{\sigma(R_i)}.$$

An equity Sharpe ratio of 0.4 requires a very volatile M, much more than consumption data suggest. This is the equity premium puzzle.

15.5 Merton's portfolio problem

Let an investor hold a fraction π_t in a GBM stock and the rest in cash:

$$\mathrm{d}X_t = X_t\big[(r + \pi_t(\mu - r))\mathrm{d}t + \pi_t\sigma\mathrm{d}W_t\big],$$

maximising $\mathbb{E}[U(X_T)]$, with CRRA utility $U(x) = x^{1-\gamma}/(1-\gamma)$.

Theorem 15.3 (Merton 1969)**.** *The optimal fraction is constant:*

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2}.$$

Proof. Let $V(t,x) = \sup_\pi \mathbb{E}_{t,x}U(X_T)$. By the dynamic programming principle, $V(t, X_t)$ is a supermartingale for every π, and a martingale at the optimum. Itô's formula gives the *Hamilton–Jacobi–Bellman equation*:

$$\partial_t V + \sup_\pi\Big\{(r + \pi(\mu - r))x\partial_x V + \tfrac{1}{2}\pi^2\sigma^2x^2\partial_{xx}V\Big\} = 0, \quad V(T,x) = U(x).$$

The first-order condition is $\pi^* = -\frac{(\mu-r)\partial_x V}{\sigma^2 x\partial_{xx}V}$. Try $V = f(t)x^{1-\gamma}/(1-\gamma)$. Then $-x\partial_{xx}V/\partial_x V = \gamma$, and so $\pi^* = (\mu - r)/\gamma\sigma^2$. The equation reduces to the ODE $f' + (1-\gamma)\big(r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\big)f = 0$, $f(T) = 1$. A verification argument (Itô's formula on V with an arbitrary π) completes the proof. □

The martingale (dual) method. In a complete market, any wealth X_T with $\mathbb{E}[M_T X_T] = X_0$ can be replicated. So optimise over terminal wealth, not strategies: maximise $\mathbb{E}U(X_T)$ subject to that budget. The first-order condition gives $U'(X_T) = yM_T$, so $X_T = (U')^{-1}(yM_T)$. Then hedge it. For CRRA, it reproduces the theorem, and the method applies where HJB is hard.

Kelly and log-optimality. For $\gamma = 1$ (log utility), $\pi^* = (\mu - r)/\sigma^2$ maximises the long-run growth rate

$$g(\pi) = \lim_{T\to\infty}\frac{1}{T}\ln\frac{X_T}{X_0} = r + \pi(\mu - r) - \tfrac{1}{2}\pi^2\sigma^2.$$

Numerical check. For $\mu = 8\%$, $\sigma = 20\%$, $r = 2\%$: Kelly is $\pi^* = 1.5$.

π/π^*	0.5	1	2	3
Growth rate	5.375%	6.500%	2.000%	−11.500%

Betting twice Kelly gets you exactly the riskless rate: all the extra return is eaten by volatility drag. Further out the growth rate turns negative, and then wealth tends to zero with probability one, despite a positive expected return. Half Kelly gives three-quarters of the excess growth at half the volatility. Since μ is poorly known, overestimating it risks being on the wrong side, and that is why practitioners stay at or below half Kelly.

Active management. Grinold's *fundamental law* says the information ratio is approximately $\text{IR} \approx \text{IC}\sqrt{\text{BR}}$, where IC is the correlation of forecasts with outcomes and BR is the number of independent bets per year. A weak signal traded widely (say IC = 0.05 on 1,000 independent bets, IR ≈ 1.6) can beat a strong one traded rarely. Breadth is the operative word: bets on correlated names are not independent.

Exercises

15.1 Prove the two-fund theorem. Show that the covariance between w_g and any frontier portfolio is $1/A$.

15.2 Find the portfolio weights maximising $\mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$ with no budget constraint. Interpret λ and the result for Σ diagonal.

15.3 With two assets of correlation ρ and equal volatility, find the GMV variance. For n assets with common correlation ρ, show $\sigma_p^2 \to \rho\sigma^2$ as $n \to \infty$. What does it say about diversification limits?

15.4 Show that the Sharpe ratio of a portfolio with leverage is unchanged, and find the relationship of optimal leverage to the Sharpe ratio under Kelly.

15.5 Solve Merton's problem with intermediate consumption $\int_0^T e^{-\delta t} U(c_t)\mathrm{d}t$ and $T = \infty$. Find c^*/X.

15.6 In the binomial world of the class notes, compute the SDF M and check $\mathbb{E}[M] = 1/(1+r)$. Using the $N = 2$ example, verify $V_0 = \mathbb{E}[MV_N]$ for a call.

15.7 A bet pays b-to-1 with probability p. Derive the Kelly fraction $f^* = p - (1-p)/b$. (This is a common interview question.)

Chapter 16

Financial Statistics and Time Series

16.1 Stylised facts

Log-returns $r_t = \ln(S_t/S_{t-1})$ share some empirical regularities across assets (Cont, 2001):

1. Linear autocorrelation of returns is close to zero beyond very short lags.
2. Unconditional distributions have fat tails, with tail index around 3 to 5.
3. *Volatility clustering:* squared and absolute returns have strong, slowly decaying autocorrelation.
4. *Leverage effect:* volatility is negatively correlated with past returns.
5. Aggregational Gaussianity: tails thin as horizon lengthens.
6. Volume and volatility are positively related, and there are intraday patterns.

Black–Scholes (i.i.d. Gaussian log returns) explains none of (2) to (6). The rest of this chapter deals with the tools for them.

16.2 Regression and inference

For $y = X\beta + \epsilon$ with full-rank X, $\hat{\beta} = (X^\top X)^{-1} X^\top y$ is the orthogonal projection of y onto the column space of X.

Theorem 16.1 (Gauss–Markov)**.** *If* $\mathbb{E}[\epsilon \mid X] = 0$ *and* $\operatorname{Var}(\epsilon \mid X) = \sigma^2 I$*, then* $\hat{\beta}$ *has the least variance among linear unbiased estimators, with* $\operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}$.

Financial data usually violate the assumptions, and the *point estimate often survives, but the standard errors do not*.

- **Heteroskedasticity:** use White's standard errors, $(X^\top X)^{-1} (\sum \hat{\epsilon}_t^2 x_t x_t^\top) (X^\top X)^{-1}$.
- **Autocorrelation:** use Newey–West. This is essential when returns overlap. Regressing 12-month returns sampled monthly inflates t-statistics hugely.
- **Regressors that are not exogenous:** a predictor like dividend yield is a persistent, lagged function of price, and this causes the Stambaugh bias in predictive regressions.
- **Errors in variables:** attenuation of the slope towards zero.

Maximum likelihood. Under regularity conditions, $\sqrt{T}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, I(\theta)^{-1})$, where I is the Fisher information, and this is asymptotically efficient. If the likelihood is misspecified (say, Gaussian likelihood for t-distributed data), we have *quasi*-MLE. It is often still consistent, but needs the sandwich covariance $H^{-1}GH^{-1}$.

What can be estimated from a price path?

Proposition 16.2. *Let* $\mathrm{d}\ln S = \nu \mathrm{d}t + \sigma \mathrm{d}W$ *be observed at* n *intervals over total time* T*. The MLE of* ν *is* $(\ln S_T - \ln S_0)/T$*, with variance* σ^2/T*,* independent of n*. The realised variance estimator* $\hat{\sigma}^2 = \frac{1}{T}\sum(\Delta \ln S)^2$ *has a variance of about* $2\sigma^4/n$*, which goes to zero with* n*.*

> **Intuition.** This is the quadratic variation theorem in statistical clothing: sampling more often over a fixed interval tells you nothing about drift (you only need the endpoints), but with continuous observation volatility is known exactly. It also explains why mean estimation is the heart of portfolio trouble. To pin down a drift of 8% with a volatility of 20% to one standard error of 2% takes 100 years.

16.3 Time series

Stationarity and ARMA. A process is weakly stationary when its mean and autocovariance $\gamma(k)$ do not depend on time. By Wold's theorem, a stationary process is a moving average of white noise, plus a deterministic part. An AR(1), $x_t = c + \phi x_{t-1} + \epsilon_t$ with $|\phi| < 1$, has $\mathbb{E}x = c/(1-\phi)$, $\operatorname{Var} x = \sigma_\epsilon^2/(1-\phi^2)$ and $\rho(k) = \phi^k$. It is exactly the discrete sampling of OU, with $\phi = e^{-\kappa h}$. ARMA(p, q) is identified by the ACF and PACF shapes, and estimated by MLE; information criteria (AIC, BIC) pick the order; and the Ljung–Box statistic tests residual autocorrelation.

Unit roots. When $\phi = 1$, x_t is a random walk: not stationary, variance growing. Estimating ϕ under the unit root has a non-standard distribution (Dickey–Fuller), biased downward, with critical values different from t. Regressing one independent random walk on another gives "significant" t-statistics with high R^2: the *spurious regression* (Granger–Newbold). So regress returns, not prices, unless there is cointegration.

Cointegration. Two $I(1)$ series are cointegrated if a linear combination $x_t - \beta y_t$ is stationary. Engle–Granger: estimate β by OLS, then run a unit-root test on the residual (with its own critical values). Johansen's test handles multiple series. This underlies pairs trading: if the spread $s_t = x_t - \beta y_t$ is OU with speed κ, the half-life is $\ln 2/\kappa$. Entry when the spread is unusually far from its mean is a standard statistical arbitrage. Note that the trade's main risk is that the relationship breaks: the OU parameters are not structural.

16.4 Conditional volatility: GARCH

Definition 16.3 (GARCH(1, 1); Bollerslev 1986)**.** $r_t = \sqrt{h_t}\, z_t$, with z_t i.i.d. zero mean and unit variance, and

$$h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}, \qquad \omega > 0,\ \alpha, \beta \geq 0.$$

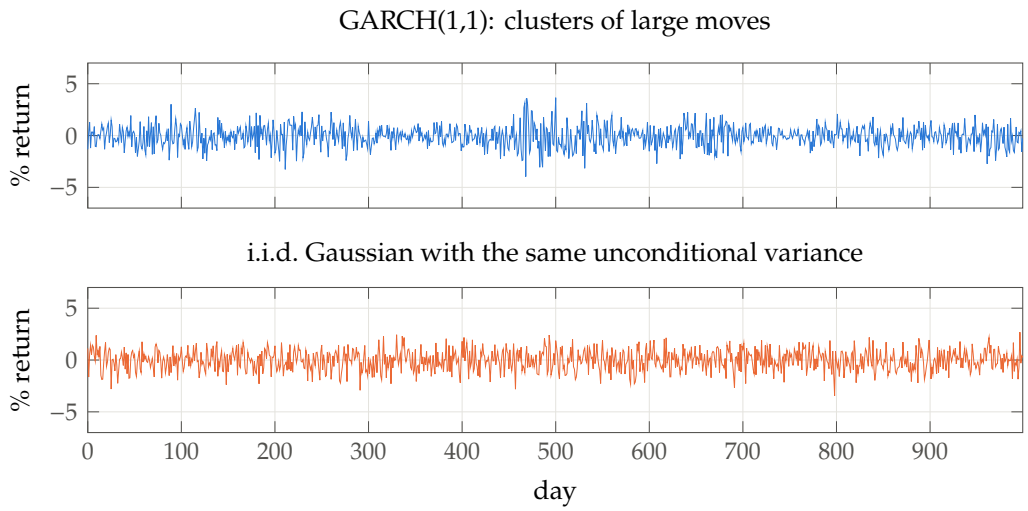


Figure 16.1: 1,000 simulated days for $\omega = 2 \times 10^{-6}$, $\alpha = 0.08$, $\beta = 0.90$ (annualised unconditional volatility 15.9%) and for a constant-volatility model.

Proposition 16.4. *If $\alpha + \beta < 1$:*

1. $\mathbb{E}r_t^2 = v := \omega/(1-\alpha-\beta)$.
2. ***Forecast:*** $\mathbb{E}_t[h_{t+k}] = v + (\alpha+\beta)^{k-1}(h_{t+1} - v)$, *with half-life* $\ln\frac{1}{2}/\ln(\alpha+\beta)$.
3. ***Fat tails*** *from Gaussian z (when $1-(\alpha+\beta)^2-2\alpha^2 > 0$): kurtosis is*

$$\frac{\mathbb{E}r^4}{(\mathbb{E}r^2)^2} = \frac{3\big(1-(\alpha+\beta)^2\big)}{1-(\alpha+\beta)^2-2\alpha^2} > 3.$$

4. ***Clustering:*** *r_t^2 follows an ARMA(1,1), with $\rho(1) = \frac{\alpha(1-\alpha\beta-\beta^2)}{1-2\alpha\beta-\beta^2}$ and $\rho(k) = (\alpha+\beta)\rho(k-1)$ for $k \geq 2$.*

Proof. (1) Taking expectations, $v = \omega + \alpha v + \beta v$. (2) follows from $h_{t+1} - v = (\alpha+\beta)(h_t - v) + \alpha h_t(z_t^2 - 1)$, and the last term has conditional mean 0. (3) Square the recursion and use $\mathbb{E}z^4 = 3$ to solve for $\mathbb{E}h^2$. Then $\mathbb{E}r^4 = 3\mathbb{E}h^2$. (4) Writing $u_t = r_t^2 - h_t$, a martingale difference, gives $r_t^2 = \omega + (\alpha+\beta)r_{t-1}^2 + u_t - \beta u_{t-1}$. □

Intuition. Today's squared shock raises tomorrow's variance. So a large move today predicts larger moves for some time. Even with Gaussian z_t, the unconditional distribution is a mixture over random h_t, hence fat-tailed. In practice $\alpha+\beta$ is close to 1 (around 0.98), which is high persistence. RiskMetrics' EWMA, $h_t = \lambda h_{t-1} + (1-\lambda)r_{t-1}^2$ with $\lambda = 0.94$, is the limit $\omega = 0$, $\alpha + \beta = 1$.

Numerical check. Simulating 2,000,000 days with these parameters: unconditional variance 1.005×10^{-4} (theory 1.000×10^{-4}); kurtosis 4.57 (theory 4.43; convergence is slow, since high moments of GARCH are heavy-tailed); autocorrelation of r^2 at lags $1, 2, 5, 10$ is $0.213, 0.217, 0.193, 0.180$ (theory $0.205, 0.201, 0.189, 0.171$).

Estimation and variants. Estimate by quasi-MLE, maximising the Gaussian log-likelihood $-\frac{1}{2}\sum[\ln h_t + r_t^2/h_t]$, which is consistent for non-Gaussian z. Variants: GJR-GARCH and EGARCH add the leverage effect with asymmetry between up and down moves. As a diffusion limit, GARCH converges to a stochastic vol model with $\mathrm{d}h = \kappa(\theta - h)\mathrm{d}t + \xi h \mathrm{d}B$ (Nelson 1990). Heston and Nandi (2000) built a GARCH with closed-form option prices, connecting the discrete and continuous worlds.

16.5 Realised volatility

Theorem 3.3 suggests estimating integrated variance with RV $= \sum_i r_i^2$ from intraday data. With continuous semimartingale prices, it converges to $\int \sigma^2 \mathrm{d}t$, with jumps adding $\sum(\Delta J)^2$. Bipower variation $\frac{\pi}{2}\sum |r_i||r_{i-1}|$ removes the jump part.

> **Pitfall.** At very high frequency, prices are not semimartingales. Bid–ask bounce adds noise: observed = true $+\epsilon$. Then $\mathbb{E}\mathrm{RV} = \int \sigma^2 + 2n \operatorname{Var}\epsilon$, which *diverges* with n. The *volatility signature plot* shows this. So practitioners sample at about five minutes, or use noise-robust estimators: two-scale (Zhang–Mykland–Aït-Sahalia), realised kernels.

Range-based estimators use more information per day: Parkinson's $\hat{\sigma}^2 = \frac{1}{4\ln 2}\left(\ln \frac{H}{L}\right)^2$ is about five times more efficient than a close-to-close squared return, as the range distribution comes from reflection (Chapter 3). Corsi's HAR model forecasts RV with a regression on daily, weekly and monthly RV, approximating the long memory seen in the data.

16.6 The Kalman filter

Consider a linear Gaussian state-space model:

$$\begin{aligned} x_t &= Fx_{t-1} + w_t, & w_t &\sim \mathcal{N}(0, Q), & &\text{(state)} \\ y_t &= Hx_t + v_t, & v_t &\sim \mathcal{N}(0, R). & &\text{(observation)} \end{aligned}$$

We want $\mathbb{E}[x_t \mid y_1, \dots, y_t]$. All joint laws are Gaussian, so Example 1.22 gives everything. Let (m_{t-1}, P_{t-1}) be the posterior mean and covariance at $t-1$.

Theorem 16.5 (Kalman)**.** Predict: $m_t^- = Fm_{t-1}$, $P_t^- = FP_{t-1}F^\top + Q$.
Update: *with* $S_t = HP_t^-H^\top + R$ *and gain* $K_t = P_t^-H^\top S_t^{-1}$,

$$m_t = m_t^- + K_t(y_t - Hm_t^-), \qquad P_t = (I - K_tH)P_t^-.$$

Proof. Given $y_{1:t-1}$, the pair (x_t, y_t) is jointly Gaussian with means (m_t^-, Hm_t^-), $\operatorname{Var} y_t = S_t$, and $\operatorname{Cov}(x_t, y_t) = P_t^-H^\top$. The conditional mean and variance of x_t given y_t are therefore exactly the formulas above, by the regression formula of Example 1.22. □

> **Intuition.** The filter is a running regression. It predicts, sees the surprise (innovation) $y_t - Hm_t^-$, and updates in proportion to how much it trusts the data: gain near 1 when state noise dominates, near 0 when observation noise dominates. For a random walk plus noise (scalar $F = H = 1$), the steady state is exponential smoothing, with a gain fixed by signal-to-noise q/R.

Numerical check. For the local level model with $Q = 0.01$, $R = 0.25$ over 500 steps: gain converges to 0.1810, equal to the steady-state Riccati solution. The filter's RMSE is 0.246, compared with 0.512 for using the raw observation.

Applications: time-varying hedge ratios in pairs trading, trend extraction, estimating unobserved volatility or the short rate from yields, and dealing with missing data. Likelihoods are produced from the innovations, so parameter estimation by MLE is cheap. For non-linear models, the extended and unscented filters use local linearisations, and particle filters use Monte Carlo.

16.7 Backtests and multiple testing

Sharpe ratio error. For i.i.d. Gaussian returns, the annualised Sharpe has an approximate standard error of $\sqrt{(1 + \frac{1}{2}\mathrm{SR}^2)/T}$ with T in years. A Sharpe of 1 over 4 years has a standard error of 0.6; a t-statistic of 1.6 is not significant. Lo (2002) gives autocorrelation corrections.

Multiple testing. Try N independent zero-alpha strategies and pick the best. Its Sharpe t-statistic has expected value about $\sqrt{2 \ln N}$. So a thousand strategies give an apparent $t \approx 3.7$ from pure noise. Published factors face this issue (Harvey–Liu–Zhu suggest a hurdle of $t > 3$). Bonferroni and Benjamini–Hochberg control false positives; the *deflated Sharpe ratio* (Bailey–López de Prado) adjusts for the number of trials. So keep a record of how many ideas you tested.

Pitfall. Common ways to fool yourself in a backtest:

- *Look-ahead bias:* features using information unavailable at the time (restated data, full-sample normalisation).
- *Survivorship bias:* a universe of today's survivors, not all names that existed at the time.
- *Costs:* ignoring spread, impact, borrow, funding.
- *Overfitting:* tuning on a test set. Purged and embargoed cross-validation addresses leakage from overlapping labels.
- *Regime dependence:* all data come from a few correlated macro regimes.

Exercises

16.1 Show that OLS on $x_t = \phi x_{t-1} + \epsilon_t$ has $\mathbb{E}\hat{\phi} \approx \phi - (1 + 3\phi)/T$ (Kendall bias). Why does this bias inflate mean reversion estimates, and overstate pairs-trading profits?

16.2 Derive the steady-state gain of the local level Kalman filter, as a function of q/R.

16.3 Derive the GARCH(1, 1) kurtosis formula from the proposition.

16.4 Simulate a Merton jump process and compare RV with bipower variation. Which one estimates σ^2?

16.5 Estimate OU parameters by regression from a discrete sample. What is the distribution of the half-life estimator for a small sample?

16.6 Show the expected range $\mathbb{E}[\max W - \min W]$ on $[0, T]$ is $\sqrt{8T/\pi}$. Using reflection, justify Parkinson's estimator: find $\mathbb{E}[(\max - \min)^2]$ and check the factor $4 \ln 2$.

16.7 Show that the expected maximum of N independent standard normals is near $\sqrt{2 \ln N}$. Interpret for multiple testing.

Chapter 17

Market Context

The earlier chapters leave out most of what makes a product tradeable. This chapter collects the essentials, in the spirit of Hull's textbook, focused on the places where they alter the mathematics.

17.1 Forwards, futures and carry

For an asset with continuous yield q (dividends, or foreign rate for FX) and funding at r, static arbitrage gives

$$F(t,T) = S_t e^{(r-q)(T-t)}.$$

- **Equity:** q includes dividends and repo (the cost of borrowing a stock to short it). Dividends are in fact discrete and uncertain. Dividend modelling matters for long-dated options, and a hard-to-borrow stock has high implied repo.
- **FX:** covered interest parity, $F = Se^{(r_d - r_f)T}$. Garman–Kohlhagen is Black–Scholes with $q = r_f$. Since 2008 the cross-currency basis has shown this relation is not costless to arbitrage, given limited bank balance sheets.
- **Commodities:** storage cost and *convenience yield* (the benefit of holding physical inventory). Curves can be in contango or backwardation, and there is seasonality.
- **Black's formula:** every European option can be priced directly on the forward, removing most of the carry problem:

$$C = D(0,T)\big[FN(d_+) - KN(d_-)\big], \qquad d_\pm = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

Put–call parity reads $C - P = D(0,T)(F - K)$. It is purely static and independent of any model. So market calls and puts at the same strike must have the same implied vol. Any deviation reflects borrow cost or wrong dividends, so parity is commonly used to back out the implied forward from listed options.

17.2 Option conventions and microstructure

- Index options (SPX) are European, cash-settled; single-stock options are American. So implied vol extraction needs an American pricer or a parity-based forward.

- Quotes come as bid and ask with a wide spread on the wings. Implied vol should be computed separately from each, or from the mid, knowing that OTM wings have large relative errors.
- Use OTM options only (calls above forward, puts below) when fitting a smile, since they are more liquid and have no intrinsic value.
- Volume concentrates in near-dated, near-the-money strikes. The surface is thin elsewhere, and interpolation is a model.
- Expiry calendars, weekly and same-day ("0DTE") options, and event days (earnings, central banks) put *kinks* in the term structure of variance. Work in *business time*, with weights by event.

Market making and optimal execution. A market maker earns the spread and bears inventory and adverse selection risk. The Avellaneda–Stoikov (2008) model uses HJB (as in Chapter 15) to set bid and ask around an inventory-adjusted reservation price, $r = S - q\gamma\sigma^2(T-t)$. Almgren–Chriss (2001) trade off market impact against timing risk for liquidating a block, giving an optimal trajectory $x(t) \propto \sinh(\kappa(T-t))$. Kyle's model (1985) relates price impact to the presence of informed traders. Empirically, impact grows roughly as the square root of the traded fraction of volume.

17.3 Funding, collateral and XVA

Pre-2008 textbooks use a single risk-free rate for funding and discounting. Today, most derivatives are collateralised, and the correct discount rate is the rate paid on collateral: *OIS* (overnight). Multi-curve frameworks use one curve for discounting and others for projecting floating rates.

Uncollateralised trades carry valuation adjustments, computed as expectations over the life of the trade.

$$\mathrm{CVA} = (1-R)\int_0^T D(0,t)\,\mathbb{E}^{\mathbb{Q}}[V_t^+]\,\mathrm{dPD}(t).$$

CVA charges for counterparty default, where R is the recovery rate, V^+ is positive exposure, and PD is the default probability. DVA is the mirror for one's own default, FVA covers funding, MVA covers initial margin, and KVA covers regulatory capital. Computing $\mathbb{E}^{\mathbb{Q}}[V_t^+]$ involves nested simulation, so regression methods (as with Longstaff–Schwartz) are the standard way.

17.4 Credit

A simple *reduced-form* model: default τ is the first jump of a Poisson process with hazard rate λ_t. Then

$$\mathbb{Q}(\tau > T) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^T \lambda_s \mathrm{d}s}\Big], \qquad \text{which is a bond price with } r \text{ replaced by } \lambda.$$

For a defaultable zero-coupon bond with recovery R and constant parameters, the credit spread is approximately $s \approx (1-R)\lambda$. A CDS is an insurance contract on default, with a running premium. Its premium leg and protection leg yield the "credit triangle" $s \approx (1-R)\lambda$

at inception. The *structural* (Merton) alternative treats equity as a call on firm value, with default when assets fall below the debt, and it links credit spreads to equity volatility.

17.5 Model risk and validation

- Fit quality is a poor check for a model: flexible models fit anything. Test instead on *hedging performance*, out of sample.
- Check all prices against bounds: arbitrage-free surface, and reproduction of the hedge instruments.
- Compare numerical methods with each other (Monte Carlo against PDE against closed form). This booklet's green boxes are a miniature example.
- Keep a model *reserve* for uncertainty on parameters that cannot be hedged, such as vol-of-vol or correlation.

Intuition. Derman's advice is worth keeping in mind. A model is not a law of nature, it is an analogy between what you want to value and what you can observe. The analogy is useful insofar as it lets you price by relating to liquid instruments, and it fails where those instruments cannot reach.

Exercises

17.1 A stock is at 100, and a 1-year forward trades at 102 with rate $r = 5\%$. What is the implied dividend yield? Set out the arbitrage if a forward trades at 106.

17.2 Use parity quotes $C(100) = 8$, $P(100) = 7.5$, $D(0,1) = 0.96$ to find the implied forward.

17.3 Derive the Garman–Kohlhagen formula by quoting in the foreign numeraire. Show that a call on EUR/USD is a put on USD/EUR.

17.4 Derive the credit triangle, assuming flat hazard and rates.

17.5 Derive the Merton structural model's credit spread for a debt with face B at T in terms of a put. Why do short-maturity spreads tend to zero in this model, unlike in the market? What would fix it?

17.6 Solve the Almgren–Chriss problem: minimise $\int \eta \dot{x}^2 + \lambda\sigma^2 x^2 \mathrm{d}t$ with $x(0) = X$, $x(T) = 0$.

Where to Go Next

This booklet is an overview; the books below go deep. They are listed in the rough order they could be read, and each is chosen for a reason.

Probability and stochastic calculus

- **Williams**, *Probability with Martingales.* Short and clear; the right first book in measure-theoretic probability.
- **Shreve**, *Stochastic Calculus for Finance I and II.* The class notes follow it. Do the exercises.
- **Øksendal**, *Stochastic Differential Equations.* SDEs, diffusions, optimal stopping and control, at a friendly pace.
- **Karatzas and Shreve**, *Brownian Motion and Stochastic Calculus.* The full treatment with complete proofs.
- **Revuz and Yor**, *Continuous Martingales and Brownian Motion.* The advanced reference, for local time and excursions.
- **Protter**, *Stochastic Integration and Differential Equations.* For general semimartingales and jumps.

Derivatives pricing

- **Hull**, *Options, Futures, and Other Derivatives.* Market context and instruments.
- **Björk**, *Arbitrage Theory in Continuous Time.* Very clean on change of numeraire and on rates.
- **Gatheral**, *The Volatility Surface.* The concise book on smile modelling.
- **Bergomi**, *Stochastic Volatility Modeling.* What a practitioner cares about: dynamics, not calibration.
- **Cont and Tankov**, *Financial Modelling with Jump Processes.* The reference on Lévy models.
- **Brigo and Mercurio**, *Interest Rate Models: Theory and Practice.* A thorough reference for short-rate models and the LMM.
- **Andersen and Piterbarg**, *Interest Rate Modeling* (3 volumes). The industrial-strength reference.
- **Duffie**, *Dynamic Asset Pricing Theory.* The rigorous general-equilibrium viewpoint.

Numerical methods

- **Glasserman**, *Monte Carlo Methods in Financial Engineering.* Essential.
- **Duffy**, *Finite Difference Methods in Financial Engineering*; or **Tavella and Randall**, *Pricing Financial Instruments: The Finite Difference Method.*
- **Kloeden and Platen**, *Numerical Solution of Stochastic Differential Equations.* The higher-order schemes.

Risk, portfolios and statistics

- **McNeil, Frey and Embrechts**, *Quantitative Risk Management.* Copulas, EVT, and coherent measures.
- **Cochrane**, *Asset Pricing.* The SDF view of everything.
- **Grinold and Kahn**, *Active Portfolio Management.* The practice of alpha and risk.
- **Tsay**, *Analysis of Financial Time Series*; **Hamilton**, *Time Series Analysis* for more depth.
- **López de Prado**, *Advances in Financial Machine Learning.* On avoiding self-deception in backtests.
- **Cartea, Jaimungal and Penalva**, *Algorithmic and High-Frequency Trading.* Market making and execution via stochastic control.

Trading perspective

- **Natenberg**, *Option Volatility and Pricing*; and **Sinclair**, *Volatility Trading.* How practitioners think about the positions.
- **Taleb**, *Dynamic Hedging.* Practical wisdom about real-world Greeks.

Suggested projects

Reading alone will not make this stick. Each of these takes from an evening to a weekend, and each teaches something a book cannot:

1. Rewrite `code/make_data.py` in your own way. Break it, then fix it.
2. Build a smile from real option quotes. Compute the implied vols, fit SVI, check butterfly and calendar arbitrage, and extract the risk-neutral density.
3. Build a Heston calibrator using the Lewis integral. Observe parameter stability from day to day.
4. Compute Dupire local vol from your fitted SVI surface. Reprice vanillas with a Monte Carlo under this local vol to check it.
5. Run a real delta-hedging experiment on an index: compare realised P&L with the gamma formula of Theorem 8.4.

6. Fit a Hull–White model to today's yield curve and price a Bermudan swaption on a tree.
7. Fit GARCH by QMLE to index returns, and compare forecasts against realised variance from intraday data.
8. Backtest a pairs strategy carefully, and then count how many ways the result can be an artefact.

Mathematical gaps to fill along the way

Linear algebra (spectral theory, Cholesky, conditioning), real analysis (Lebesgue, Fourier transforms), PDEs (the maximum principle, the heat kernel, viscosity solutions for HJB), and convex optimisation (duality). Each appears here only as far as it was needed, and each repays further study.